2026 Fall Linear Algebraic Groups Note

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Introduction

1. [Mil17] J. S. Milne, *Algebraic Groups*.
2. [Wat79] W. C. Waterhouse, *Introduction to Affine Group Schemes*, GTM 66.
3. [Dem70] *SGA III: Schémas en groupes*.
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Goal. Understand the structure of linear algebraic groups (LAG) over a field k.

Remark. Faithfully follow Miaofen Chen's lecture in Fudan University, 2026 Fall.

1 Definition and basic properties

1.1 Definitions and conventions

Let k be a field. For a scheme X of finite type over k, let

$$X^{\mathrm{cl}} := \{\text{closed points of } X\}, \qquad j : X^{\mathrm{cl}} \hookrightarrow X, \qquad \mathcal{O}_{X^{\mathrm{cl}}} := j^{-1}\mathcal{O}_X.$$

The locally ringed spaces obtained in this way are the *ultra-schemes* of [Gro66, §10.9] For an affine scheme,

$$(\operatorname{Spec} A)^{\mathrm{cl}} = \operatorname{Spm} A,$$

the maximal spectrum of A.

Proposition 1.1. [Gro66, 10.9.6] The functor $X \mapsto X^{\mathrm{cl}}$, with the induced structure sheaf, gives an equivalence between the category of Jacobson schemes and the category of ultra-schemes.

Convention. From now on, schemes over k are schemes of finite type over k. Under the preceding equivalence, affine schemes are written as $\operatorname{Spm} A$. Thus a scheme over k admits a finite affine covering

$$X = \bigcup_{i=1}^{r} \operatorname{Spm} A_i.$$

Definition 1.2. An *algebraic group over k* is a scheme G/k together with morphisms

$$m : G \times_k G \longrightarrow G, \qquad e : \operatorname{Spec} k \longrightarrow G, \qquad i : G \longrightarrow G,$$

called multiplication, unit, and inverse, satisfying the following properties.

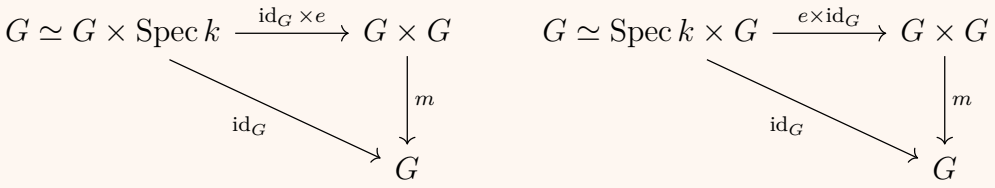
(1) Associativity.

$$\begin{array}{ccc} G \times G \times G & \xrightarrow{m \times \mathrm{id}_G} & G \times G \\ {\scriptstyle \mathrm{id}_G \times m}\downarrow & & \downarrow{\scriptstyle m} \\ G \times G & \xrightarrow[m]{} & G \end{array}$$

(2) Existence of a unit. Writing $p_G : G \to \operatorname{Spec} k$ for the structure morphism,

$$m \circ (\mathrm{id}_G, e \circ p_G) = \mathrm{id}_G = m \circ (e \circ p_G, \mathrm{id}_G).$$

Equivalently, the two diagrams

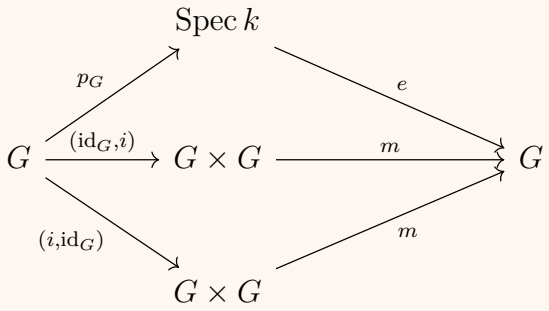


commute.

(3) Inverse property.

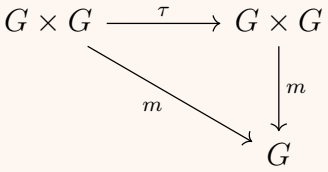
$$m \circ (\mathrm{id}_G, i) = e \circ p_G = m \circ (i, \mathrm{id}_G).$$

Equivalently,



Definition 1.3. A *linear algebraic group over* k is an algebraic group over k whose underlying scheme is affine.

Definition 1.4. An algebraic group G is *commutative* if the diagram



$$\big(\tau(x, y) = (y, x)\big)$$

commutes.

Remark. Let $G = \operatorname{Spm} A$, where A is a finitely generated commutative k-algebra. The structure morphisms of G correspond contravariantly to

$$m^{\#} : A \longrightarrow A \otimes_k A, \qquad e^{\#} : A \longrightarrow k, \qquad i^{\#} : A \longrightarrow A.$$

These are called, respectively, the *comultiplication*, *counit*, and *coinverse* (or *antipode*).

Moreover,

$$A \simeq \mathrm{Hom}_{k\text{-schemes}}(G, \mathbb{A}^1_k).$$

If

$$m^{\#}(f) = \sum_j f_j \otimes g_j,$$

then for $x, y \in G$,

$$f\big(m(x,y)\big) = \sum_j f_j(x) g_j(y).$$

The group axioms translate into the following identities for $m^{\#}, e^{\#}, i^{\#}$.

Coassociativity.

$$(m^{\#} \otimes \mathrm{id}_A) \circ m^{\#} = (\mathrm{id}_A \otimes m^{\#}) \circ m^{\#}.$$

Equivalently,

$$\begin{array}{ccc} A \otimes A \otimes A & \xleftarrow{\mathrm{id}_A \otimes m^{\#}} & A \otimes A \\ \uparrow{\scriptstyle m^{\#} \otimes \mathrm{id}_A} & & \uparrow{\scriptstyle m^{\#}} \\ A \otimes A & \xleftarrow[m^{\#}]{} & A \end{array}$$

Counit. Using $k \otimes A \simeq A \simeq A \otimes k$,

$$(e^{\#} \otimes \mathrm{id}_A) \circ m^{\#} = \mathrm{id}_A = (\mathrm{id}_A \otimes e^{\#}) \circ m^{\#}.$$

Equivalently,

$$\begin{array}{ccc} A & \xleftarrow{\mathrm{id}_A \otimes e^{\#}} & A \otimes A \\ \uparrow{\scriptstyle e^{\#} \otimes \mathrm{id}_A} & & \uparrow{\scriptstyle m^{\#}} \\ A \otimes A & \xleftarrow[m^{\#}]{} & A \end{array}$$

Antipode identity. Let

$$\mu_A : A \otimes_k A \longrightarrow A$$

be multiplication in A, and let

$$\eta_A : k \longrightarrow A$$

be the unit map. Then

$$\mu_A \circ (i^{\#} \otimes \mathrm{id}_A) \circ m^{\#} = \eta_A \circ e^{\#} = \mu_A \circ (\mathrm{id}_A \otimes i^{\#}) \circ m^{\#}.$$

Definition 1.5. A commutative k-algebra A, together with homomorphisms

$$m^{\#} : A \to A \otimes_k A, \qquad e^{\#} : A \to k, \qquad i^{\#} : A \to A$$

satisfying the preceding coassociativity, counit, and antipode identities, is called a *Hopf algebra*.

Remark (Anti-equivalence of categories)**.** There is an anti-equivalence

$$\left\{ \begin{array}{c} \text{finitely generated commutative} \\ \text{Hopf } k\text{-algebras} \end{array} \right\} \simeq \{\text{linear algebraic groups over } k\},$$

sending A to $\operatorname{Spm} A$.

Definition 1.6. A *homomorphism of algebraic groups* $\varphi : G \to G'$ is a morphism of schemes such that

$$\begin{array}{ccc} G \times G & \xrightarrow{(\varphi,\varphi)} & G' \times G' \\ \downarrow{\scriptstyle m_G} & & \downarrow{\scriptstyle m_{G'}} \\ G & \xrightarrow[\varphi]{} & G' \end{array}$$

commutes.

Definition 1.7. An *algebraic subgroup* H/k of an algebraic group G/k is an algebraic group H/k such that

1. H is a closed k-subscheme of G;
2. the inclusion $H \hookrightarrow G$ is a homomorphism of algebraic groups.

Remark. In the preceding definition, the word "closed" in (1) is redundant for algebraic subgroup schemes of finite type over a field.

1.1.1 Algebraic groups as functors

Let Alg_k denote the category of finitely generated commutative k-algebras.

Proposition 1.8 (Yoneda)**.** The functor-of-points construction

$$X \longmapsto \big(R \longmapsto X(R) := \mathrm{Hom}_{\mathrm{Sch}/k}(\operatorname{Spec} R, X)\big)$$

gives a fully faithful functor

$$\{\text{schemes over } k\} \longrightarrow \{\text{functors } \mathrm{Alg}_k \to \mathbf{Set}\}.$$

The functors in its essential image are called *representable functors*.

Now let G be an algebraic group over k, with multiplication m, unit e, and inverse i. For every $R \in \mathrm{Alg}_k$, these induce

$$m_R : G(R) \times G(R) \longrightarrow G(R), \qquad e_R := e \circ p_R \in G(R), \qquad i_R : G(R) \longrightarrow G(R),$$

where $p_R : \operatorname{Spec} R \to \operatorname{Spec} k$ is the structure morphism. The commutative diagrams in the definition of an algebraic group imply that $(G(R), m_R)$ is a group.

Hence an algebraic group gives a functor

$$\widetilde{G} : \mathrm{Alg}_k \longrightarrow \mathbf{Grp}, \qquad R \longmapsto G(R).$$

Proposition 1.9 (Functorial definition)**.** An algebraic group over k can equivalently be described as a functor

$$\widetilde{G} : \mathrm{Alg}_k \longrightarrow \mathbf{Grp}$$

such that the composite

$$\mathrm{Alg}_k \xrightarrow{\widetilde{G}} \mathbf{Grp} \longrightarrow \mathbf{Set}$$

is representable.

Exercise 1.10. Check directly that the scheme-theoretic and functorial definitions of an algebraic group coincide.

Remark. For an algebraic group, the unit e and inverse i are uniquely determined by the multiplication m.

Remark. An algebraic group G is commutative if and only if $G(R)$ is a commutative group for every $R \in \mathrm{Alg}_k$.

If k is perfect and G is reduced (equivalently, by the smoothness criterion below, G is smooth), then it is enough to test this on a separable closure k^s:

$$G \text{ is commutative} \iff G(k^s) \text{ is commutative.}$$

Example 1.11 (Basic examples)**.** 1. The trivial group:

$$G = \operatorname{Spm} k.$$

2. The additive group:

$$\mathbb{G}_a : \mathrm{Alg}_k \longrightarrow \mathbf{Grp}, \qquad R \longmapsto (R, +),$$

with

$$\mathbb{G}_a = \operatorname{Spm} k[T].$$

Its Hopf algebra structure is

$$m^{\#} : k[T] \longrightarrow k[T] \otimes_k k[T], \qquad T \longmapsto T \otimes 1 + 1 \otimes T,$$

$$e^{\#} : k[T] \longrightarrow k, \qquad T \longmapsto 0,$$

and

$$i^{\#} : k[T] \longrightarrow k[T], \qquad T \longmapsto -T.$$

Thus, if $m^{\#}(T) = \sum_j f_j \otimes g_j$, then for $x, y \in \mathbb{G}_a$,

$$\sum_j f_j(x) g_j(y) = x + y.$$

3. The multiplicative group:

$$\mathbb{G}_m : \mathrm{Alg}_k \longrightarrow \mathbf{Grp}, \qquad R \longmapsto (R^{\times}, \cdot),$$

with

$$\mathbb{G}_m = \operatorname{Spm} k[T, T^{-1}].$$

Its Hopf algebra structure is

$$m^{\#} : k[T, T^{-1}] \longrightarrow k[T, T^{-1}] \otimes_k k[T, T^{-1}], \qquad T \longmapsto T \otimes T,$$

$$e^{\#} : k[T, T^{-1}] \longrightarrow k, \qquad T \longmapsto 1,$$

and

$$i^{\#} : k[T, T^{-1}] \longrightarrow k[T, T^{-1}], \qquad T \longmapsto T^{-1}.$$

4. Let Γ be a finite group. The associated constant algebraic group is

$$\Gamma = \coprod_{\gamma\in\Gamma} \operatorname{Spec} k = \operatorname{Spm}\left(\bigoplus_{\gamma\in\Gamma} k e_\gamma\right),$$

where the e_γ are orthogonal idempotents. Writing $\mathcal{O}(\Gamma) = \bigoplus_{\gamma\in\Gamma} k e_\gamma$, one has

$$m^{\#} : \mathcal{O}(\Gamma) \longrightarrow \mathcal{O}(\Gamma) \otimes_k \mathcal{O}(\Gamma), \qquad e_x \longmapsto \sum_{g\in\Gamma} e_g \otimes e_{g^{-1}x},$$

$$e^{\#} : \mathcal{O}(\Gamma) \longrightarrow k, \qquad e_x \longmapsto \begin{cases} 1, & x = e, \\ 0, & x \neq e, \end{cases}$$

and

$$i^{\#} : \mathcal{O}(\Gamma) \longrightarrow \mathcal{O}(\Gamma), \qquad e_x \longmapsto e_{x^{-1}}.$$

5. For $n \in \mathbb{N}$,

$$\mathrm{GL}_n \supseteq B_n \supseteq U_n, \qquad T_n \subseteq B_n.$$

As functors on Alg_k,

$$\mathrm{GL}_n(R) = \left\{(x_{ij}) \in M_n(R) \,\middle|\, \det(x_{ij}) \in R^\times\right\},$$

$$B_n(R) = \left\{(x_{ij}) \in \mathrm{GL}_n(R) \,\middle|\, x_{ij} = 0 \text{ for } i > j\right\},$$

$$U_n(R) = \left\{(x_{ij}) \in B_n(R) \,\middle|\, x_{ii} = 1 \text{ for all } i\right\},$$

and

$$T_n(R) = \left\{(x_{ij}) \in B_n(R) \,\middle|\, x_{ij} = 0 \text{ for } i < j\right\}.$$

Thus the typical matrix shapes are

$$B_n : \begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{pmatrix}, \qquad U_n : \begin{pmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{pmatrix}, \qquad T_n : \begin{pmatrix} * & 0 & 0 \\ 0 & * & 0 \\ 0 & 0 & * \end{pmatrix}.$$

Here GL_n is a reductive group, B_n is a solvable (Borel, hence parabolic) subgroup of GL_n, U_n is a unipotent group, and T_n is a torus.

1.2 Basic properties

Proposition 1.12. Algebraic groups are separated.

Proof. It is enough to show that the diagonal

$$\Delta_G = \{(x, x) \mid x \in G\} \subseteq G \times G$$

is closed. Consider the morphism

$$G \times G \xrightarrow{(\mathrm{id}_G, i)} G \times G \xrightarrow{m} G, \qquad (x, y) \longmapsto xy^{-1}.$$

Then

$$\Delta_G = \big(m \circ (\mathrm{id}_G, i)\big)^{-1}(e),$$

and the identity section $e : \operatorname{Spec} k \to G$ is a closed point. Hence Δ_G is closed. □

Proposition 1.13. Let G be an algebraic group over k. Then

$$G \text{ is geometrically reduced} \quad \Longleftrightarrow \quad G \text{ is smooth over } k.$$

Proof. The implication "smooth ⇒ geometrically reduced" is immediate.

For the converse, after extending scalars to an algebraic closure $\overline{k}$, we may assume $k = \overline{k}$. Then G is reduced. The smooth locus of the finite-type k-scheme G is a nonempty open dense subset. Choose a smooth point $g \in G$. For any $h \in G$, left translation by hg^{-1} is an automorphism of G sending g to h. Smoothness is preserved by isomorphisms, so every point of G is smooth. Thus G is smooth. □

Remark. If $\operatorname{char} k = 0$, then every algebraic group over k is smooth.[1]

Example 1.14. Assume $\operatorname{char} k = p > 0$. Define

$$\mu_p := \ker\Big(\mathbb{G}_m \xrightarrow{[p]} \mathbb{G}_m\Big), \qquad x \longmapsto x^p.$$

[1] 2026.9.7 stopped at here.

Equivalently, μ_p is the fiber product

$$\begin{array}{ccc} \mu_p & \longrightarrow & \operatorname{Spec} k \\ \downarrow & & \downarrow{\scriptstyle e} \\ \mathbb{G}_m & \xrightarrow{[p]} & \mathbb{G}_m. \end{array}$$

On coordinate rings, the p-power map corresponds to

$$k[T, T^{-1}] \longrightarrow k[T, T^{-1}], \qquad T \longmapsto T^p,$$

while the identity section corresponds to

$$k[T, T^{-1}] \longrightarrow k, \qquad T \longmapsto 1.$$

Hence

$$\mu_p \simeq \operatorname{Spm} \frac{k[T]}{(T^p - 1)} = \operatorname{Spm} \frac{k[T]}{((T-1)^p)}.$$

Thus μ_p is nonreduced, and therefore it is not smooth.

Remark (Reduced subscheme)**.** For a scheme X, write X_{red} for its reduced closed subscheme. If $X = \operatorname{Spec} A$, then

$$X_{\mathrm{red}} = \operatorname{Spec}\big(A/\sqrt{(0)}\big) \hookrightarrow \operatorname{Spec} A.$$

The universal property used below is that every morphism from a reduced scheme to X factors uniquely through X_{red}.

Recall. A morphism $\varphi : G \to H$ of algebraic groups is a homomorphism precisely when the multiplication square commutes:

$$\begin{array}{ccc} G \times G & \xrightarrow{m_G} & G \\ {\scriptstyle \varphi\times\varphi}\downarrow & & \downarrow{\scriptstyle \varphi} \\ H \times H & \xrightarrow[m_H]{} & H. \end{array}$$

Idea. Restrict the multiplication of G to

$$G_{\mathrm{red}} \times G_{\mathrm{red}} \longrightarrow G.$$

To factor this morphism through G_{red}, it is enough that $G_{\mathrm{red}} \times G_{\mathrm{red}}$ be reduced. In particular, this holds when G_{red} is geometrically reduced.

Proposition 1.15. Let G be an algebraic group over k. If G_{red} is geometrically reduced, then G_{red} is an algebraic subgroup of G. In particular, this holds whenever k is perfect.

Proof. Let $j : G_{\mathrm{red}} \hookrightarrow G$ be the canonical closed immersion.

Since $\operatorname{Spec} k$ is reduced, the identity morphism $e : \operatorname{Spec} k \to G$ factors uniquely through G_{red}:

$$\begin{array}{ccc} \operatorname{Spec} k & \xrightarrow{\quad e \quad} & G \\ & {\scriptstyle e_{\mathrm{red}}} \searrow \quad \nearrow {\scriptstyle j} & \\ & G_{\mathrm{red}} & \end{array}$$

Since G_{red} is reduced, the inverse morphism $\iota \circ j : G_{\mathrm{red}} \to G$ factors uniquely through G_{red}.

Finally, because G_{red} is geometrically reduced, $G_{\mathrm{red}} \times_k G_{\mathrm{red}}$ is reduced. Hence

$$m_G \circ (j \times j) : G_{\mathrm{red}} \times_k G_{\mathrm{red}} \longrightarrow G$$

factors uniquely through G_{red}:

$$\begin{array}{ccc} G_{\mathrm{red}} \times G_{\mathrm{red}} & \overset{m_{\mathrm{red}}}{\dashrightarrow} & G_{\mathrm{red}} \\ {\scriptstyle j\times j}\downarrow & & \downarrow{\scriptstyle j} \\ G \times G & \xrightarrow[m_G]{} & G. \end{array}$$

Thus the identity, inverse, and multiplication of G restrict to G_{red}, so G_{red} is an algebraic subgroup of G.

If k is perfect, every reduced finite-type k-scheme is geometrically reduced by [GW20, Proposition 5.49], so the hypothesis is automatic. □

Example 1.16 (Why geometric reducedness matters)**.** Let

$$k = \mathbb{F}_p(t), \qquad K = k[x]/(x^p - t), \qquad X = \operatorname{Spec} K.$$

Since $t \notin k^p$, the polynomial $x^p - t$ is irreducible over k, so K is a field and X is reduced. If $\beta \in \overline{k}$ satisfies $\beta^p = t$, then

$$X_{\overline{k}} \cong \operatorname{Spec} \frac{\overline{k}[y]}{(y^p - t)} = \operatorname{Spec} \frac{\overline{k}[y]}{(y - \beta)^p},$$

which is not reduced. Thus X is reduced but not geometrically reduced.

If $\alpha \in K$ is the class of x, so that $\alpha^p = t$, then

$$X \times_k X \cong \operatorname{Spec}(K \otimes_k K) \cong \operatorname{Spec} \frac{K[y]}{(y^p - t)} = \operatorname{Spec} \frac{K[y]}{(y - \alpha)^p},$$

which is not reduced.

Definition 1.17 (Kernel)**.** Let $\varphi : G \to H$ be a homomorphism of algebraic groups. Its kernel is the fiber product

$$\ker(\varphi) = G \times_{H,e} \operatorname{Spec} k,$$

i.e.

$$\begin{array}{ccc} \ker(\varphi) & \longrightarrow & \operatorname{Spec} k \\ \downarrow & & \downarrow e \\ G & \xrightarrow[\varphi]{} & H. \end{array}$$

Definition 1.18 (Normal algebraic subgroup)**.** Let G be an algebraic group over k, and let $H \subseteq G$ be an algebraic subgroup. Then H is *normal* if, for every $R \in \mathrm{Alg}_k$,

$$H(R) \trianglelefteq G(R).$$

Remark. Let $\varphi : G \to H$ be a homomorphism. For every $R \in \mathrm{Alg}_k$,

$$(\ker \varphi)(R) = \ker\big(G(R) \longrightarrow H(R)\big).$$

Hence $\ker \varphi$ is a normal algebraic subgroup of G.

Remark. If G and H are affine, then $\ker \varphi$ is affine. Writing

$$I_H = \ker\big(e^\# : \mathcal{O}(H) \to k\big), \qquad \varphi^\# : \mathcal{O}(H) \to \mathcal{O}(G),$$

one has

$$\mathcal{O}(\ker \varphi) \cong \mathcal{O}(G) \otimes_{\mathcal{O}(H)} k \cong \frac{\mathcal{O}(G)}{\mathcal{O}(G)\, \varphi^\#(I_H)}.$$

Definition 1.19 (Identity component)**.** Let G be an algebraic group over k. The connected component of G containing the identity e is called the *identity component* of G, and is denoted by G°.

Proposition 1.20 (Largest finite étale subalgebra)**.** Let X be a finite-type k-scheme. Then there exists a largest finite étale k-subalgebra

$$\pi^0(X) \subseteq \mathcal{O}(X).$$

Proof. A finite étale k-algebra has the form

$$\prod_{i=1}^{r} L_i,$$

with each L_i/k a finite separable field extension. A quotient of a finite étale k-algebra is again finite étale.

If $A_1, A_2 \subseteq \mathcal{O}(X)$ are finite étale k-subalgebras, then their compositum inside $\mathcal{O}(X)$ is the image of

$$A_1 \otimes_k A_2 \longrightarrow \mathcal{O}(X),$$

hence is again finite étale. It therefore suffices to bound $\dim_k A$ for all finite étale k-subalgebras $A \subseteq \mathcal{O}(X)$.

Let k^s be a separable closure of k, and put $n = \dim_k A$. Then

$$A \otimes_k k^s \cong (k^s)^n.$$

Thus

$$1 = e_1 + \cdots + e_n,$$

where the e_i are pairwise orthogonal idempotents. Viewed as functions on X_{k^s}, these idempotents determine n pairwise disjoint nonempty open-and-closed subsets. Consequently,

$$n \leq \#\{\text{connected components of } X_{k^s}\},$$

which is finite because X is of finite type over k.

Choose a finite étale subalgebra of maximal possible k-dimension. Taking its compositum with any other finite étale subalgebra cannot increase the dimension, so it already contains every other one. This gives the largest such subalgebra $\pi^0(X)$. □

Definition 1.21 (The scheme of connected components)**.** Define

$$\pi_0(X) = \operatorname{Spec} \pi^0(X).$$

The inclusion

$$\pi^0(X) \hookrightarrow \mathcal{O}(X)$$

induces a canonical morphism

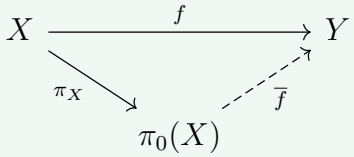
$$\pi_X : X \longrightarrow \pi_0(X).$$

The scheme $\pi_0(X)$ is finite étale over k.

Proposition 1.22 (Universal property)**.** The morphism $\pi_X : X \to \pi_0(X)$ is universal for morphisms from X to finite étale k-schemes. Namely, for every morphism $f : X \to Y$ with Y finite étale over k, there is a unique morphism $\overline{f} : \pi_0(X) \to Y$ such that

$$f = \overline{f} \circ \pi_X.$$

Equivalently,



commutes.

Proof. The map f induces

$$f^{\#} : \mathcal{O}(Y) \longrightarrow \mathcal{O}(X).$$

Its image is a quotient of a finite étale k-algebra, hence is finite étale and therefore lies in $\pi^0(X)$. This gives the factorization, and maximality of $\pi^0(X)$ gives uniqueness. □

Proposition 1.23 (Basic properties of π_0)**.** Let X, Y be finite-type k-schemes.

1. For every field extension k'/k,

$$\pi_0(X_{k'}) \cong \pi_0(X)_{k'}.$$

2.

$$\pi_0(X \times_k Y) \cong \pi_0(X) \times_k \pi_0(Y).$$

3. The fibers of $\pi_X : X \to \pi_0(X)$ are the connected components of X.

4. For every $x \in \pi_0(X)$, the fiber X_x is geometrically connected over $\kappa(x)$ (not

necessarily over k).

5. The morphism
$$X \longrightarrow \pi_0(X)$$
is faithfully flat.

Remark. See [Mil17, Propositions 1.30, 1.31, 1.33].

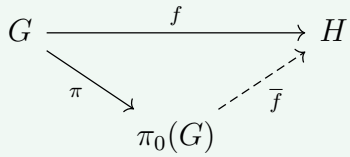
Corollary 1.24. Let X be a connected finite-type k-scheme such that $X(k) \neq \varnothing$. Then X is geometrically connected. Moreover, for every connected finite-type k-scheme Y, the product $X \times_k Y$ is connected.

Proof. Since X is connected,
$$\pi_0(X) = \operatorname{Spec} L$$
for some finite separable extension L/k. A k-rational point of X gives a composite
$$\operatorname{Spec} k \longrightarrow X \longrightarrow \operatorname{Spec} L,$$
hence a k-algebra map $L \to k$. Therefore $L = k$. By the geometric-connectedness property of the fibers, X is geometrically connected over k.

For connected Y,
$$\pi_0(X \times_k Y) \cong \pi_0(X) \times_k \pi_0(Y) = \operatorname{Spec} k \times_k \pi_0(Y) \cong \pi_0(Y),$$
which is single point. Hence $X \times_k Y$ is connected. □

Remark (The hypothesis $X(k) \neq \varnothing$ is essential)**.** Let
$$X = \operatorname{Spec} L,$$
where L/k is a finite Galois extension with $L \neq k$. Then X is connected, but
$$X_{\overline{k}} = \operatorname{Spec}(L \otimes_k \overline{k}) \cong \coprod_{[L:k]} \operatorname{Spec} \overline{k},$$
so X is not geometrically connected. Also,
$$X \times_k X = \operatorname{Spec}(L \otimes_k L) \cong \coprod_{[L:k]} \operatorname{Spec} L,$$
so $X \times_k X$ is not connected.

Proposition 1.25. Let G be an algebraic group over k. Then:

1. $\pi_0(G)$ is a finite étale algebraic group over k.
2. The canonical morphism
$$\pi : G \longrightarrow \pi_0(G)$$
is a homomorphism of algebraic groups, and it is universal among homomorphisms from G to finite étale algebraic groups. Thus, if H is finite étale and $f : G \to H$ is a homomorphism, there is a unique homomorphism $\overline{f} : \pi_0(G) \to H$ such that


commutes.
3. $$\ker\big(G \longrightarrow \pi_0(G)\big) = G^\circ.$$
In particular, G° is an algebraic subgroup of G.
4. For every field extension k'/k,
$$\pi_0(G_{k'}) \cong \pi_0(G)_{k'}, \qquad (G_{k'})^\circ \cong (G^\circ)_{k'}.$$
In particular, G° is geometrically connected.
5. The fibers of $G \to \pi_0(G)$ are the connected components of G.
6. For algebraic groups G, G',
$$\pi_0(G \times_k G') \cong \pi_0(G) \times_k \pi_0(G'), \qquad (G \times_k G')^\circ \cong G^\circ \times_k (G')^\circ.$$

Proof. For the multiplication, consider the composite

$$G \times G \xrightarrow{m_G} G \xrightarrow{\pi} \pi_0(G).$$

By the universal property of π_0, it factors uniquely through

$$\pi_0(G \times G) \cong \pi_0(G) \times \pi_0(G),$$

giving a multiplication

$$m_0 : \pi_0(G) \times \pi_0(G) \longrightarrow \pi_0(G)$$

such that

$$\begin{array}{ccc} G \times G & \xrightarrow{m_G} & G \\ {\scriptstyle \pi\times\pi}\downarrow & & \downarrow{\scriptstyle \pi} \\ \pi_0(G) \times \pi_0(G) & \xrightarrow[m_0]{} & \pi_0(G) \end{array}$$

commutes. The identity and inverse descend in the same way:

$$\operatorname{Spec} k \xrightarrow{e} G \xrightarrow{\pi} \pi_0(G), \qquad \begin{array}{ccc} G & \xrightarrow{\iota} & G \\ {\scriptstyle \pi}\downarrow & & \downarrow{\scriptstyle \pi} \\ \pi_0(G) & \xrightarrow[\iota_0]{} & \pi_0(G). \end{array}$$

These maps satisfy the group axioms, and π is a homomorphism. The universal property among homomorphisms follows from the universal property of π_0.

For (iii), the kernel is the fiber of π over the identity of $\pi_0(G)$; by the fiber description above, this is exactly the connected component G°.

For (iv), using the identity sections of the component groups,

$$\begin{aligned} (G_{k'})^\circ &= G_{k'} \times_{\pi_0(G_{k'})} \operatorname{Spec} k' \\ &\cong \left(G \times_{\pi_0(G)} \operatorname{Spec} k\right) \times_k k' \\ &= (\ker(G \to \pi_0(G)))_{k'} \\ &= (G^\circ)_{k'}. \end{aligned}$$

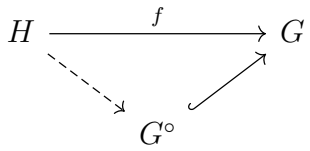
The remaining assertions follow from the basic properties of π_0. □

Definition 1.26 (Component group)**.** The finite étale algebraic group $\pi_0(G)$ is called the *component group*, or the *group of connected components*, of G.

Lemma 1.27. Let G be an algebraic group over k.

1. G is connected if and only if
$$\pi_0(G) = \operatorname{Spec} k.$$
2. If $f : H \to G$ is a homomorphism and H is connected, then f factors through

G°:



3. The identity component G° is geometrically connected, but the other connected components of G need not be geometrically connected over k.

Proof. For (1), a connected algebraic group has the k-rational identity point, so the preceding corollary gives geometric connectedness and forces its finite étale component scheme to be $\operatorname{Spec} k$. The converse is immediate from the description of the fibers of $G \to \pi_0(G)$.

For (2), the composite

$$H \longrightarrow G \longrightarrow \pi_0(G)$$

has connected image and contains the identity, so it is the identity morphism into the identity component of the finite étale scheme $\pi_0(G)$, namely the single point $\operatorname{Spec} k$. Hence $H \to G$ factors through the kernel G°.

Part (3) follows from base change for G°; the following example shows the second assertion. □

Example 1.28. Let $k = \mathbb{Q}$, and consider

$$\mu_3(R) = \{x \in R^\times : x^3 = 1\}, \qquad R \in \mathrm{Alg}_{\mathbb{Q}}.$$

Then

$$\mu_3 = \operatorname{Spec} \frac{\mathbb{Q}[T]}{(T^3-1)} \cong \operatorname{Spec} \mathbb{Q} \coprod \operatorname{Spec} \frac{\mathbb{Q}[T]}{(T^2+T+1)}.$$

The first component contains the identity. The second component has no $\mathbb{Q}$-point and is not geometrically connected: after base change to $\overline{\mathbb{Q}}$, it splits into two points. Thus connected components other than G° need not be geometrically connected over the ground field.

Definition 1.29 (Exact sequence of algebraic groups)**.** A sequence

$$e \longrightarrow N \xrightarrow{i} G \xrightarrow{q} Q \longrightarrow e$$

is called exact if:

1. i identifies N isomorphically with $\ker(q : G \to Q)$;

2. q is faithfully flat.

Example 1.30. The sequence

$$e \longrightarrow G^\circ \longrightarrow G \longrightarrow \pi_0(G) \longrightarrow e$$

is exact.

Proposition 1.31. Every connected algebraic group is irreducible. In fact, it is geometrically irreducible.

Proof. It is enough to prove that G is geometrically irreducible. Since a connected algebraic group is geometrically connected, after base change we may assume $k = \overline{k}$.

The reduced closed subscheme $G_{\mathrm{red}} \subseteq G$ has the same underlying topological space:

$$|G_{\mathrm{red}}| = |G|.$$

Since k is perfect, G_{red} is an algebraic subgroup of G, and a reduced algebraic group over a perfect field is smooth. Thus, for the topological question of irreducibility, we may replace G by the smooth reduced group G_{red}.

Suppose G had more than one irreducible component. Because G is connected, two irreducible components must meet. Choose a point x in such an intersection. Then the local ring $\mathcal{O}_{G,x}$ has at least two distinct minimal prime ideals. On the other hand, smoothness implies that $\mathcal{O}_{G,x}$ is a regular local ring, hence an integral domain, so it has only one minimal prime. This is a contradiction. □

Proposition 1.32. Let $\varphi : G \to H$ be a homomorphism of algebraic groups. Then the image $\varphi(|G|) \subseteq |H|$ is closed.

Proof. Step 1. It is enough to reduce to the case $k = \overline{k}$.

First, for any finite-type k-scheme X, the projection

$$p_X : |X_{\overline{k}}| \longrightarrow |X|$$

is a quotient map of topological spaces.

Second, if

$$p_H : |H_{\overline{k}}| \longrightarrow |H|$$

is the projection, then

$$\varphi_{\overline{k}}\big(|G_{\overline{k}}|\big) = p_H^{-1}\big(\varphi(|G|)\big).$$

Therefore $\varphi(|G|)$ is closed in $|H|$ if and only if the image after base change to $\overline{k}$ is closed in $|H_{\overline{k}}|$. Hence the proposition reduces to the algebraically closed case. [2]

Step 2. It is enough to work after passing to an algebraic closure, so assume $k = \overline{k}$. Then $|G| = G(k)$ on closed points. Put

$$N := \varphi(G(k)) \subseteq H(k),$$

and let $\overline{N}$ denote its closure in $|H|$. We show first that $\overline{N}$ is a subgroup of $H(k)$.

Let

$$i : H \xrightarrow{\sim} H, \qquad x \longmapsto x^{-1}.$$

Since i is a homeomorphism and $N^{-1} = N$,

$$(\overline{N})^{-1} = i(\overline{N}) = \overline{i(N)} = \overline{N}.$$

For $x \in N$, left translation

$$\ell_x : H \xrightarrow{\sim} H, \qquad y \longmapsto xy$$

satisfies

$$\ell_x(\overline{N}) = \overline{\ell_x(N)} = \overline{xN} = \overline{N}.$$

Hence $N\overline{N} \subseteq \overline{N}$. Now let $x \in \overline{N}$. Right translation

$$r_x : H \xrightarrow{\sim} H, \qquad y \longmapsto yx$$

gives

$$r_x(\overline{N}) = \overline{r_x(N)} = \overline{Nx} \subseteq \overline{N},$$

because $Nx \subseteq N\overline{N} \subseteq \overline{N}$. Thus

$$\overline{N}\,\overline{N} \subseteq \overline{N},$$

so $\overline{N}$ is a subgroup of $H(k)$.

Remark. This is the same idea as in the standard topological-group argument.

By Chevalley's theorem below, there exists a nonempty open dense subset

$$U \subseteq \overline{N}$$

[2] 2026.9.10 stopped at here.

with $U \subseteq N$. For any $x \in \overline{N}$, both U and $x^{-1}U$ are open dense subsets of $\overline{N}$. Hence

$$U \cap x^{-1}U \neq \varnothing.$$

Choose u in this intersection. Then $u \in U$ and $xu \in U$, so

$$x = (xu)u^{-1} \in UU^{-1} \subseteq N.$$

Therefore $\overline{N} \subseteq N$, hence $N = \overline{N}$ is closed. □

Theorem 1.33 (Chevalley)**.** Let $f : Y \to X$ be a finite-type morphism of noetherian schemes. Then $f(|Y|)$ is constructible in $|X|$. In particular, $f(|Y|)$ contains an open dense subset of its closure.

Corollary 1.34. If $\varphi : G \to H$ is a dominant homomorphism of algebraic groups, then φ is surjective on underlying topological spaces.

Proof. By the proposition, $\varphi(|G|)$ is closed in $|H|$. Dominance says that it is dense, hence it equals $|H|$. □

Proposition 1.35. Let $\varphi : G \to H$ be a surjective homomorphism of algebraic groups. If H is reduced, then φ is faithfully flat.

Proof. By generic flatness there is an open dense subset $U \subseteq H$ such that

$$\varphi^{-1}(U) \longrightarrow U$$

is flat. After extending scalars to an algebraic closure $\overline{k}$, the corresponding morphism over $U_{\overline{k}}$ is still flat.

For $g \in G(\overline{k})$, left translations give a commutative diagram

$$\begin{array}{ccc} G_{\overline{k}} & \xrightarrow[\sim]{L_g} & G_{\overline{k}} \\ {\scriptstyle \varphi_{\overline{k}}}\downarrow & & \downarrow{\scriptstyle \varphi_{\overline{k}}} \\ H_{\overline{k}} & \xrightarrow[\sim]{L_{\varphi(g)}} & H_{\overline{k}}. \end{array}$$

Thus $\varphi_{\overline{k}}$ is flat over every translate $\varphi(g)U_{\overline{k}}$. Surjectivity gives

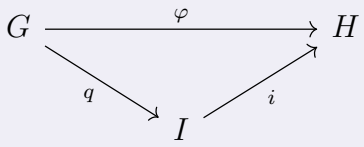
$$H_{\overline{k}} = \bigcup_{g \in G(\overline{k})} \varphi(g)U_{\overline{k}},$$

so $\varphi_{\overline{k}}$ is flat everywhere. Flatness descends along the faithfully flat extension $\overline{k}/k$, hence φ is flat. Since φ is surjective, it is faithfully flat. □

1.3 The image of a homomorphism

Motivation. For a homomorphism of abstract groups $G \xrightarrow{\varphi} H$, one factors φ through its image $\varphi(G)$. The analogous statement for algebraic groups is the following.

Theorem 1.36. Let $\varphi : G \to H$ be a homomorphism of algebraic groups, and suppose that G is smooth. Then φ factors as



where q is faithfully flat and i is a closed immersion.

Remark. The role of the smoothness hypothesis in this proof is to ensure that $G_{\overline{k}}$ is reduced; this forces the reduced image I to be geometrically reduced, which is what allows the group operations to restrict to I.

Proof. Let

$$I := \varphi(G)_{\mathrm{red}}$$

with its reduced induced closed-subscheme structure in H. Since G is reduced, φ factors through I; write the factorization as $G \xrightarrow{q} I \xrightarrow{i} H$. The map q is surjective, and I is reduced, so Proposition 1.35 shows that q is faithfully flat.

It remains to see that I is an algebraic subgroup of H. It is enough to prove that I is geometrically reduced. After base change to $\overline{k}$,

$$q_{\overline{k}} : G_{\overline{k}} \longrightarrow I_{\overline{k}}$$

is faithfully flat. Let $x \in |I_{\overline{k}}|$ and choose $g \in |G_{\overline{k}}|$ mapping to x. Then

$$\mathcal{O}_{I_{\overline{k}},x} \longrightarrow \mathcal{O}_{G_{\overline{k}},g}$$

is a faithfully flat local homomorphism, hence injective. Since G is smooth, $G_{\overline{k}}$ is reduced, so $\mathcal{O}_{G_{\overline{k}},g}$ is reduced. Therefore $\mathcal{O}_{I_{\overline{k}},x}$ is reduced. As x was arbitrary, $I_{\overline{k}}$ is reduced, hence I is geometrically reduced.

Now the multiplication and inverse maps of H carry the underlying sets $|I| \times |I|$ and $|I|$ into $|I|$. Since I and $I \times I$ are geometrically reduced, these morphisms factor through the

reduced closed subscheme I. Thus I inherits an algebraic-group structure and i is a closed immersion of algebraic groups. □

Definition 1.37. The algebraic subgroup I above is called the *image* of φ.

Remark. For linear algebraic groups, an image factorization of this form exists without assuming that G is smooth.

1.4 Group actions

Definition 1.38. An action of an algebraic group G on a k-scheme X is a morphism

$$\varphi : G \times_k X \longrightarrow X$$

such that the following two conditions hold.

1. If $e : \operatorname{Spec} k \to G$ is the identity section, then

$$\begin{array}{ccc} X = \operatorname{Spec} k \times X & \xrightarrow{e \times \mathrm{id}_X} & G \times X \\ & \searrow^{\mathrm{id}_X} & \downarrow{\scriptstyle \varphi} \\ & & X \end{array}$$

   commutes.

2. If $m : G \times G \to G$ is multiplication, then

$$\begin{array}{ccc} G \times G \times X & \xrightarrow{m \times \mathrm{id}_X} & G \times X \\ {\scriptstyle \mathrm{id}_G \times \varphi}\downarrow & & \downarrow{\scriptstyle \varphi} \\ G \times X & \xrightarrow[\varphi]{} & X \end{array}$$

   commutes.

Remark. By Yoneda's lemma, an action of G on X is equivalently a natural transformation

$$\widetilde{G} \times \widetilde{X} \longrightarrow \widetilde{X}$$

such that $G(R)$ acts on $X(R)$ for every k-algebra R.

Example 1.39. 1. The trivial action:

$$G \times X \xrightarrow{\mathrm{pr}_X} X, \qquad g \cdot x = x.$$

2. For $X = G$, left multiplication:

$$G \times G \xrightarrow{m} G, \qquad g \cdot x = gx.$$

3. For $X = G$, right multiplication written as a left action:

$$G \times G \longrightarrow G, \qquad g \cdot x = xg^{-1}.$$

4. For $X = G$, conjugation:

$$G \times G \longrightarrow G, \qquad g \cdot x = gxg^{-1}.$$

Definition 1.40. A linear representation of G is a homomorphism of algebraic groups

$$\rho : G \longrightarrow \mathrm{GL}_n$$

for some $n \geq 1$.

Remark. If $\rho : G \to \mathrm{GL}_n$ is a representation, then ρ gives an action of G on $\mathbb{A}^n$. On R-points,

$$G(R) \times \mathbb{A}^n(R) \longrightarrow \mathbb{A}^n(R), \qquad \left(g, \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right) \longmapsto \rho(g) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

for every k-algebra R.

Lemma 1.41. Let $\varphi : G \times X \to X$ be an action. Then φ is faithfully flat. Moreover, φ is smooth if G is smooth over k, and φ is finite if G is finite over k.

Proof. The map

$$\alpha : G \times X \xrightarrow{\sim} G \times X, \qquad (g, x) \longmapsto (g, gx)$$

is an isomorphism, with inverse $(g, y) \mapsto (g, g^{-1}y)$. It satisfies

$$\varphi = \mathrm{pr}_X \circ \alpha.$$

Thus every property listed in the statement for φ is the corresponding property of the projection $\mathrm{pr}_X : G \times X \to X$, which is the base change of $G \to \operatorname{Spec} k$. □

Definition 1.42. Let $\varphi : G \times X \to X$ be an action and let $x \in X(k)$. The *orbit map* is

$$\varphi_x : G \simeq G \times \operatorname{Spec} k \xrightarrow{\mathrm{id}_G \times x} G \times X \xrightarrow{\varphi} X, \qquad g \longmapsto gx.$$

The orbit of x is the image of $|\varphi_x|$ and is denoted by $G \cdot x$.

Lemma 1.43. The orbit $G \cdot x$ is a locally closed subset of $|X|$.

Proof. As in the earlier reduction, it is enough to work after passing to an algebraic closure, so assume $k = \overline{k}$; the same reduction uses that $|X_{\overline{k}}| \to |X|$ is a quotient map. By Chevalley's theorem applied to $\varphi_x : G \to X$, there is a nonempty subset

$$U \subseteq G \cdot x$$

which is open and dense in $\overline{G \cdot x}$. Since U is nonempty and contained in the orbit,

$$G \cdot x = G \cdot U = \bigcup_{g \in G(k)} gU.$$

For every $g \in G(k)$, translation by g preserves $\overline{G \cdot x}$, and gU is open in $\overline{G \cdot x}$. Hence $G \cdot x$ is open in its closure, i.e. it is locally closed. □

Remark. It is possible that $X(k) = \varnothing$. For example, let

$$k = \mathbb{Q}, \qquad G = (\mathbb{Z}/2\mathbb{Z})_k, \qquad X = \operatorname{Spec} \mathbb{Q}(\sqrt{2}).$$

Then G acts on X by Galois conjugation, while $X(\mathbb{Q}) = \varnothing$.

Proposition 1.44. Assume $k = \overline{k}$ and that G acts on X. Every orbit of minimum dimension is closed. In particular, if $X \neq \varnothing$, a closed orbit exists.

Proof. For a subset $X_1 \subseteq X$, write

$$G \cdot X_1 := \{gx \in X : g \in G(k),\ x \in X_1\}.$$

First note that the closure of an orbit is G-stable, hence is a union of orbits. Indeed, for every $g \in G(k)$, the map

$$\ell_g : X \xrightarrow{\sim} X, \qquad x \longmapsto gx$$

is an isomorphism, and therefore

$$\ell_g(\overline{G \cdot x}) = \overline{\ell_g(G \cdot x)} = \overline{g(G \cdot x)} = \overline{G \cdot x}.$$

Thus $G \cdot \overline{G \cdot x} = \overline{G \cdot x}$.

Now let

$$y \in \overline{G \cdot x} \setminus G \cdot x.$$

Then

$$G \cdot y \subseteq \overline{G \cdot x} \setminus G \cdot x.$$

By Lemma 1.43, $G \cdot x$ is open and dense in $\overline{G \cdot x}$. Hence its boundary contains no irreducible component of $\overline{G \cdot x}$, and therefore has strictly smaller dimension. Since $G \cdot y$ is contained in the boundary,

$$\dim(G \cdot y) < \dim(G \cdot x).$$

Therefore an orbit of minimum dimension has empty boundary and is closed. □

1.5 Structure of general algebraic groups

Remark. Reference: [Mil17, Chapter 8].

Recall. For an algebraic group G over k, there is an exact sequence

$$e \longrightarrow G^\circ \longrightarrow G \longrightarrow \pi_0(G) \longrightarrow e,$$

where G° is the connected identity component and $\pi_0(G)$ is the finite étale component group.

Theorem 1.45 (Barsotti–Chevalley)**.** Let G be a connected algebraic group over a field k. Then there exists an exact sequence

$$e \longrightarrow N \longrightarrow G \longrightarrow Q \longrightarrow e,$$

where N is a connected affine normal algebraic subgroup and Q is an abelian variety. If, in addition, G is smooth and k is perfect, then N is the unique connected affine normal subgroup variety with abelian-variety quotient.

Remark. This is the pair of statements referred to [Mil17, Theorem 8.27 and 8.28].

Recall. An abelian variety is an algebraic group that is connected, smooth, and complete. In particular, every abelian variety is commutative.

Theorem 1.46. Every algebraic group G over k has a greatest (maximal) affine algebraic quotient

$$G \longrightarrow G_{\mathrm{aff}},$$

universal among homomorphisms from G to affine algebraic groups.[3]

Remark. More generally, every algebraic group G has a maximal affine quotient. Write

$$G_{\mathrm{aff}} := \operatorname{Spm} \mathcal{O}(G).$$

The first question is whether $\mathcal{O}(G)$ carries the expected Hopf algebra structure. The multiplication, identity, and inverse of G induce

$$m^{\#} : \mathcal{O}(G) \longrightarrow \mathcal{O}(G \times G) \cong \mathcal{O}(G) \otimes_k \mathcal{O}(G), \qquad e^{\#} : \mathcal{O}(G) \longrightarrow k, \qquad i^{\#} : \mathcal{O}(G) \longrightarrow \mathcal{O}(G),$$

and these maps make $\mathcal{O}(G)$ a Hopf k-algebra.

Theorem 1.47. Let G be an algebraic group over k. Then:

(i) $\mathcal{O}(G)$ is a finitely generated k-algebra. In particular, $G_{\mathrm{aff}} = \operatorname{Spm} \mathcal{O}(G)$ is a linear algebraic group.

(ii) The natural morphism

$$\phi : G \longrightarrow G_{\mathrm{aff}}$$

is universal among homomorphisms from G to linear algebraic groups: if $\phi' : G \to H$ is a homomorphism and H is a linear algebraic group, then there is a unique homomorphism $\psi : G_{\mathrm{aff}} \to H$ such that $\phi' = \psi \circ \phi$.

(iii) The morphism ϕ is faithfully flat.

(iv) The kernel $\ker(\phi)$ is anti-affine, i.e.

$$\mathcal{O}(\ker \phi) = k.$$

In particular, $\ker(\phi)$ is commutative.

Remark. Reference. [Mil17, Proposition 8.36, Corollary 8.13] and [Dem70, Exp. VI, Theorem 12.2].

[3] 2026.9.14 stopped at here.

Consequence. There is an exact sequence

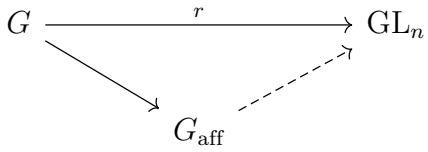
$$e \longrightarrow G_{\text{ant}} \longrightarrow G \longrightarrow G_{\text{aff}} \longrightarrow e,$$

where G_{ant} is anti-affine (hence commutative) and G_{aff} is a linear algebraic group.

Consequence. For every linear representation

$$r : G \longrightarrow \mathrm{GL}_n,$$

there is a unique factorization



Thus the study of linear representations of G is equivalent to the study of linear representations of G_{aff}. In what follows, we focus on linear algebraic groups.

2 Linear representations of linear algebraic groups

2.1 Hopf algebras

Recall. Let $G = \operatorname{Spm} A$ be a linear algebraic group over k. Then A has a Hopf algebra structure

$$m^{\#} : A \longrightarrow A \otimes_k A, \qquad e^{\#} : A \longrightarrow k, \qquad i^{\#} : A \longrightarrow A.$$

Geometrically, if

$$m^{\#}(f) = \sum_i g_i \otimes h_i,$$

then for every $R \in \mathrm{Alg}_k$ and $x, y \in G(R)$,

$$f(xy) = \sum_i g_i(x) h_i(y) \in R.$$

Moreover,

$$e^{\#}(f) = f(e), \qquad i^{\#}(f)(x) = f(x^{-1}).$$

Definition 2.1. Let A and B be Hopf k-algebras. A *homomorphism of Hopf algebras* $f : A \to B$ is a homomorphism of k-algebras such that

$$\begin{array}{ccc} A & \xrightarrow{\ f\ } & B \\ {\scriptstyle m_A^{\#}}\downarrow & & \downarrow{\scriptstyle m_B^{\#}} \\ A \otimes_k A & \xrightarrow[f \otimes f]{} & B \otimes_k B \end{array}$$

commutes.

Remark. For such an f, the counit and antipode are automatically compatible as well:

$$e_B^{\#} \circ f = e_A^{\#}, \qquad i_B^{\#} \circ f = f \circ i_A^{\#}.$$

Remark. Homomorphisms of linear algebraic groups

$$\operatorname{Spm} B \longrightarrow \operatorname{Spm} A$$

correspond contravariantly to homomorphisms of Hopf algebras

$$A \longrightarrow B.$$

Definition 2.2. Let B be a Hopf algebra over k. A k-subalgebra $A \subseteq B$ is called a *Hopf subalgebra* if

$$m_B^{\#}(A) \subseteq A \otimes_k A, \qquad i_B^{\#}(A) \subseteq A.$$

Proposition 2.3. Let $A \subseteq B$ be a Hopf subalgebra. Then A is a Hopf algebra with

$$m_A^{\#} = m_B^{\#}|_A, \qquad e_A^{\#} = e_B^{\#}|_A, \qquad i_A^{\#} = i_B^{\#}|_A.$$

Moreover, if $f : A \to B$ is a homomorphism of Hopf algebras, then $f(A)$ is a Hopf subalgebra of B.

Definition 2.4. Let A be a Hopf k-algebra. A *Hopf ideal* is an ideal $\mathfrak{a} \subseteq A$ such that

$$m^{\#}(\mathfrak{a}) \subseteq \mathfrak{a} \otimes_k A + A \otimes_k \mathfrak{a}, \qquad i^{\#}(\mathfrak{a}) \subseteq \mathfrak{a}, \qquad e^{\#}(\mathfrak{a}) = 0.$$

Equivalently, the composite

$$\mathfrak{a} \hookrightarrow A \xrightarrow{m^{\#}} A \otimes_k A \longrightarrow (A/\mathfrak{a}) \otimes_k (A/\mathfrak{a})$$

is zero, together with the antipode and counit conditions above.

Lemma 2.5. If $f : A \to B$ is a homomorphism of Hopf algebras, then $\ker f$ is a Hopf ideal of A.

Proof. Let $\mathfrak{a} = \ker f$. If $a \in \mathfrak{a}$, then

$$(f \otimes f)m_A^{\#}(a) = m_B^{\#}(f(a)) = 0.$$

Over the field k,

$$\ker(f \otimes f) = \mathfrak{a} \otimes_k A + A \otimes_k \mathfrak{a},$$

so

$$m_A^{\#}(\mathfrak{a}) \subseteq \mathfrak{a} \otimes_k A + A \otimes_k \mathfrak{a}.$$

The compatibility of f with the antipode and counit gives $i_A^{\#}(\mathfrak{a}) \subseteq \mathfrak{a}$ and $e_A^{\#}(\mathfrak{a}) = 0$. □

Proposition 2.6. Let $\mathfrak{a}$ be a Hopf ideal of a Hopf k-algebra A.

(i) The quotient $A/\mathfrak{a}$ has a Hopf algebra structure for which $A \to A/\mathfrak{a}$ is a homo-

morphism of Hopf algebras.

(ii) If $f : A \to B$ is a homomorphism of Hopf k-algebras, then the canonical map induces an isomorphism of Hopf algebras

$$A/\ker f \xrightarrow{\sim} f(A).$$

In particular, f factors as

$$A \twoheadrightarrow A/\ker f \xrightarrow{\sim} f(A) \hookrightarrow B.$$

Fact. Let $A \subseteq B$ be finitely generated Hopf k-algebras. Then B is faithfully flat over A. See [Mil17, Theorem 3.31].

Corollary 2.7. Every homomorphism

$$\varphi : G \longrightarrow H$$

of linear algebraic groups factors as

$$G \xrightarrow{q} I \xrightarrow{\iota} H,$$

where q is faithfully flat and ι is a closed immersion.

Corollary 2.8. Let $\varphi : G \to H$ be a homomorphism of linear algebraic groups. If

$$\ker \varphi = \operatorname{Spm} k,$$

then φ is a closed immersion.

Proof. By the preceding corollary, it is enough to treat the faithfully flat factor q. Thus we may assume that φ itself is faithfully flat, and it remains to show that φ is an isomorphism.

Since $\ker \varphi = \operatorname{Spm} k$, for every $R \in \operatorname{Alg}_k$ the map

$$\varphi(R) : G(R) \longrightarrow H(R)$$

is injective. We show that it is also surjective.

Take $y \in H(R)$. Since φ is faithfully flat, there is a faithfully flat R-algebra R' and an

element $x \in G(R')$ whose image in $H(R')$ is the pullback of y. The descent diagram is

$$\begin{array}{ccccc} G(R) & \longrightarrow & G(R') & \rightrightarrows & G(R' \otimes_R R') \\ {\scriptstyle \varphi(R)}\downarrow & & {\scriptstyle \varphi(R')}\downarrow & & \downarrow{\scriptstyle \varphi(R' \otimes_R R')} \\ H(R) & \longrightarrow & H(R') & \rightrightarrows & H(R' \otimes_R R'). \end{array}$$

The two pullbacks of x to $G(R' \otimes_R R')$ have the same image under $\varphi(R' \otimes_R R')$, because both come from y. Since $\varphi(R' \otimes_R R')$ is injective, the two pullbacks of x agree. Faithfully flat descent for the affine scheme G therefore gives a unique element of $G(R)$ descending to x, and its image under $\varphi(R)$ is y.

Thus $\varphi(R)$ is bijective for every $R \in \mathrm{Alg}_k$. By Yoneda, φ is an isomorphism. □

Remark. A faithfully flat homomorphism of linear algebraic groups $\varphi : G \to H$ need not be surjective on R-points for every $R \in \mathrm{Alg}_k$ when $\ker \varphi \neq \operatorname{Spm} k$.

Proof. Take $k = \mathbb{R}$, $G = H = \mathbb{G}_m$, and

$$\varphi : \mathbb{G}_m \longrightarrow \mathbb{G}_m, \qquad x \longmapsto x^2.$$

On coordinate rings,

$$\varphi^{\#} : \mathbb{R}[T, T^{-1}] \longrightarrow \mathbb{R}[T, T^{-1}], \qquad T \longmapsto T^2,$$

and

$$\varphi^{\#}\big(\mathbb{R}[T, T^{-1}]\big) = \mathbb{R}[T^2, T^{-2}].$$

The ring $\mathbb{R}[T, T^{-1}]$ is a free $\mathbb{R}[T^2, T^{-2}]$-module with basis $\{1, T\}$, so φ is faithfully flat. However,

$$\varphi(\mathbb{R}) : \mathbb{R}^\times \longrightarrow \mathbb{R}^\times, \qquad x \longmapsto x^2,$$

is not surjective. □

Let V be a k-vector space. Define the group-valued functor

$$\mathrm{GL}_V : \mathrm{Alg}_k \longrightarrow \mathrm{Grp}, \qquad R \longmapsto \mathrm{GL}_V(R) := \mathrm{Aut}_R(V \otimes_k R).$$

If V is finite-dimensional, then after choosing a basis of V,

$$\mathrm{GL}_V \cong \mathrm{GL}_n, \qquad n = \dim_k V,$$

where the displayed isomorphism depends on the choice of basis.

2.2 Linear representations

Definition 2.9. Let G be a linear algebraic group over k and let V be a k-vector space. A *linear representation* of G on V is a homomorphism of algebraic groups

$$r : G \longrightarrow \mathrm{GL}_V .$$

The representation r is *faithful* if

$$\ker r = \mathrm{Spm}\, k,$$

and it is *trivial* if its image is the trivial group $\mathrm{Spm}\, k$.

Remark. For a representation $r : G \to \mathrm{GL}_V$, the following are equivalent:

r is faithful;
$r(R) : G(R) \longrightarrow \mathrm{GL}_V(R)$ is injective for every $R \in \mathrm{Alg}_k$;
r is a closed immersion.

Construction. Let $r : G \to \mathrm{GL}_V$ be a representation and write $G = \mathrm{Spm}\, A$. Since

$$G(A) = \mathrm{Hom}_{k\text{-alg}}(A, A),$$

the identity map id_A defines an A-point of G. Hence

$$r(A)(\mathrm{id}_A) \in \mathrm{Aut}_A(V \otimes_k A).$$

Define a k-linear map

$$\rho_r : V \longrightarrow V \otimes_k A, \qquad \rho_r(v) := r(A)(\mathrm{id}_A)(v \otimes 1).$$

Remark. Let $R \in \mathrm{Alg}_k$ and $x \in G(R)$, corresponding to a k-algebra map

$$x^{\#} : A \longrightarrow R.$$

Then naturality gives

$$\begin{array}{ccc} V & \xrightarrow{\ \rho_r\ } & V \otimes_k A \\ & \searrow^{v \mapsto r(R)(x)(v \otimes 1)} & \downarrow{\scriptstyle \mathrm{id}_V \otimes x^{\#}} \\ & & V \otimes_k R. \end{array}$$

Equivalently, the R-linearization of

$$(\mathrm{id}_V \otimes x^{\#}) \circ \rho_r : V \longrightarrow V \otimes_k R$$

is exactly $r(R)(x) : V \otimes_k R \to V \otimes_k R$.

2.3 Comodules

Definition 2.10. A *right $\mathcal{O}(G)$-comodule* is a k-vector space V together with a k-linear map

$$\rho : V \longrightarrow V \otimes_k \mathcal{O}(G)$$

such that

$$(\mathrm{id}_V \otimes e^{\#}) \circ \rho = \mathrm{id}_V$$

and

$$(\mathrm{id}_V \otimes m^{\#}) \circ \rho = (\rho \otimes \mathrm{id}_{\mathcal{O}(G)}) \circ \rho.$$

Proposition 2.11. The map ρ_r satisfies

$$(\mathrm{id}_V \otimes e^{\#}) \circ \rho_r = \mathrm{id}_V$$

and

$$(\rho_r \otimes \mathrm{id}_A) \circ \rho_r = (\mathrm{id}_V \otimes m^{\#}) \circ \rho_r.$$

Hence (V, ρ_r) is a right A-comodule.

Proof. The first identity is the coordinate-ring form of the identity action $e \cdot v = v$. The second is the coordinate-ring form of associativity of the action,

$$g_1\big(g_2 v\big) = (g_1 g_2)v.$$

Indeed, applying the construction to the universal points and using that r is a group homomorphism gives exactly the two displayed identities.[4] □

Definition 2.12. Let (V, ρ) be an $\mathcal{O}(G)$-comodule. An $\mathcal{O}(G)$-subcomodule of V is a

[4] 2026.9.17 stopped at here.

k-subspace $W \subseteq V$ such that

$$\rho(W) \subseteq W \otimes \mathcal{O}(G).$$

Then $(W, \rho|_W)$ is again an $\mathcal{O}(G)$-comodule.

Lemma 2.13. For a k-vector space V, there is a bijection

$$\left\{ \text{ representations } r : G \to \mathrm{GL}_V \right\} \longleftrightarrow \left\{ \mathcal{O}(G)\text{-comodule structures on } V \right\},$$

sending r to ρ_r.

Proof. It remains to define the inverse map. Let

$$\rho : V \longrightarrow V \otimes \mathcal{O}(G)$$

be a comodule map. For $R \in \mathrm{Alg}_k$ and

$$g \in G(R) = \mathrm{Hom}_{k\text{-alg}}(\mathcal{O}(G), R),$$

consider

$$V \xrightarrow{\rho} V \otimes \mathcal{O}(G) \xrightarrow{\mathrm{id}_V \otimes g} V \otimes R.$$

Its R-linearization is an element of

$$\mathrm{Aut}_R(V \otimes R) = \mathrm{GL}_V(R).$$

This defines $r_\rho(R) : G(R) \to \mathrm{GL}_V(R)$. The comodule identities give the identity and multiplication laws, and the two constructions are inverse to each other. □

Example 2.14 (The standard representation of GL_n)**.** Let $G = \mathrm{GL}_n$, let $V = k^n$ with basis $\{e_1, \ldots, e_n\}$, and let

$$r : \mathrm{GL}_n \xrightarrow{\sim} \mathrm{GL}_V$$

be the standard representation. Write

$$\mathcal{O}(G) = k[T_{ij}, \det(T)^{-1}].$$

If

$$\rho_r(e_i) = \sum_{j=1}^{n} e_j \otimes f_{ji},$$

then for $R \in \mathrm{Alg}_k$ and $g = (a_{st}) \in G(R)$,

$$\sum_{j=1}^{n} a_{ji} e_j = g e_i = \sum_{j=1}^{n} f_{ji}(g) e_j.^{5}$$

Hence $f_{ji}(g) = a_{ji}$, so $f_{ji} = T_{ji}$. Therefore

$$\rho_r(e_i) = \sum_{j=1}^{n} e_j \otimes T_{ji}.$$

Exercise 2.15. Compute the $\mathcal{O}(G)$-comodule structures on $V^\vee$ and on $V \otimes V$.

Proposition 2.16. Every $\mathcal{O}(G)$-comodule (V, ρ) is a filtered union of its finite-dimensional subcomodules.

Proof. It is enough to show that every $v \in V$ is contained in a finite-dimensional subcomodule. Let $\{f_i\}_{i \in I}$ be a k-basis of $\mathcal{O}(G)$. Write

$$\rho(v) = \sum_{i \in I_0} v_i \otimes f_i$$

for a finite subset $I_0 \subset I$, and let

$$W = \mathrm{span}_k\{v_i : i \in I_0\}.$$

The counit identity gives

$$v = (\mathrm{id}_V \otimes e^{\#})\rho(v) = \sum_{i \in I_0} v_i \, e^{\#}(f_i) \in W,$$

and $\dim_k W < \infty$.

[5] The second equality comes from another description of ρ_r which can be proved by $(\mathrm{id}_V \otimes g^{\#}) \circ \rho_r = r(R)(g)(- \circ 1)$.

It remains to show that $\rho(W) \subseteq W \otimes \mathcal{O}(G)$. For each $i \in I_0$, write

$$m^{\#}(f_i) = \sum_{j,\ell \in I} a_{ij\ell} f_j \otimes f_\ell,$$

where only finitely many coefficients occur. Coassociativity gives the commutative square

$$\begin{array}{ccc} V & \xrightarrow{\ \rho\ } & V \otimes \mathcal{O}(G) \\ {\scriptstyle \rho}\downarrow & & \downarrow{\scriptstyle \mathrm{id}_V \otimes m^{\#}} \\ V \otimes \mathcal{O}(G) & \xrightarrow[\rho \otimes \mathrm{id}]{} & V \otimes \mathcal{O}(G) \otimes \mathcal{O}(G). \end{array}$$

Applying the two paths to v, one obtains

$$\sum_i v_i \otimes m^{\#}(f_i) = \sum_{i,j,\ell} a_{ij\ell} v_i \otimes f_j \otimes f_\ell$$

and

$$\sum_\ell \rho(v_\ell) \otimes f_\ell.$$

Comparing coefficients of the basis elements f_ℓ in the third tensor factor gives

$$\rho(v_\ell) = \sum_{i,j} a_{ij\ell} v_i \otimes f_j \in W \otimes \mathcal{O}(G).$$

Thus W is a finite-dimensional subcomodule containing v. Since the sum of two finite-dimensional subcomodules is again a finite-dimensional subcomodule, these subcomodules form a filtered system. □

Corollary 2.17. Every representation of G is a filtered union of its finite-dimensional subrepresentations.

Proof. This is the preceding proposition under the representation–comodule correspondence. □

2.4 LAG is linear

Proposition 2.18 (Affine right actions and comodule algebras)**.** Let X be an affine k-scheme. A k-algebra homomorphism

$$\rho : \mathcal{O}(X) \longrightarrow \mathcal{O}(X) \otimes \mathcal{O}(G)$$

which makes $\mathcal{O}(X)$ an $\mathcal{O}(G)$-comodule corresponds contravariantly to a right action

$$a : X \times G \longrightarrow X.$$

Proof. The counit condition is the commutativity of

$$\begin{array}{ccc} \mathcal{O}(X) & \xrightarrow{\rho} & \mathcal{O}(X) \otimes \mathcal{O}(G) \\ & \searrow^{\mathrm{id}} & \downarrow{\scriptstyle \mathrm{id} \otimes e^{\#}} \\ & & \mathcal{O}(X), \end{array}$$

which is dual to the commutative triangle

$$\begin{array}{ccc} X \times G & \xrightarrow{a} & X \\ \uparrow{\scriptstyle (\mathrm{id}_X, e)} & \nearrow_{\mathrm{id}_X} & \\ X & & \end{array}$$

for the identity axiom of a right action. The coassociativity condition is

$$\begin{array}{ccc} \mathcal{O}(X) & \xrightarrow{\rho} & \mathcal{O}(X) \otimes \mathcal{O}(G) \\ \downarrow{\scriptstyle \rho} & & \downarrow{\scriptstyle \mathrm{id} \otimes m^{\#}} \\ \mathcal{O}(X) \otimes \mathcal{O}(G) & \xrightarrow[\rho \otimes \mathrm{id}]{} & \mathcal{O}(X) \otimes \mathcal{O}(G) \otimes \mathcal{O}(G), \end{array}$$

which is dual to

$$\begin{array}{ccc} X \times G \times G & \xrightarrow{a \times \mathrm{id}_G} & X \times G \\ \downarrow{\scriptstyle \mathrm{id}_X \times m} & & \downarrow{\scriptstyle a} \\ X \times G & \xrightarrow[a]{} & X. \end{array}$$

Thus the two sets of axioms are equivalent. □

Remark. If $r : G \to \mathrm{GL}_V$ is a representation, then for every $R \in \mathrm{Alg}_k$, the group $G(R)$ acts

on $V \otimes R$ on the left. The action extends to $\mathrm{Sym}(V)$, and because Spm is contravariant, G acts on $\operatorname{Spm}\mathrm{Sym}(V)$ on the right. Equivalently, the comodule map on V extends to a comodule-algebra map

$$\mathrm{Sym}(V) \longrightarrow \mathrm{Sym}(V) \otimes \mathcal{O}(G).$$

Definition 2.19. A representation V of G is *simple* if $V \neq 0$ and V has no nontrivial subrepresentations.

Corollary 2.20. Every simple representation of a linear algebraic group is finite-dimensional.

Proof. A nonzero vector is contained in a finite-dimensional subrepresentation by the preceding local-finiteness proposition. Simplicity forces that subrepresentation to be all of V. □

Let $X = G$, and let

$$m : G \times G \longrightarrow G$$

be right multiplication. The induced map

$$m^{\#} : \mathcal{O}(G) \longrightarrow \mathcal{O}(G) \otimes \mathcal{O}(G)$$

gives $\mathcal{O}(G)$ its regular $\mathcal{O}(G)$-comodule structure.

Definition 2.21. The representation

$$r_G : G \longrightarrow \mathrm{GL}_{\mathcal{O}(G)}$$

associated with the comodule $(\mathcal{O}(G), m^{\#})$ is called the *regular representation* of G.

Theorem 2.22. The regular representation of G has a faithful finite-dimensional subrepresentation. In particular, the regular representation is faithful.

Proof. Let $V \subseteq \mathcal{O}(G)$ be a finite-dimensional subcomodule. The subcomodule condition is expressed by the commutative diagram

$$\begin{array}{ccc} V & \hookrightarrow & \mathcal{O}(G) \\ \downarrow{\scriptstyle \rho} & & \downarrow{\scriptstyle m^{\#}} \\ V \otimes \mathcal{O}(G) & \hookrightarrow & \mathcal{O}(G) \otimes \mathcal{O}(G). \end{array}$$

Choose a basis $f_1, \dots, f_n$ of V. Write

$$m^{\#}(f_j) = \sum_{i=1}^{n} f_i \otimes b_{ij}, \qquad b_{ij} \in \mathcal{O}(G).$$

Then the corresponding representation

$$G \longrightarrow \mathrm{GL}_V \simeq \mathrm{GL}_n$$

has matrix $(b_{ij}(g))$ at g, and hence its comorphism is

$$\mathcal{O}(\mathrm{GL}_n) = k[T_{ij}, \det(T)^{-1}] \longrightarrow \mathcal{O}(G), \qquad T_{ij} \longmapsto b_{ij}.$$

This representation is faithful exactly when this map is surjective, that is, when the b_{ij} generate $\mathcal{O}(G)$ as a k-algebra.

Applying $e^{\#} \otimes \mathrm{id}$ to $m^{\#}(f_j) = \sum_i f_i \otimes b_{ij}$ gives

$$f_j = \sum_{i=1}^{n} f_i(e) b_{ij}.$$

Therefore it is enough to choose V so that

(1) V generates $\mathcal{O}(G)$ as a k-algebra;

(2) V is an $\mathcal{O}(G)$-subcomodule;

(3) $\dim_k V < \infty$.

Because $\mathcal{O}(G)$ is a finitely generated k-algebra, choose finitely many algebra generators. By local finiteness, each generator is contained in a finite-dimensional subcomodule. The sum of those finitely many subcomodules satisfies all three conditions. Hence the resulting $G \to \mathrm{GL}_V$ is faithful. □

Corollary 2.23. If G is a linear algebraic group, then G is isomorphic to an algebraic subgroup of GL_n for some n.

Proof. Take the faithful finite-dimensional subrepresentation furnished by the theorem. A faithful finite-dimensional representation of a linear algebraic group is a closed immersion. □

2.5 Constructing all finite-dimensional representations

Definition 2.24. For a k-vector space V, the comodule

$$V \otimes \mathcal{O}(G)$$

with coaction

$$\mathrm{id}_V \otimes m^{\#} : V \otimes \mathcal{O}(G) \longrightarrow V \otimes \mathcal{O}(G) \otimes \mathcal{O}(G)$$

is called the *free comodule on* V.

Remark. If $\dim_k V = n$ and a basis of V is fixed, then

$$V \otimes \mathcal{O}(G) \simeq \mathcal{O}(G)^n,$$

with the componentwise free-comodule structure.

Proposition 2.25. Every finite-dimensional $\mathcal{O}(G)$-comodule V is a subcomodule of

$$\left(\mathcal{O}(G)^n, (m^{\#})^n\right)$$

for $n = \dim_k V$.

Proof. The coaction

$$\rho : V \longrightarrow V \otimes \mathcal{O}(G)$$

is injective, because

$$(\mathrm{id}_V \otimes e^{\#}) \circ \rho = \mathrm{id}_V .$$

Moreover, the diagram

$$\begin{array}{ccc} V & \xrightarrow{\quad\rho\quad} & V \otimes \mathcal{O}(G) \\ {\scriptstyle\rho}\downarrow & & \downarrow{\scriptstyle \mathrm{id}_V \otimes m^{\#}} \\ V \otimes \mathcal{O}(G) & \xrightarrow[\rho\otimes\mathrm{id}]{} & V \otimes \mathcal{O}(G) \otimes \mathcal{O}(G) \end{array}$$

commutes. Hence ρ is a comodule morphism from V into the free comodule $V \otimes \mathcal{O}(G)$. After choosing a basis of V,

$$V \otimes \mathcal{O}(G) \simeq \mathcal{O}(G)^n.$$

□

Theorem 2.26. Let

$$r : G \longrightarrow \mathrm{GL}_V$$

be a faithful finite-dimensional representation of G. Then every finite-dimensional representation W of G can be constructed from V by taking tensor products, finite direct sums, duals, and subquotients.

Proof. Step 1: reduction to $W \subseteq \mathcal{O}(G)$. By the preceding proposition, we may embed

$$W \hookrightarrow \mathcal{O}(G)^n$$

for some n. Let

$$W_i = \mathrm{im}\Big(W \hookrightarrow \mathcal{O}(G)^n \xrightarrow{\mathrm{pr}_i} \mathcal{O}(G)\Big).$$

Then each W_i is a finite-dimensional subcomodule of $\mathcal{O}(G)$, and the map

$$W \longrightarrow \bigoplus_i W_i$$

is injective. Hence it is enough to treat finite-dimensional subcomodules $W \subseteq \mathcal{O}(G)$.
Step 2: reduction to $G = \mathrm{GL}_V$. Since r is faithful, its comorphism

$$r^{\#} : \mathcal{O}(\mathrm{GL}_V) \longrightarrow \mathcal{O}(G)$$

is surjective.

If

$$\widetilde{W} \subseteq \mathcal{O}(\mathrm{GL}_V)$$

is a finite-dimensional $\mathcal{O}(\mathrm{GL}_V)$-subcomodule, then the Hopf-algebra identity

$$m_G^{\#} \circ r^{\#} = (r^{\#} \otimes r^{\#}) \circ m_{\mathrm{GL}_V}^{\#}$$

shows that

$$r^{\#}(\widetilde{W}) \subseteq \mathcal{O}(G)$$

is an $\mathcal{O}(G)$-subcomodule. Equivalently, one has the commutative diagram

$$\begin{array}{ccccc}
\widetilde{W} & \hookrightarrow & \mathcal{O}(\mathrm{GL}_V) & \xrightarrow{m^{\#}_{\mathrm{GL}_V}} & \mathcal{O}(\mathrm{GL}_V)\otimes\mathcal{O}(\mathrm{GL}_V) \\
\downarrow{\scriptstyle r^{\#}|_{\widetilde{W}}} & & \downarrow{\scriptstyle r^{\#}} & & \downarrow{\scriptstyle r^{\#}\otimes r^{\#}} \\
r^{\#}(\widetilde{W}) & \hookrightarrow & \mathcal{O}(G) & \xrightarrow[m^{\#}_{G}]{} & \mathcal{O}(G)\otimes\mathcal{O}(G).
\end{array}$$

Conversely, given a finite-dimensional subcomodule $W \subseteq \mathcal{O}(G)$, choose lifts in $\mathcal{O}(\mathrm{GL}_V)$ of a basis of W. By local finiteness of the regular representation of GL_V, those lifts are contained in a finite-dimensional subcomodule

$$\widetilde{W} \subseteq \mathcal{O}(\mathrm{GL}_V)$$

such that

$$r^{\#}(\widetilde{W}) \supseteq W.$$

Thus it is enough to treat $G = \mathrm{GL}_V$ case. [6]

Step 3: the required finite-dimensional subrepresentation. It suffices to find a finite-dimensional subrepresentation

$$\widetilde{W} \subseteq \mathcal{O}(\mathrm{GL}(V))$$

such that:

(1) $\widetilde{W}$ is constructed from V by tensor products, finite direct sums, duals, and subquotients;

(2) $\widetilde{W}$ generates $\mathcal{O}(\mathrm{GL}(V))$ as a k-algebra.

Indeed, condition (2) gives a surjective homomorphism of comodule algebras

$$\mathrm{Sym}(\widetilde{W}) \longrightarrow \mathcal{O}(\mathrm{GL}(V)).$$

The algebra $\mathrm{Sym}(\widetilde{W})$ is the filtered union of the finite-dimensional representations

$$\bigoplus_{q=0}^{d} \mathrm{Sym}^q(\widetilde{W}), \qquad d \geq 0.$$

Each symmetric power is a quotient of a tensor power. Hence every finite-dimensional subrepresentation of $\mathcal{O}(\mathrm{GL}(V))$ is a subquotient of a representation constructed from V by

[6] 2026.9.21 stopped at here.

the permitted operations.

Step 4: construction of $\widetilde{W}$. Choose a basis $e_1, \dots, e_n$ of V. Then

$$\mathrm{GL}(V) \simeq \mathrm{GL}_n, \qquad \mathcal{O}(\mathrm{GL}_n) = k[T_{ij} \mid 1 \le i, j \le n, \det(T)^{-1}].$$

Set

$$\widetilde{W} = \left(\bigoplus_{1 \le i,j \le n} kT_{ij} \right) \oplus k \det(T)^{-1}.$$

The comultiplication is given by

$$m^{\#}(T_{ij}) = \sum_{\ell=1}^{n} T_{i\ell} \otimes T_{\ell j}, \qquad m^{\#}(\det(T)^{-1}) = \det(T)^{-1} \otimes \det(T)^{-1}.$$

For $1 \le i \le n$, let

$$V_i = \bigoplus_{j=1}^{n} kT_{ij}, \qquad V_0 = k \det(T)^{-1}.$$

Then

$$\widetilde{W} = V_0 \oplus V_1 \oplus \cdots \oplus V_n,$$

and all the V_i are subrepresentations of the regular representation. The standard coaction on V is

$$\rho \colon V \longrightarrow V \otimes \mathcal{O}(\mathrm{GL}(V)), \qquad e_j \longmapsto \sum_{i=1}^{n} e_i \otimes T_{ij}.$$

It follows that

$$V_i \simeq V \quad (1 \le i \le n), \qquad V_0 \simeq \left(\bigwedge^{n} V \right)^{\vee}.$$

Thus $\widetilde{W}$ has property (1), and its displayed basis shows that it generates $\mathcal{O}(\mathrm{GL}(V))$, giving property (2). □

2.6 Chevalley's theorem

Let $r \colon G \to \mathrm{GL}(V)$ be a finite-dimensional representation, and let $W \subseteq V$ be a k-linear subspace. For a k-algebra R, write $V_R = V \otimes_k R$ and $W_R = W \otimes_k R$.

Proposition 2.27. The functor

$$\mathrm{Stab}_G(W) \colon \mathrm{Alg}_k \longrightarrow \mathrm{Grp}, \qquad R \longmapsto \{g \in G(R) \mid r_R(g)(W_R) = W_R\},$$

is represented by an algebraic subgroup of G.

Proof. Let $(e_j)_{j\in J}$ be a basis of W and extend it to a basis $(e_i)_{i\in I\sqcup J}$ of V. Write the coaction as

$$\rho(e_j) = \sum_{i\in I\sqcup J} e_i \otimes f_{ij}, \qquad f_{ij} \in \mathcal{O}(G).$$

For every $R \in \mathrm{Alg}_k$ and $g \in G(R)$, one has

$$\begin{aligned} g \in \mathrm{Stab}_G(W)(R) &\iff r_R(g)(W_R) = W_R \\ &\iff r_R(g)(e_j) \in W_R \quad \text{for every } j \in J \\ &\iff f_{ij}(g) = 0 \quad \text{for every } i \in I,\ j \in J. \end{aligned}$$

Here containment of $r_R(g)(W_R)$ in W_R is equivalent to equality: in the chosen basis, $r_R(g)$ is then block upper triangular, and its invertibility forces its diagonal blocks to be invertible. Therefore

$$\mathrm{Stab}_G(W) \simeq \mathrm{Spm}\left(\mathcal{O}(G)/(f_{ij} \mid i \in I,\ j \in J)\right),$$

which is a closed subgroup of G. □

Definition 2.28. The algebraic subgroup

$$G_W := \mathrm{Stab}_G(W)$$

is called the *stabilizer of W in G*.

Theorem 2.29 (Chevalley)**.** Every algebraic subgroup $H \subseteq G$ is the stabilizer of a one-dimensional subspace L in a finite-dimensional representation of G.

Remark. This theorem is one of the main ingredients in proving the existence of the quotient G/H as a scheme.

Proof. Step 1: realize H as the stabilizer of a finite-dimensional subspace. Let

$$\mathfrak{a} = \ker\big(\mathcal{O}(G) \longrightarrow \mathcal{O}(H)\big).$$

This is a Hopf ideal. Choose a finite-dimensional subcomodule $V \subseteq \mathcal{O}(G)$ containing a set of generators of $\mathfrak{a}$ as an ideal, and put

$$W = V \cap \mathfrak{a}.$$

Choose a basis $(f_j)_{j\in J}$ of W and extend it to a basis $(f_i)_{i\in I\sqcup J}$ of V. Write

$$m^{\#}(f_j) = \sum_{i\in I\sqcup J} f_i \otimes a_{ij}, \qquad j \in J,$$

and set

$$\mathfrak{a}' = (a_{ij} \mid i \in I,\ j \in J).$$

By the preceding proposition,

$$\mathrm{Stab}_G(W) = \mathrm{Spm}\big(\mathcal{O}(G)/\mathfrak{a}'\big).$$

Because $\mathfrak{a}$ is a Hopf ideal,

$$m^{\#}(\mathfrak{a}) \subseteq \mathfrak{a} \otimes \mathcal{O}(G) + \mathcal{O}(G) \otimes \mathfrak{a}.$$

The images of $(f_i)_{i\in I}$ in $\mathcal{O}(G)/\mathfrak{a}$ are linearly independent. Reducing the displayed expression for $m^{\#}(f_j)$ modulo

$$\mathfrak{a} \otimes \mathcal{O}(G) + \mathcal{O}(G) \otimes \mathfrak{a}$$

therefore gives $a_{ij} \in \mathfrak{a}$ for $i \in I$ and $j \in J$. Hence $\mathfrak{a}' \subseteq \mathfrak{a}$.

Conversely, applying $e^{\#} \otimes \mathrm{id}$ to $m^{\#}(f_j)$ gives

$$f_j = \sum_{i\in I\sqcup J} f_i(e)a_{ij} = \sum_{i\in I} f_i(e)a_{ij} \in \mathfrak{a}',$$

because $f_i(e) = 0$ for $i \in J$. Thus $W \subseteq \mathfrak{a}'$. Since W contains a generating set of $\mathfrak{a}$ as an ideal, we have $\mathfrak{a} \subseteq \mathfrak{a}'$. It follows that $\mathfrak{a} = \mathfrak{a}'$ and hence $H = \mathrm{Stab}_G(W)$.

Step 2: replace W by a line. Let $d = \dim_k W$, set

$$\widetilde{V} = \bigwedge^d V, \qquad L = \bigwedge^d W \subseteq \widetilde{V}.$$

We show that $\mathrm{Stab}_G(L) = H$. It is enough to prove that, for every k-algebra R and every $\alpha \in \mathrm{GL}(V_R)$,

$$\alpha(W_R) = W_R \quad\Longleftrightarrow\quad \left(\bigwedge^d \alpha\right)(L_R) = L_R.$$

Choose a basis $e_1, \dots, e_d$ of W and extend it to a basis of V; put $w = e_1 \wedge \cdots \wedge e_d$. The

forward implication is immediate. For the reverse implication, suppose

$$\left(\bigwedge^{d} \alpha\right)(w) = cw \qquad \text{for some } c \in R^{\times}.$$

Since

$$W_R = \{v \in V_R \mid w \wedge v = 0\},$$

every $v \in W_R$ satisfies

$$0 = \left(\bigwedge^{d+1} \alpha\right)(w \wedge v) = cw \wedge \alpha(v).$$

As c is a unit, $\alpha(v) \in W_R$. Applying the same argument to α^{-1} gives the reverse containment, so $\alpha(W_R) = W_R$. This completes the proof. □

2.7 Characters and eigenspaces

Definition 2.30. A *character* of a linear algebraic group G over k is a homomorphism

$$\chi \colon G \longrightarrow \mathbb{G}_m.$$

Remark. The character χ is determined by its comorphism

$$\chi^{\#} \colon k[T, T^{-1}] \longrightarrow \mathcal{O}(G),$$

which fits into the commutative diagram

$$\begin{array}{ccc} k[T,T^{-1}] & \xrightarrow{\quad\chi^{\#}\quad} & \mathcal{O}(G) \\ \downarrow{\scriptstyle m^{\#}} & & \downarrow{\scriptstyle m^{\#}} \\ k[T,T^{-1}] \otimes k[T,T^{-1}] & \xrightarrow[\chi^{\#}\otimes\chi^{\#}]{} & \mathcal{O}(G) \otimes \mathcal{O}(G). \end{array}$$

Since $m^{\#}(T) = T \otimes T$, one has

$$m^{\#}\big(\chi^{\#}(T)\big) = \chi^{\#}(T) \otimes \chi^{\#}(T).$$

An element $a \in \mathcal{O}(G)^{\times}$ satisfying

$$m^{\#}(a) = a \otimes a$$

is called *group-like*. There is a bijection

$$\{\text{characters of } G \text{ over } k\} \longleftrightarrow \{\text{group-like elements of } \mathcal{O}(G)\}, \qquad \chi \longmapsto a(\chi) := \chi^{\#}(T).$$

The trivial character corresponds to $1 \in \mathcal{O}(G)$. For every $R \in \mathrm{Alg}_k$, the induced map is

$$\chi(R)\colon G(R) \longrightarrow R^{\times}, \qquad g \longmapsto a(\chi)(g).$$

Thus characters of G are precisely its one-dimensional representations.

Example 2.31. The characters of $\mathbb{G}_m$ are

$$\chi_n\colon \mathbb{G}_m \longrightarrow \mathbb{G}_m, \qquad x \longmapsto x^n, \qquad n \in \mathbb{Z}.$$

Proposition 2.32. Distinct group-like elements of $\mathcal{O}(G)$ are linearly independent over k.

Proof. Suppose that $a_1, \ldots, a_n$ are distinct group-like elements satisfying a nontrivial linear relation

$$\sum_{i=1}^{n} c_i a_i = 0, \qquad c_i \in k,$$

and choose such a relation with n minimal. Then every c_i is nonzero. Since $a_1 \neq a_2$, there is a k-linear functional $\lambda\colon \mathcal{O}(G) \to k$ such that $\lambda(a_1) \neq \lambda(a_2)$. Applying $\lambda \otimes \mathrm{id}$ to the comultiplication of the relation gives

$$\sum_{i=1}^{n} c_i \lambda(a_i) a_i = 0.$$

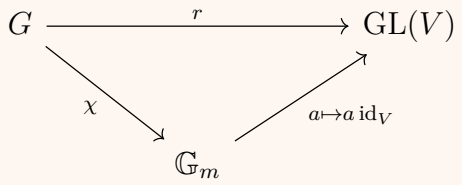
Subtracting $\lambda(a_1)$ times the original relation yields

$$\sum_{i=2}^{n} c_i \big(\lambda(a_i) - \lambda(a_1)\big) a_i = 0.$$

The coefficient of a_2 is nonzero, so this is a nontrivial relation involving fewer than n group-like elements, contradicting minimality. □

Corollary 2.33. Distinct characters $\chi_1, \ldots, \chi_n$ of G are linearly independent over k, when identified with the elements $a(\chi_1), \ldots, a(\chi_n) \in \mathcal{O}(G)$.

Definition 2.34. Let $r\colon G \to \mathrm{GL}(V)$ be a representation and let $\chi\colon G \to \mathbb{G}_m$ be a character. We say that G *acts on* V *through* χ if r factors as



Equivalently, the coaction satisfies

$$\rho(v) = v \otimes a(\chi) \qquad \text{for every } v \in V.$$

Definition 2.35. Let $r\colon G \to \mathrm{GL}(V)$ be a representation and let $W \subseteq V$ be a G-stable subspace. We say that G *acts on* W *through* χ if the restricted representation on W acts through χ. The largest subspace of V on which G acts through χ is called the χ-*eigenspace* (or χ-*weight space*) and is denoted by V_χ. Explicitly,

$$V_\chi = \{v \in V \mid \rho(v) = v \otimes a(\chi)\}^{7}.$$

Proposition 2.36. If V is a sum of character eigenspaces, then the sum is direct:

$$V = \sum_\chi V_\chi \qquad \Longrightarrow \qquad V = \bigoplus_\chi V_\chi.$$

Proof. Suppose that $v_1 + \cdots + v_s = 0$, with $v_i \in V_{\chi_i}$ and the χ_i pairwise distinct. Applying the coaction gives

$$0 = \sum_{i=1}^{s} \rho(v_i) = \sum_{i=1}^{s} v_i \otimes a(\chi_i).$$

For every $f \in V^\vee = \mathrm{Hom}_k(V, k)$, applying $f \otimes \mathrm{id}_A$ yields $\sum_i f(v_i)a(\chi_i) = 0$. The elements $a(\chi_i)$ are linearly independent, so $f(v_i) = 0$ for every f and every i. Hence all $v_i = 0$. □

Remark. If $G = \mathbb{G}_m^n$, every finite-dimensional representation of G is a direct sum of its character eigenspaces.

[7] 2026.9.24 stopped at here.

Example 2.37. Consider the two-dimensional representation of $\mathbb{G}_a$ given by

$$t \longmapsto \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

If χ_0 is the trivial character, then $V_{\chi_0} = ke_1$ is the only nonzero character eigenspace of this representation; it is one dimensional. In particular, the preceding corollary does not extend to arbitrary G.

2.8 Reconstructing the group from representations

Motivation. For a commutative group G, consider its group of continuous characters $G^\vee = \mathrm{Hom}_{\mathrm{cont}}(G, S^1)$. For locally compact Hausdorff abelian groups, Pontryagin duality gives $G^{\vee\vee} \simeq G$. This suggests the question: can a linear algebraic group be recovered from its representations? The answer is yes. A reference is Deligne–Milne, *Tannakian Categories*.

Definition 2.38. $\mathrm{Rep}_k(G)$ is the category of representations of G on finite-dimensional k-vector spaces. It is a tensor category: for $V, W \in \mathrm{Rep}_k(G)$, also $V \otimes_k W \in \mathrm{Rep}_k(G)$. The forgetful functor is

$$\omega : \mathrm{Rep}_k(G) \longrightarrow \mathrm{Vec}_k, \qquad (V, r) \longmapsto V,$$

where Vec_k denotes finite-dimensional k-vector spaces.

Definition 2.39. Define the group-valued functor

$$\mathrm{Aut}^{\otimes}(\omega) : \mathrm{Alg}_k \longrightarrow \mathrm{Grp}, \qquad R \longmapsto \mathrm{Aut}^{\otimes}(\omega)(R).$$

An element $\lambda = (\lambda_X)_X$ of $\mathrm{Aut}^{\otimes}(\omega)(R)$ assigns to each $X \in \mathrm{Rep}_k(G)$ an R-linear automorphism

$$\lambda_X : \omega(X) \otimes_k R \xrightarrow{\sim} \omega(X) \otimes_k R$$

such that:

(1) $\lambda_{\mathbf{1}} = \mathrm{id}_R$ for the trivial representation $\mathbf{1}$;

(2) $\lambda_{X \otimes Y} = \lambda_X \otimes_R \lambda_Y$ for all X, Y;

(3) for every morphism $\alpha : X \to Y$ in $\mathrm{Rep}_k(G)$, the following diagram commutes:

$$\begin{array}{ccc} \omega(X)\otimes_k R & \xrightarrow{\lambda_X} & \omega(X)\otimes_k R \\ {\scriptstyle \omega(\alpha)\otimes 1}\downarrow & & \downarrow{\scriptstyle \omega(\alpha)\otimes 1} \\ \omega(Y)\otimes_k R & \xrightarrow[\lambda_Y]{} & \omega(Y)\otimes_k R. \end{array}$$

Multiplication in $\mathrm{Aut}^{\otimes}(\omega)(R)$ is componentwise composition.

Every $g \in G(R)$ gives an element of $\mathrm{Aut}^{\otimes}(\omega)(R)$: for $X = (V, r)$ take $\lambda_X = r_R(g)$ on $V \otimes_k R$. The three conditions above hold, giving a natural homomorphism of group-valued functors

$$G \longrightarrow \mathrm{Aut}^{\otimes}(\omega), \qquad g \longmapsto \big(r_R(g)\big)_{(V,r)}.$$

Theorem 2.40 (Reconstruction)**.** For a linear algebraic group G over k, the natural homomorphism

$$G \xrightarrow{\sim} \mathrm{Aut}^{\otimes}(\omega)$$

is an isomorphism of group-valued functors.

Remark. The corresponding statement for a general algebraic group cannot recover the entire group in this way; its finite-dimensional linear representations recover its affinization G^{aff}.

Let $R \in \mathrm{Alg}_k$. Put $G_R = G \times_k \operatorname{Spec} R$ and $A_R = \mathcal{O}(G_R) = A \otimes_k R$. The right regular representation is

$$r_{A,R'}(g)f(x) = f(xg) \quad (g, x \in G(R'),\ f \in A_R \otimes_R R'),$$

for each R-algebra R'. When $R = k$ we simply write r_A.

Lemma 2.41 (Translations)**.** Let $u : A_R \to A_R$ be an R-algebra endomorphism. If, with $\Delta_R = m^* : A_R \to A_R \otimes_R A_R$,

$$\begin{array}{ccc} A_R & \xrightarrow{\Delta_R} & A_R \otimes_R A_R \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle \mathrm{id}\otimes u} \\ A_R & \xrightarrow[\Delta_R]{} & A_R \otimes_R A_R \end{array}$$

commutes, then $u = r_{A,R}(g)$ for some $g \in G(R)$.

Proof. Let $\varphi : G_R \to G_R$ correspond to u. For every R-algebra R', the displayed identity says $\varphi_{R'}(xy) = x\varphi_{R'}(y)$ for $x, y \in G(R')$. Taking $y = e$ and setting $g = \varphi_R(e) \in G(R)$ gives $\varphi_{R'}(x) = xg_{R'}$ for all R'. Hence u is right translation by g. □

Proof of the reconstruction theorem. Fix $R \in \mathrm{Alg}_k$. We prove that

$$G(R) \longrightarrow \mathrm{Aut}^{\otimes}(\omega)(R)$$

is bijective.

Step 1: Extend λ. Let $\lambda = (\lambda_X)_X \in \mathrm{Aut}^{\otimes}(\omega)(R)$. The conditions defining λ also apply to an infinite-dimensional rational representation, because every such representation is the directed union of its finite-dimensional subrepresentations. Naturality on their inclusions makes the maps λ_X compatible, so they define a map on the union. In particular λ acts on the regular representation A. For any rational representation $V = \bigcup_i V_i$ with finite-dimensional subrepresentations V_i, the resulting automorphism of $V \otimes_k R$ is characterized by

$$\lambda_V|_{V_i \otimes_k R} = \lambda_{V_i} \qquad \text{for all } i.$$

Its inverse is obtained by extending the inverses $\lambda_{V_i}^{-1}$.

Step 2: Recover a point from the regular representation. Set $u = \lambda_A : A_R \to A_R$. The unit $k \to A$ and multiplication $m : A \otimes_k A \to A$ are equivariant for the right regular action (with diagonal action on $A \otimes_k A$). Tensor compatibility and naturality give $u(1) = 1$ and $u(fh) = u(f)u(h)$. Thus u is an R-algebra endomorphism.

The comultiplication $\Delta = m^* : A \to A \otimes_k A$ is equivariant when the first factor on the right is given the *trivial* G-action and the second factor has the right regular action: indeed $\Delta(r_A(g)f) = (\mathrm{id} \otimes r_A(g))\Delta(f)$. A tensor automorphism of the forgetful functor acts as the identity on trivial representations. Naturality of Δ therefore gives

$$\begin{array}{ccc} A_R & \xrightarrow{\Delta_R} & A_R \otimes_R A_R \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle \mathrm{id}\otimes u} \\ A_R & \xrightarrow[\Delta_R]{} & A_R \otimes_R A_R. \end{array}$$

By the translation lemma, $u = r_{A,R}(g)$ for some $g \in G(R)$.

Step 3: Control every representation. For $X = (V, r) \in \mathrm{Rep}_k(G)$, choose a basis of V and regard $A \otimes_k V$ as a finite direct sum of copies of the regular representation. The matrix-coefficient map

$$\iota : V \longrightarrow A \otimes_k V, \qquad \iota(v)(x) = r(x)v$$

is G-equivariant and is injective, since evaluation at e sends $\iota(v)$ to v. Thus V embeds in $A^{\oplus \dim V}$. Functoriality gives the commutative diagram

$$\begin{array}{ccc} V \otimes_k R & \xrightarrow{\iota\otimes 1} & A_R \otimes_R (V \otimes_k R) \\ {\scriptstyle \lambda_V}\downarrow & & \downarrow{\scriptstyle u\otimes \mathrm{id}} \\ V \otimes_k R & \xrightarrow[\iota\otimes 1]{} & A_R \otimes_R (V \otimes_k R). \end{array}$$

Since $u = r_{A,R}(g)$ and ι is equivariant, the same diagram commutes with $r_R(g)$ in place of λ_V. The top arrow remains injective after base change to R; hence $\lambda_V = r_R(g)$.

Step 4: Uniqueness. There is a finite-dimensional faithful representation $r : G \hookrightarrow \mathrm{GL}(V)$ which is a closed immersion. If two points of $G(R)$ give the same tensor automorphism, their images under r_R coincide, and so the points coincide. This proves bijectivity for every R, naturally in R. □

Remark. Reconstruction allows us to replace G by GL_n in some situations.

2.9 Characterization of categories of representations

Theorem 2.42 (Neutral Tannakian reconstruction)**.** Let $(\mathcal{C}, \otimes)$ be a neutral Tannakian category over k, and let $\omega : \mathcal{C} \to \mathrm{Vec}_k$ be a k-valued fibre functor. Then $\mathrm{Aut}^{\otimes}(\omega)$ is represented by an affine group scheme G over k, and the functor

$$\mathcal{C} \longrightarrow \mathrm{Rep}_k(G)$$

defined by ω is an equivalence of tensor categories.

References. Milne, *Tannakian Categories*, Section 9.47; Deligne–Milne, *Tannakian Categories*, Theorem 2.11.[8]

Definition 2.43 (Neutral Tannakian category)**.** A category $(\mathcal{C}, \otimes, \phi, \psi)$ is called a *neutral Tannakian category over k* if the following hold.

(1) $\mathcal{C}$ is a k-linear abelian category.

(2) $\otimes : \mathcal{C} \times \mathcal{C} \longrightarrow \mathcal{C}$ is a k-bilinear functor.

[8] 2026.9.28 stopped at here.

(3) There are functorial isomorphisms

$$\phi_{X,Y,Z} : X \otimes (Y \otimes Z) \xrightarrow{\sim} (X \otimes Y) \otimes Z, \qquad \psi_{X,Y} : X \otimes Y \xrightarrow{\sim} Y \otimes X,$$

for all $X, Y, Z \in \mathrm{Ob}(\mathcal{C})$, satisfying the usual associativity and symmetry coherence identities. There is an identity object $\mathbf{1}$, with compatible unit isomorphisms

$$\mathbf{1} \otimes X \simeq X \simeq X \otimes \mathbf{1}.$$

In particular, the functor $X \longmapsto \mathbf{1} \otimes X$ is an equivalence of categories.

(4) Every $X \in \mathrm{Ob}(\mathcal{C})$ has a dual $X^\vee$, equipped with the evaluation morphism

$$\mathrm{ev}_X : X^\vee \otimes X \longrightarrow \mathbf{1},$$

satisfying: for every $T \in \mathrm{Ob}(\mathcal{C})$, evaluation induces a natural isomorphism

$$\begin{aligned} \mathrm{Hom}_{\mathcal{C}}(T, X^\vee) &\xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}}(T \otimes X, \mathbf{1}), \\ \alpha &\longmapsto \left(T \otimes X \xrightarrow{\alpha \otimes \mathrm{id}_X} X^\vee \otimes X \xrightarrow{\mathrm{ev}_X} \mathbf{1} \right). \end{aligned}$$

(5) There is a k-valued fibre functor

$$\omega : (\mathcal{C}, \otimes) \longrightarrow \underline{\mathrm{Vect}}_k;$$

that is, an exact, faithful, k-linear functor preserving the tensor structure, including its associativity, symmetry, and unit constraints.

2.10 Jordan decomposition

Motivation: Jordan normal form

Let $x \in \mathrm{GL}_n(\overline{k})$. Its Jordan normal form is

$$x = g^{-1} \operatorname{diag}\big(B_{a_1}^{n_1}, \ldots, B_{a_t}^{n_t}\big) g, \qquad g \in \mathrm{GL}_n(\overline{k}),$$

where each $a_i \neq 0$ and

$$B_a^r = \begin{pmatrix} a & 1 & & 0 \\ & a & \ddots & \\ & & \ddots & 1 \\ 0 & & & a \end{pmatrix}_{r \times r}$$

is a Jordan block. Put

$$x_s = g^{-1}\operatorname{diag}\big(a_1 I_{n_1}, \dots, a_t I_{n_t}\big) g.$$

Then x_s is semisimple.

Put

$$x_u = g^{-1}\operatorname{diag}\big(a_1^{-1} B_{a_1}^{n_1}, \dots, a_t^{-1} B_{a_t}^{n_t}\big) g.$$

Then x_u is unipotent and

$$x = x_s x_u = x_u x_s.$$

This is the Jordan decomposition of x.

Remark. We now replace $\overline{k}$ by k and assume that k is perfect. The difficulty is that the eigenvalues of x may not lie in k, and how to replace GL_n by an arbitrary linear algebraic group G.

Let V be a finite-dimensional k-vector space and $\alpha \in \mathrm{End}_k(V)$.

$$P_\alpha(T) = \det(T \operatorname{id}_V - \alpha) \in k[T]$$

is the characteristic polynomial for α.

Definition 2.44 (Eigenvalues lying in k; primary spaces)**.** We say that the eigenvalues of α lie in k if $P_\alpha(T)$ splits over k, so that

$$P_\alpha(T) = \prod_{i=1}^{t} (T - a_i)^{m_i}, \qquad a_i \in k,$$

where the a_i are distinct. For an eigenvalue $a \in k$, its *primary space* is

$$V^a = \{v \in V \mid (\alpha - a \operatorname{id}_V)^N v = 0 \text{ for some } N > 0\}.$$

When the eigenvalues lie in k, the primary decomposition is

$$V = \bigoplus_{i=1}^{t} V^{a_i}.$$

Definition 2.45 (Semisimple and unipotent endomorphisms)**.** Let $\alpha \in \mathrm{End}_k(V)$.

(1) α is *semisimple* if $\alpha_{\overline{k}} \in \mathrm{End}_{\overline{k}}(V \otimes_k \overline{k})$ is diagonalizable.

(2) α is *unipotent* if $(\alpha - \operatorname{id}_V)^N = 0$ for some $N > 0$. Equivalently, all its eigenvalues

in $\overline{k}$ are equal to 1.

Theorem 2.46 (Multiplicative Jordan decomposition)**.** Let V be a finite-dimensional vector space over a perfect field k, and let $\alpha \in \mathrm{Aut}_k(V)$. There exist unique automorphisms α_s, α_u of V such that

(a) $\alpha = \alpha_s\alpha_u = \alpha_u\alpha_s$;

(b) α_s is semisimple and α_u is unipotent.

Moreover, $\alpha_s, \alpha_u \in k[\alpha]$: each is a polynomial in α with coefficients in k.

Proof. The assertion is immediate if $V = 0$, so assume $V \neq 0$.

Case 1: the eigenvalues of α lie in k. Write

$$P_\alpha(T) = \prod_{i=1}^{t}(T - a_i)^{m_i}, \qquad V = \bigoplus_{i=1}^{t} V^{a_i}.$$

Since α is invertible, every a_i is nonzero. Define

$$\alpha_s\big|_{V^{a_i}} = a_i \,\mathrm{id}_{V^{a_i}}, \qquad \alpha_u = \alpha\alpha_s^{-1}.$$

Then α_s is semisimple and commutes with α. On V^{a_i},

$$(\alpha_u - \mathrm{id}_V)\big|_{V^{a_i}} = a_i^{-1}(\alpha - a_i \,\mathrm{id}_V)\big|_{V^{a_i}},$$

which is nilpotent. Thus α_u is unipotent, and (a) and (b) hold.

Polynomiality. To obtain $Q(\alpha) = \alpha_s$, it is enough that

$$Q(T) \equiv a_i \pmod{(T - a_i)^{m_i}} \qquad \text{for every } i.$$

Indeed, $\alpha|_{V^{a_i}}$ has characteristic polynomial $(T-a_i)^{m_i}$, so Cayley–Hamilton gives the required equality on each primary space. The factors $(T - a_i)^{m_i}$ are pairwise coprime; the Chinese remainder theorem therefore supplies such a $Q \in k[T]$. Similarly, choose $\widetilde{Q} \in k[T]$ with

$$\widetilde{Q}(T) \equiv a_i^{-1} \pmod{(T - a_i)^{m_i}} \qquad \text{for every } i.$$

Then

$$\alpha_s = Q(\alpha), \qquad \alpha_s^{-1} = \widetilde{Q}(\alpha), \qquad \alpha_u = \alpha\widetilde{Q}(\alpha) \in k[\alpha].$$

Uniqueness. Suppose also that $\alpha = \beta_s\beta_u = \beta_u\beta_s$, where β_s is semisimple and β_u is unipotent. β_s and β_u commutes with α, hence with α_s and α_u, since the latter are polynomials in α. Consequently,

$$\beta_s^{-1}\alpha_s = \beta_u\alpha_u^{-1}.$$

The left-hand side is semisimple and the right-hand side is unipotent: commuting semisimple maps are simultaneously diagonalizable over $\overline{k}$, and commuting unipotent maps are simultaneously upper triangular with all diagonal entries equal to 1. An endomorphism which is both semisimple and unipotent is the identity. Thus $\alpha_s = \beta_s$ and $\alpha_u = \beta_u$.

Case 2: the general case. Because k is perfect, there is a finite Galois extension k'/k over which all eigenvalues of α lie. Choose a k-basis of V, and let $A \in \mathrm{GL}_n(k)$ be the matrix of α.

By Case 1, over k' there is a unique decomposition

$$A = A_sA_u = A_uA_s, \qquad A_s, A_u \in \mathrm{GL}_n(k'),$$

with A_s semisimple and A_u unipotent. For every $\sigma \in \mathrm{Gal}(k'/k)$,

$$A = \sigma(A) = \sigma(A_s)\sigma(A_u) = \sigma(A_u)\sigma(A_s).$$

This is another decomposition of the same kind. By uniqueness,

$$\sigma(A_s) = A_s, \qquad \sigma(A_u) = A_u.$$

Thus $A_s, A_u \in \mathrm{GL}_n(k)$, defining $\alpha_s, \alpha_u \in \mathrm{Aut}_k(V)$ satisfying (a) and (b). Uniqueness over k follows from uniqueness over k'.

Polynomiality over k. Let $\mu_A(T)$ be the minimal polynomial of A over k' and set $d = \deg \mu_A$. Choose a polynomial $Q \in k'[T]$ of least degree such that $A_s = Q(A)$. Reduction modulo μ_A gives $\deg Q < d$. Write

$$Q(T) = \sum_{i=0}^{d-1} c_iT^i, \qquad c_i \in k'.$$

For every $\sigma \in \mathrm{Gal}(k'/k)$, the equalities $\sigma(A) = A$ and $\sigma(A_s) = A_s$ give

$$0 = \sum_{i=0}^{d-1} (\sigma(c_i) - c_i)A^i.$$

By the definition of the minimal polynomial, $I, A, \ldots, A^{d-1}$ are linearly independent over k'. Hence $\sigma(c_i) = c_i$ for all i and all σ, so $c_i \in k$. Therefore $Q \in k[T]$ and $\alpha_s \in k[\alpha]$. The same

argument applies to A_u, giving $\alpha_u \in k[\alpha]$. □

Definition 2.47 (Semisimple and unipotent parts)**.** The automorphisms α_s and α_u are called the *semisimple part* and *unipotent part* of α. The factorization

$$\alpha = \alpha_s \alpha_u = \alpha_u \alpha_s$$

is its *multiplicative Jordan decomposition.*

Example 2.48 (A nonperfect field)**.** Suppose $\operatorname{char} k = 2$ and k is not perfect. Choose $a \in k \setminus k^2$, and put

$$x = \begin{pmatrix} 0 & 1 \\ a & 0 \end{pmatrix} \in \mathrm{GL}_2(k).$$

Its characteristic polynomial is

$$P_x(T) = T^2 - a = (T - \sqrt{a})^2.$$

Over $\overline{k}$, its Jordan decomposition is

$$\begin{pmatrix} 0 & 1 \\ a & 0 \end{pmatrix} = \underbrace{\begin{pmatrix} \sqrt{a} & 0 \\ 0 & \sqrt{a} \end{pmatrix}}_{\text{semisimple}} \underbrace{\begin{pmatrix} 0 & 1/\sqrt{a} \\ \sqrt{a} & 0 \end{pmatrix}}_{\text{unipotent}}.$$

The second factor U satisfies $(U - I)^2 = 0$ in characteristic 2. Neither factor belongs to $\mathrm{GL}_2(k)$. If x had a Jordan decomposition over k, base change to $\overline{k}$ would contradict uniqueness there. Thus the perfectness assumption in the theorem is necessary.

Proposition 2.49. Let V, W be finite-dimensional vector spaces over a perfect field k, and let $\alpha \in \mathrm{Aut}_k(V)$ and $\beta \in \mathrm{Aut}_k(W)$.

(1) *Subspaces and quotients.* If $W \subseteq V$ is α-stable, then W is stable under α_s and α_u, and

$$\alpha|_W = (\alpha_s|_W)(\alpha_u|_W) = (\alpha_u|_W)(\alpha_s|_W)$$

is the Jordan decomposition of $\alpha|_W$. Likewise, the automorphisms induced by α_s and α_u on V/W are the semisimple and unipotent parts of the automorphism induced by α.

(2) *Homomorphisms.* If a k-linear map $\varphi : W \longrightarrow V$ satisfies $\varphi\beta = \alpha\varphi$, then

$$\varphi\beta_s = \alpha_s\varphi, \qquad \varphi\beta_u = \alpha_u\varphi.$$

Equivalently, from the commutative square

$$\begin{array}{ccc} W & \xrightarrow{\varphi} & V \\ {\scriptstyle\beta}\downarrow & & \downarrow{\scriptstyle\alpha} \\ W & \xrightarrow{\varphi} & V \end{array}$$

we obtain the two commutative squares

$$\begin{array}{ccc} W & \xrightarrow{\varphi} & V \\ {\scriptstyle\beta_s}\downarrow & & \downarrow{\scriptstyle\alpha_s} \\ W & \xrightarrow{\varphi} & V \end{array} \qquad \begin{array}{ccc} W & \xrightarrow{\varphi} & V \\ {\scriptstyle\beta_u}\downarrow & & \downarrow{\scriptstyle\alpha_u} \\ W & \xrightarrow{\varphi} & V \end{array}$$

(3) *Tensor products.* For $\alpha \otimes \beta \in \mathrm{Aut}_k(V \otimes_k W)$,

$$(\alpha \otimes \beta)_s = \alpha_s \otimes \beta_s, \qquad (\alpha \otimes \beta)_u = \alpha_u \otimes \beta_u.$$

Proof. Semisimplicity, unipotence, and the identities in question can be checked after extending scalars. By uniqueness, extending scalars preserves Jordan decomposition. We may therefore assume that all eigenvalues of α and β lie in k.

(1) Since $\alpha_s, \alpha_u \in k[\alpha]$, every α-stable subspace W is stable under both factors. The restrictions remain semisimple and unipotent, respectively, and commute. Hence

$$\alpha|_W = (\alpha_s|_W)(\alpha_u|_W)$$

is its Jordan decomposition. The same reasoning applies to the induced automorphisms on V/W.

(2) Factor φ through its image:

$$W \xrightarrow{\varphi} \varphi(W) \hookrightarrow V.$$

Both $\ker\varphi$ and $\varphi(W)$ are stable under the corresponding automorphisms. The original

commutative diagram factors as

$$\begin{array}{ccccc} W & \xrightarrow{\varphi} & \varphi(W) & \hookrightarrow & V \\ \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \alpha|_{\varphi(W)}} & & \downarrow{\scriptstyle \alpha} \\ W & \xrightarrow{\varphi} & \varphi(W) & \hookrightarrow & V \end{array}$$

The surjective map identifies $W/\ker\varphi$ with $\varphi(W)$. Uniqueness of Jordan decomposition preserves its factors under this identification. Applying (1) to the quotient and the inclusion proves the two asserted identities.

(3) Write the primary decompositions as

$$V = \bigoplus_i V^{a_i}, \qquad W = \bigoplus_j W^{b_j}.$$

Then

$$V \otimes_k W = \bigoplus_{i,j} V^{a_i} \otimes_k W^{b_j},$$

and $\alpha_s \otimes \beta_s$ acts on $V^{a_i} \otimes_k W^{b_j}$ by scalar multiplication by $a_i b_j$. In particular, it is semisimple. The automorphism $\alpha_u \otimes \beta_u$ is unipotent: in bases where α_u and β_u are upper triangular with diagonal entries 1, their tensor product has the same property. The two factors commute, and

$$\alpha \otimes \beta = (\alpha_s \otimes \beta_s)(\alpha_u \otimes \beta_u) = (\alpha_u \otimes \beta_u)(\alpha_s \otimes \beta_s).$$

Uniqueness gives the claimed semisimple and unipotent parts.[9] □

Definition 2.50. Let V be a k-vector space and let $\alpha\colon V \xrightarrow{\sim} V$ be a k-linear automorphism.

1. We say that α is *locally finite* if V is a union of finite-dimensional α-stable subspaces.
2. Suppose that α is locally finite. We say that α is *semisimple* (respectively, *locally unipotent*) if, for every finite-dimensional α-stable subspace $W \subseteq V$, the restriction $\alpha|_W$ is semisimple (respectively, unipotent). Here semisimple means diagonalizable after extension to an algebraic closure.

[9] 2026.10.8 stopped at here.

Remark. Local unipotence does not require a single integer $N > 0$ such that $(\alpha - \mathrm{id}_V)^N = 0$ on all of V. The nilpotence exponent may depend on the finite-dimensional stable subspace; there need not be a uniform upper bound.

Example 2.51. Let G be a linear algebraic group and let $r_V \colon G \to \mathrm{GL}_V$ be a rational representation. For every $g \in G(k)$, the automorphism $r_V(g) \in \mathrm{Aut}_k(V)$ is locally finite, since V is a union of finite-dimensional G-stable subspaces.

Corollary 2.52 (Locally finite Jordan decomposition)**.** Let $\alpha \in \mathrm{Aut}_k(V)$ be locally finite. There are unique locally finite automorphisms α_s, α_u of V, called the *semisimple part* and the *locally unipotent part*, such that

1. $\alpha = \alpha_s \alpha_u = \alpha_u \alpha_s$;
2. α_s is semisimple and α_u is locally unipotent.

This factorization is the *Jordan decomposition* of α.

Proof. Write $V = \bigcup_i W_i$, where the W_i are finite-dimensional α-stable subspaces, and define

$$\alpha_s|_{W_i} := (\alpha|_{W_i})_s, \qquad \alpha_u|_{W_i} := (\alpha|_{W_i})_u.$$

Claim. These automorphisms are well defined and glue together. Indeed, compatibility of finite-dimensional Jordan decomposition with restriction gives

$$(\alpha|_{W_i})_s|_{W_i \cap W_j} = (\alpha|_{W_i \cap W_j})_s = (\alpha|_{W_j})_s\big|_{W_i \cap W_j},$$

and the same equality holds for the unipotent parts. The resulting maps are linear: any two vectors lie in the finite-dimensional stable subspace $W_i + W_j$. Their inverses glue in the same way, so they are automorphisms. The factorization and commutativity can be checked on each W_i. Moreover, every finite-dimensional α_s-stable subspace is contained in some finite-dimensional α-stable subspace, on which α_s is semisimple. The analogous assertion holds for α_u.

For uniqueness, suppose $\alpha = \beta_s \beta_u = \beta_u \beta_s$ is another such decomposition. Every vector lies in a common finite-dimensional stable subspace for the commuting locally finite maps β_s, β_u: take its β_s-stable span and then the β_u-stable span of a basis. This subspace is also α-stable, so finite-dimensional uniqueness gives $\beta_s = \alpha_s$ and $\beta_u = \alpha_u$ on it, hence on V. $\square$

Theorem 2.53 (Jordan decomposition in a linear algebraic group)**.** Let G be a linear algebraic group over a perfect field k, and let $g \in G(k)$. There are unique elements $g_s, g_u \in G(k)$ such that

1. $g = g_s g_u = g_u g_s$;
2. for every $(V, r_V) \in \mathrm{Rep}_k(G)$,

$$r_V(g_s) = r_V(g)_s, \qquad r_V(g_u) = r_V(g)_u.$$

Here $\mathrm{Rep}_k(G)$ denotes the category of finite-dimensional rational representations of G.

Definition 2.54.
1. The elements g_s and g_u are the *semisimple part* and the *unipotent part* of g. The factorization $g = g_s g_u$ is the *Jordan decomposition* of g.
2. An element $g \in G(k)$ is *semisimple* if $g = g_s$, and *unipotent* if $g = g_u$.

Remark (The fibre functor and tensor automorphisms)**.** Recall the fibre functor

$$\omega \colon \mathrm{Rep}_k(G) \longrightarrow \mathrm{Vect}_k, \qquad (V, r_V) \longmapsto V.$$

The group-valued functor $\mathrm{Aut}^{\otimes}(\omega) \colon \mathrm{Alg}_k \to \mathrm{Grp}$ assigns to a k-algebra R the families $\lambda = (\lambda_V)_{(V,r_V)}$ of R-linear automorphisms

$$\lambda_V \colon V \otimes_k R \xrightarrow{\sim} V \otimes_k R$$

satisfying the following properties:

1. $\lambda_{\mathbf{1}} = \mathrm{id}_R$, where $\mathbf{1}$ is the trivial one-dimensional representation;
2. $\lambda_{V \otimes W} = \lambda_V \otimes_R \lambda_W$;
3. for every homomorphism of representations $\varphi \colon V \to W$, the following diagram commutes:

$$\begin{array}{ccc} V \otimes_k R & \xrightarrow{\varphi \otimes 1} & W \otimes_k R \\ {\scriptstyle \lambda_V}\downarrow & & \downarrow{\scriptstyle \lambda_W} \\ V \otimes_k R & \xrightarrow[\varphi \otimes 1]{} & W \otimes_k R. \end{array}$$

Remark (Reconstruction)**.** The natural map

$$G \xrightarrow{\sim} \mathrm{Aut}^{\otimes}(\omega)$$

is an isomorphism of group-valued functors on Alg_k. In particular,

$$G(k) \xrightarrow{\sim} \mathrm{Aut}^{\otimes}(\omega)(k), \qquad g \longmapsto \big(r_V(g)\big)_{(V,r_V)}.$$

Proof of the theorem. For every $(V, r_V) \in \mathrm{Rep}_k(G)$, form $r_V(g)_s, r_V(g)_u \in \mathrm{Aut}_k(V)$. *Claim.* The families

$$\big(r_V(g)_s\big)_{(V,r_V)}, \qquad \big(r_V(g)_u\big)_{(V,r_V)}$$

belong to $\mathrm{Aut}^{\otimes}(\omega)(k)$.

We verify the three properties above.

1. Since $r_{\mathbf{1}}(g) = \mathrm{id}_k$, both its semisimple and unipotent parts are id_k.

2. Compatibility of Jordan decomposition with tensor products gives

$$r_{V\otimes W}(g)_s = \big(r_V(g) \otimes r_W(g)\big)_s = r_V(g)_s \otimes r_W(g)_s,$$
$$r_{V\otimes W}(g)_u = \big(r_V(g) \otimes r_W(g)\big)_u = r_V(g)_u \otimes r_W(g)_u.$$

3. If $\varphi\colon V \to W$ is a homomorphism of representations, the square

$$\begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ {\scriptstyle r_V(g)}\downarrow & & \downarrow{\scriptstyle r_W(g)} \\ V & \xrightarrow[\varphi]{} & W \end{array}$$

commutes. Compatibility of Jordan decomposition with homomorphisms gives the two commutative squares

$$\begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ {\scriptstyle r_V(g)_s}\downarrow & & \downarrow{\scriptstyle r_W(g)_s} \\ V & \xrightarrow[\varphi]{} & W \end{array} \qquad \begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ {\scriptstyle r_V(g)_u}\downarrow & & \downarrow{\scriptstyle r_W(g)_u} \\ V & \xrightarrow[\varphi]{} & W. \end{array}$$

The claim follows. Reconstruction therefore supplies unique $g_s, g_u \in G(k)$ with the required identities in every representation.

For the factorization and commutativity, take a faithful representation $r_V\colon G \hookrightarrow \mathrm{GL}_V$. Then

$$r_V(g_s g_u) = r_V(g)_s r_V(g)_u = r_V(g),$$
$$r_V(g_u g_s) = r_V(g)_u r_V(g)_s = r_V(g).$$

Faithfulness gives $g = g_s g_u = g_u g_s$. Uniqueness also follows from this faithful representation and uniqueness of the Jordan decomposition in GL_V. □

Corollary 2.55. Let $\phi\colon G \to H$ be a homomorphism of linear algebraic groups over a perfect field k. For every $g \in G(k)$,

$$\phi(g_s) = \phi(g)_s, \qquad \phi(g_u) = \phi(g)_u.$$

Proof. Choose a faithful representation $r_V\colon H \hookrightarrow \mathrm{GL}_V$. The composite $(V, r_V \circ \phi)$ is a representation of G.

Applying the theorem to this composite representation and to H gives

$$\begin{aligned} r_V\big(\phi(g_s)\big) &= (r_V \circ \phi)(g_s) = \big((r_V \circ \phi)(g)\big)_s = r_V\big(\phi(g)_s\big), \\ r_V\big(\phi(g_u)\big) &= (r_V \circ \phi)(g_u) = \big((r_V \circ \phi)(g)\big)_u = r_V\big(\phi(g)_u\big). \end{aligned}$$

The conclusion follows from faithfulness of r_V. □

3 Quotient of Linear Algbraic Groups

3.1 Preliminaries

3.1.1 Density of points

Reference: [Mil17, Chapter I, §§1.9–1.18, pp. 9–11]. In this subsection, X is a scheme of finite type over a field k. Write $|X|$ for its underlying topological space.

Definition 3.1 (Schematic density)**.** Let $S \subseteq X(k)$. We say that S is *schematically dense* in X if the only closed subscheme $Z \subseteq X$ such that $S \subseteq Z(k)$ is X itself. In particular, schematic density of S implies that X is reduced.

Remark (The affine criterion)**.** Let $X = \operatorname{Spec} A$ and identify $X(k)$ with the maximal ideals $\mathfrak{m} \subset A$ whose residue field is k. For a closed subscheme $Z = \operatorname{Spec}(A/\mathfrak{a})$,

$$S \subseteq Z(k) \quad \Longleftrightarrow \quad \mathfrak{a} \subseteq \mathfrak{m} \text{ for every } \mathfrak{m} \in S.$$

Consequently,

$$S \text{ is schematically dense in } X \quad \Longleftrightarrow \quad \bigcap_{\mathfrak{m} \in S} \mathfrak{m} = 0.$$

Proposition 3.2. Let $S \subseteq X(k) \subseteq |X|$. The following are equivalent:

1. S is schematically dense in X;
2. X is reduced and S is dense in $|X|$;
3. writing $i_s \colon \operatorname{Spec} k \to X$ for each $s \in S$, the homomorphism of sheaves

$$\mathcal{O}_X \longrightarrow \prod_{s \in S} (i_s)_* \mathcal{O}_{\operatorname{Spec} k}$$

is injective. Equivalently, for every open subset $U \subseteq X$,

$$f \in \mathcal{O}_X(U), \qquad f(s) = 0 \text{ for all } s \in S \cap U \quad \Longrightarrow \quad f = 0.$$

In particular, schematic density is a local property on X.

Proposition 3.3. 1. A schematically dense subset $S \subseteq X(k)$ remains schematically dense after extension of the base field: for every field extension k'/k, its image in

$X_{k'}(k')$ is schematically dense in $X_{k'}$. This implies X is geometrically reduced.

2. In particular, if $X(k)$ is schematically dense in X, then X is geometrically reduced.

Remark. The converse of (2) does not hold. For example,

$$k = \mathbb{R}, \qquad X = \operatorname{Spec}\big(\mathbb{R}[x,y]/(x^2+y^2+1)\big)$$

is geometrically reduced but has no real points. Also, for $k = \mathbb{F}_p$ and $X = \mathbb{A}^1_k$, all k-points lie in the proper closed subscheme

$$Z = \operatorname{Spec}\big(k[T]/(T^p - T)\big) \subsetneq X.$$

Thus $X(k)$ is not schematically dense in X.

Definition 3.4 (Density of points over an extension)**.** Let k'/k be a field extension. We say that $X(k')$ is *schematically dense in* X if the only closed subscheme $Z \subseteq X$ such that $Z(k') = X(k')$ is X itself.

Proposition 3.5. Let k^s be a separable closure of k. If X is geometrically reduced, then $X(k^s)$ is schematically dense in X.

Example 3.6. If X is nonreduced, then $X(k^s)$ is not schematically dense in X: the proper closed subscheme X_{red} has the same k^s-points as X.

3.1.2 Frobenius

In this subsection, k is a field of characteristic $p > 0$.

Definition 3.7 (Absolute and relative Frobenius)**.** For every $R \in \mathrm{Alg}_k$, the map

$$f_R \colon R \longrightarrow R, \qquad x \longmapsto x^p$$

is a ring homomorphism. It need not be a homomorphism of k-algebras. Write $f = f_k$ and $\sigma_k \colon \operatorname{Spec} k \to \operatorname{Spec} k$ for the corresponding morphism.
For a scheme X over k, its *absolute Frobenius* $\sigma_X \colon X \to X$ is the identity on $|X|$ and

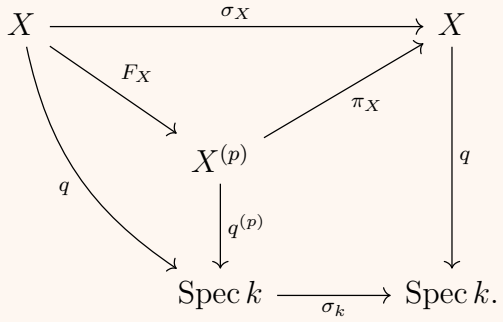
sends local sections a of $\mathcal{O}_X$ to a^p. If $q\colon X \to \operatorname{Spec} k$ is the structure morphism,

$$\begin{array}{ccc} X & \xrightarrow{\sigma_X} & X \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle q} \\ \operatorname{Spec} k & \xrightarrow[\sigma_k]{} & \operatorname{Spec} k \end{array}$$

commutes.
The *Frobenius twist* is

$$X^{(p)} := X \times_{\operatorname{Spec} k,\, \sigma_k} \operatorname{Spec} k.$$

The *relative Frobenius* $F_X\colon X \to X^{(p)}$ is the unique k-morphism in the diagram



Here the square formed by $X^{(p)}$, X, and the two copies of $\operatorname{Spec} k$ is cartesian, and $\pi_X F_X = \sigma_X$.

For $R \in \operatorname{Alg}_k$, let ${}_fR$ be R as a ring, with k-algebra structure

$$k \xrightarrow{f} k \longrightarrow R.$$

Then, naturally in R,

$$X^{(p)}(R) = X({}_fR).$$

The map

$$R \longrightarrow {}_fR, \qquad x \longmapsto x^p$$

is a homomorphism of k-algebras, and it induces

$$F_X(R)\colon X(R) \longrightarrow X^{(p)}(R) = X({}_fR).$$

For $X = \operatorname{Spec} A$, this corresponds to

$$F_X^{\#}\colon A \otimes_{k,f} k \longrightarrow A, \qquad a \otimes c \longmapsto c\, a^p.$$

Proposition 3.8 (Basic facts about relative Frobenius)**.**

1. *Functoriality.* For every k-morphism $\varphi\colon X \to Y$, the diagram

$$\begin{array}{ccc} X & \xrightarrow{\varphi} & Y \\ {\scriptstyle F_X}\downarrow & & \downarrow{\scriptstyle F_Y} \\ X^{(p)} & \xrightarrow[\varphi^{(p)}]{} & Y^{(p)} \end{array}$$

   commutes.

2. *Products.* Under the canonical isomorphism $(X \times_k Y)^{(p)} \simeq X^{(p)} \times_k Y^{(p)}$,

$$F_{X\times_k Y} = F_X \times_k F_Y.$$

3. *Base change.* For every field extension k'/k, the canonical identification

$$(X^{(p)})_{k'} \simeq (X_{k'})^{(p)}$$

   identifies $(F_X)_{k'}$ with $F_{X_{k'}}$.

Remark. If G is an algebraic group over k, then $G^{(p)}$ is an algebraic group over k: it represents the group-valued functor

$$\mathrm{Alg}_k \longrightarrow \mathrm{Grp}, \qquad R \longmapsto G({}_fR) = G^{(p)}(R).$$

Proposition 3.9. The relative Frobenius $F_G\colon G \to G^{(p)}$ is a homomorphism of algebraic groups.

Proof. Functoriality and compatibility with products give the commutative diagram

$$\begin{array}{ccc} G \times_k G & \xrightarrow{m} & G \\ {\scriptstyle F_G\times F_G}\downarrow & & \downarrow{\scriptstyle F_G} \\ G^{(p)} \times_k G^{(p)} & \xrightarrow[m^{(p)}]{} & G^{(p)}. \end{array}$$

Thus F_G preserves multiplication. Functoriality applied to the unit and inverse morphisms gives their compatibility as well. □

Proposition 3.10. If G is smooth and connected, then $G^{(p)}$ is smooth and connected. Moreover, $F_G\colon G \to G^{(p)}$ is faithfully flat.

Proof. By compatibility with base change and descent, we may assume that $k = \overline{k}$. The map $f\colon k \to k$, $a \mapsto a^p$, is then an isomorphism. Hence the projection $G^{(p)} \to G$ is an isomorphism of underlying schemes, so $G^{(p)}$ is connected; it is smooth over k by base change of the smooth structure morphism of G.

The map $k \to {}_fk$, $a \mapsto a^p$, is an isomorphism of k-algebras. Hence $F_G\colon G(k) \to G^{(p)}(k)$ is bijective. All closed points of $G^{(p)}$ therefore lie in the image. The image of a homomorphism of algebraic groups is closed, so F_G is surjective. Since the target $G^{(p)}$ is smooth, hence reduced, the faithful-flatness Proposition 1.35 for surjective homomorphisms of algebraic groups applies. Thus F_G is faithfully flat.[10] □

In particular, if $G = \operatorname{Spm} A$ is a linear algebraic group, then

$$A^{(p)} = A \otimes_{k,F_k} k, \qquad F_G^{\#} : A^{(p)} \longrightarrow A, \qquad a \otimes b \longmapsto a^p b.$$

This map is a homomorphism of Hopf algebras. When k is perfect, the image of $F_G^{\#}$ is A^p.

For every $n \geq 1$, define

$$F_G^n = F_{G^{(p^{n-1})}} \circ \cdots \circ F_{G^{(p)}} \circ F_G : G \longrightarrow G^{(p^n)},$$

where $(G^{(p^i)})^{(p)} = G^{(p^{i+1})}$. Each F_G^n is a homomorphism of algebraic groups.

Proposition 3.11. Let G be a linear algebraic group over a perfect field k of characteristic $p > 0$, and put $A = \mathcal{O}(G)$. For $n \geq 0$, the image

$$A^{p^n} = \operatorname{im}\big((F_G^n)^{\#} : \mathcal{O}(G^{(p^n)}) \longrightarrow A\big)$$

is a Hopf subalgebra of A. For all sufficiently large n, A^{p^n} is geometrically reduced.

Proof. The image is a Hopf subalgebra because F_G^n is a homomorphism of algebraic groups. Let $\mathfrak{n} = \sqrt{0} \subset A$ be the nilradical. Since A is finitely generated, there is an integer n_0 with $\mathfrak{n}^{n_0} = 0$. Choose n so that $p^n \geq n_0$. If $a \in \mathfrak{n} \cap A^{p^n}$, write $a = b^{p^n}$ for some $b \in A$. Then b is nilpotent, hence $b \in \mathfrak{n}$, and therefore $a = b^{p^n} = 0$. Thus A^{p^n} is reduced. A finitely generated reduced algebra over a perfect field is geometrically reduced. □

[10] 2026.10.12 stopped at here.

Remark. Recall that: every homomorphism $\varphi : G \to H$ of linear algebraic groups factors as

$$G \longrightarrow I_\varphi \hookrightarrow H,$$

where the first map is faithfully flat and the second is a closed immersion. On coordinate rings,

$$\mathcal{O}(I_\varphi) = \mathrm{im}\big(\varphi^\# : \mathcal{O}(H) \longrightarrow \mathcal{O}(G)\big).$$

In particular, for G a linear algebraic group and the homomorphism $F_G^n : G \to G^{(p^n)}$, by Proposition 3.10, $I_{F_G^n} = G^{(p^n)}$. By Proposition 3.11 and Proposition 1.13, for n sufficiently large, $I_{F_G^n} = G^{(p^n)}$ is smooth.

3.1.3 Smoothness of algebraic groups over k with $\mathrm{char}\, k = 0$

For $x \in |G|$, let $\mathfrak{m}_x \subset \mathcal{O}(G)$ be the maximal ideal corresponding to x. The Zariski tangent space is

$$T_x G = \mathrm{Hom}_k(\mathfrak{m}_x/\mathfrak{m}_x^2, k).$$

One has

$$\dim_k T_x G \geq \dim \mathcal{O}_{G,x},$$

and x is smooth precisely when equality holds. Translations give

$$\dim G = \dim \mathcal{O}_{G,x} \qquad (x \in |G|),$$

so G is smooth if and only if it is smooth at the identity e, equivalently if and only if

$$\dim_k T_e G = \dim G.$$

Lemma 3.12. Let G be a linear algebraic group over $k = \overline{k}$. If every nilpotent element of $\mathcal{O}(G)$ lies in $\mathfrak{m}_e^2$, then G is smooth.

Proof. Let $A = \mathcal{O}(G)$, let $\mathfrak{m} = \mathfrak{m}_e$, and let $N = \sqrt{0}$. Put $A_{\mathrm{red}} = A/N$ and $\overline{\mathfrak{m}} = \mathfrak{m}/N$. The canonical map

$$\mathfrak{m}/\mathfrak{m}^2 \longrightarrow \overline{\mathfrak{m}}/\overline{\mathfrak{m}}^2$$

is surjective, with kernel $(N + \mathfrak{m}^2)/\mathfrak{m}^2$. Under the hypothesis the kernel is zero. Hence

$$\dim_k T_e G = \dim_k T_e G_{\mathrm{red}}.$$

The reduced algebraic group G_{red} over the algebraically closed field k is smooth, and $\dim G = \dim G_{\mathrm{red}}$. Therefore G is smooth at e, and hence smooth. □

Theorem 3.13 (Cartier)**.** Every linear algebraic group over a field of characteristic 0 is smooth.

Proof. After extending the ground field, we may assume $k = \overline{k}$. Write $G = \operatorname{Spm} A$, let $e^{\#} : A \to k$ be the counit, and put $\mathfrak{p} = \ker(e^{\#})$. By the lemma it is enough to show that every nilpotent $a \in A$ belongs to $\mathfrak{p}^2$.

Choose n minimal such that $a^n = 0$ in $A_{\mathfrak{p}}$ and $a^{n-1} \neq 0$ in $A_{\mathfrak{p}}$. Multiplying a by an element of $A \setminus \mathfrak{p}$ if necessary, we may assume in addition that $a^n = 0$ in A.

Let $m^{\#} : A \to A \otimes_k A$ be the comultiplication. The counit identities imply

$$m^{\#}(a) \equiv 1 \otimes a \pmod{\mathfrak{p} \otimes A}, \qquad m^{\#}(a) \equiv a \otimes 1 \pmod{A \otimes \mathfrak{p}}.$$

Consequently

$$m^{\#}(a) = a \otimes 1 + 1 \otimes a + y \qquad \text{for some } y \in \mathfrak{p} \otimes \mathfrak{p}.$$

Reducing $m^{\#}(a)^n = m^{\#}(a^n) = 0$ modulo $A \otimes \mathfrak{p}^2$ gives

$$n a^{n-1} \otimes a \in a^{n-1}\mathfrak{p} \otimes A + A \otimes \mathfrak{p}^2.$$

We claim that $a^{n-1} \notin a^{n-1}\mathfrak{p}$. Otherwise $a^{n-1} = a^{n-1}z$ for some $z \in \mathfrak{p}$, so $a^{n-1}(1 - z) = 0$, which contradicts $a^{n-1} \neq 0$ in $A_{\mathfrak{p}}$ because $1 - z$ is a unit there. Since $n \neq 0$ in k, the displayed containment, after passing to the tensor product of the k-vector spaces $A/a^{n-1}\mathfrak{p}$ and $A/\mathfrak{p}^2$, forces the class of a in $A/\mathfrak{p}^2$ to be zero. Thus $a \in \mathfrak{p}^2$, and the lemma proves the theorem. □

Corollary 3.14. Every algebraic group over a field of characteristic 0 is smooth.

Proof. Smoothness may be checked after extension to an algebraic closure, so we may assume $k = \overline{k}$. By translation between the connected components, it suffices to treat $G = G^{\circ}$. Recall the exact sequence supplied by the structure theorem for connected algebraic groups:

$$e \longrightarrow G_{\text{aff}} \longrightarrow G \longrightarrow A \longrightarrow e,$$

where G_{aff} is affine and A is an abelian variety. Cartier's theorem makes G_{aff} smooth, while A is smooth by definition. The map $G \to A$ is a torsor under the smooth group G_{aff}, hence is smooth. It follows that G is smooth. □

3.2 Quotient maps

The quotient problem starts with an abstract group G and a subgroup H. If H is normal in a linear algebraic group, one wants G/H to exist as a scheme, and, in the normal

case, as a quotient group with an exact sequence

$$e \longrightarrow H \longrightarrow G \longrightarrow G/H \longrightarrow e.$$

Definition 3.15. A homomorphism of algebraic groups is called a *quotient map* if it is faithfully flat.

Proposition 3.16. Let $q : G \to Q$ be a homomorphism of linear algebraic groups, with $G = \operatorname{Spm} B$ and $Q = \operatorname{Spm} A$. Then q is a quotient map if and only if

$$q^{\#} : A \longrightarrow B$$

is injective.

Proof. Use the fact [Mil17, Theorem 3.31] that if $A \subseteq B$ is an inclusion of finitely generated Hopf algebras over k, then B is faithfully flat over A. □

Theorem 3.17 (Universal property of a quotient map)**.** Let $q : G \to Q$ be a quotient map of linear algebraic groups and let $N = \ker q$. If $\varphi : G \to G'$ is a homomorphism with $N \subseteq \ker \varphi$, then there is a unique homomorphism $\overline{\varphi} : Q \to G'$ such that

$$\varphi = \overline{\varphi} \circ q.$$

Proof. Uniqueness follows from the injectivity of $q^{\#}$.

For existence, write $G = \operatorname{Spm} B$, $Q = \operatorname{Spm} A$, and $G' = \operatorname{Spm} C$. The canonical map

$$G \times N \longrightarrow G \times_Q G, \qquad (g, n) \longmapsto (g, gn),$$

is an isomorphism: on R-points its inverse sends (g_1, g_2) with $q(g_1) = q(g_2)$ to $(g_1, g_1^{-1} g_2)$.

Since $N \subseteq \ker \varphi$, the two projections $G \times_Q G \rightrightarrows G$ have the same composite with φ. On coordinate rings, $\varphi^{\#} : C \to B$ therefore equalizes the two maps

$$B \rightrightarrows B \otimes_A B, \qquad b \longmapsto b \otimes 1, \quad b \longmapsto 1 \otimes b.$$

Faithful flatness gives the equalizer identity

$$A = \operatorname{Eq}\big(B \rightrightarrows B \otimes_A B\big),$$

so $\varphi^{\#}(C) \subseteq A$. Thus $\varphi^{\#}$ factors uniquely through a map $C \to A$, which is the comorphism

of the required homomorphism $\overline{\varphi} : Q \to G'$. □

Corollary 3.18. A quotient map $q : G \to Q$ is determined by its kernel up to a unique isomorphism of quotient groups.

Proof. If $q : G \to Q$ and $q' : G \to Q'$ are quotient maps with the same kernel, the universal property gives homomorphisms $Q \to Q'$ and $Q' \to Q$ whose composites with q and q' agree. Applying the universal property once more shows that the two composites are identities. □

Definition 3.19. For a normal algebraic subgroup $N \triangleleft G$, the quotient map with kernel N is denoted

$$G \longrightarrow G/N,$$

and G/N is called *the quotient of G by N*.

3.3 Existence of quotients

Definition 3.20. Let G be a linear algebraic group and let $\rho : G \to \mathrm{GL}(V)$ be a finite-dimensional representation. Define

$$V^G = \{v \in V \mid g \cdot (v \otimes 1) = v \otimes 1 \text{ in } V \otimes_k R,\ \forall R \in \mathrm{Alg}_k,\ \forall g \in G(R)\}.$$

Lemma 3.21. For every $R \in \mathrm{Alg}_k$, put $V_R = V \otimes_k R$ and $G_R = G \times_{\mathrm{Spec}\, k} \mathrm{Spec}\, R$. Then

$$(V_R)^{G_R} = V^G \otimes_k R,$$

where the left-hand side means

$$(V_R)^{G_R} = \{v' \in V_R \mid g \cdot (v' \otimes_R 1) = v' \otimes_R 1 \text{ in } V \otimes_k S,\ \forall S \in \mathrm{Alg}_R,\ \forall g \in G(S)\}.$$

Proof. The inclusion $V^G \otimes_k R \subseteq (V_R)^{G_R}$ is immediate from the definition. Conversely, choose a k-basis $(e_i)_{i \in I}$ of R and write

$$v' = \sum_{i \in I} v_i \otimes e_i,$$

with only finitely many nonzero v_i. Let $S \in \mathrm{Alg}_k$ and $g \in G(S)$. Along $S \to S \otimes_k R$, $G(S) \to G(S \otimes_k R)$ maps g to g'. Invariance of v' gives

$$\sum_i (g \cdot (v_i \otimes 1)) \otimes e_i = \sum_i g' \cdot (v_i \otimes 1 \otimes e_i) = \sum_i v_i \otimes 1 \otimes e_i \qquad \text{in } V \otimes_k (S \otimes_k R).$$

So $g(v_i \otimes 1) = v_i \otimes 1$ for every i. Hence each $v_i \in V^G$, and therefore $v' \in V^G \otimes_k R$.[11] □

Lemma 3.22. Let G be a linear algebraic group, let $N \trianglelefteq G$ be a normal algebraic subgroup, and let $(V, r) \in \mathrm{Rep}_k G$. Then V^N is stable under G.

Proof. For every $R \in \mathrm{Alg}_k$, $g \in G(R)$, and $v \in V^N$, we must show that $gv_R \in V^N \otimes_k R$. By Lemma 3.21,

$$(V^N)_R = (V_R)^{N_R}.$$

It therefore suffices to check that, for every R-algebra R' and every $n' \in N(R')$, the vector $(gv_R)_{R'}$ is fixed by n'. Let g' be the image of g in $G(R')$. Normality gives $g'^{-1}n'g' \in N(R')$, and hence

$$n'(gv_R)_{R'} = n'g'v_{R'} = g'(g'^{-1}n'g')v_{R'} = g'v_{R'} = (gv_R)_{R'}.$$

□

Recall (Theorem 2.29)**.** Let G be a linear algebraic group and $H \subseteq G$ a closed algebraic subgroup. There exist $(V, r_V) \in \mathrm{Rep}_k G$ and a one-dimensional subspace $L \subseteq V$ such that

$$H = \mathrm{Stab}_G(L).$$

Proposition 3.23. Assume $k = \overline{k}$. Let G be a linear algebraic group and $H \trianglelefteq G$ a normal algebraic subgroup. There exist $(V, r_V) \in \mathrm{Rep}_k G$ and a one-dimensional subspace $L \subseteq V$ such that

$$H = \mathrm{Stab}_G(L), \qquad H \text{ acts trivially on } L.$$

Theorem 3.24 (Existence of quotients)**.** Every normal algebraic subgroup H of a linear algebraic group G is the kernel of a quotient map

$$q\colon G \longrightarrow Q$$

with Q a linear algebraic group. Thus the quotient G/H exists and is isomorphic to Q.

Proof of Proposition 3.23 $\Longrightarrow$ Theorem 3.24, when $k = \overline{k}$. Choose (V, r_V) and $L \subseteq V$ as in Proposition 3.23. Lemma 3.22 gives a representation V^H of G, and $L \subseteq V^H$ because H acts

[11] 2026.10.15 stopped at here.

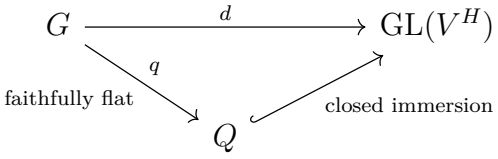
trivially on L. Consider the homomorphism

$$d = r_V|_{V^H} : G \longrightarrow \mathrm{GL}(V^H).$$

We claim that $\ker d = H$. Indeed, as subgroup schemes,

$$H \subseteq \ker d \subseteq \mathrm{Stab}_G(L) = H.$$

Factor d through its scheme-theoretic image:



Then Q is a linear algebraic group and $\ker q = \ker d = H$. Consequently $Q \simeq G/H$. □

Lemma 3.25. Assume $k = \overline{k}$. Let G be a linear algebraic group and $H \trianglelefteq G$ a normal algebraic subgroup. If a character χ of H occurs in a representation of G, then there is an integer $m > 0$ such that $-m\chi$ occurs in a representation of G, possibly a different one. Here occurrence means that the restricted representation contains an H-stable line on which H acts through the specified character.

Remark. The normality hypothesis in Lemma 3.25 is essential.

Proof of Lemma 3.25 $\Longrightarrow$ Proposition 3.23. By Chevalley's theorem, choose $(V, r_V) \in \mathrm{Rep}_k G$ and a line $L \subseteq V$ such that $H = \mathrm{Stab}_G(L)$. Let χ be the character through which H acts on L. If $\chi = 0$, the assertion already holds.

Otherwise Lemma 3.25 gives $(W, r_W) \in \mathrm{Rep}_k G$ and an H-stable line $L_1 \subseteq W$ on which H acts through $-m\chi$ for some $m > 0$. The line

$$L^{\otimes m} \otimes_k L_1 \subseteq V^{\otimes m} \otimes_k W$$

is fixed pointwise by H. By the tensor-product stabilizer fact below,

$$\begin{aligned} \mathrm{Stab}_G(L^{\otimes m} \otimes_k L_1) &= \mathrm{Stab}_G(L) \cap \mathrm{Stab}_G(L_1) \\ &= H \cap \mathrm{Stab}_G(L_1) = H, \end{aligned}$$

where the last equality uses $H \subseteq \mathrm{Stab}_G(L_1)$. □

Fact. Let $(V_1, r_1), (V_2, r_2) \in \mathrm{Rep}_k G$, and let $0 \neq L_1 \subseteq V_1$ and $0 \neq L_2 \subseteq V_2$ be subspaces. For the tensor-product representation on $V_1 \otimes_k V_2$,

$$\mathrm{Stab}_G(L_1 \otimes_k L_2) = \mathrm{Stab}_G(L_1) \cap \mathrm{Stab}_G(L_2).$$

Lemma 3.26. Let G be a linear algebraic group over k, and let $(V, r_V) \in \mathrm{Rep}_k G$. Suppose

$$V = \sum_{i \in I} S_i,$$

where the S_i are simple subrepresentations. For every subrepresentation $W \subseteq V$, there exists $J \subseteq I$ such that

$$V = W \oplus \bigoplus_{j \in J} S_j.$$

In particular, V and W are direct sums of simple subrepresentations, and W is a direct summand of V.

Proof. Using Zorn's lemma, choose $J \subseteq I$ maximal subject to the conditions that

$$S_J := \sum_{j \in J} S_j = \bigoplus_{j \in J} S_j, \qquad W \cap S_J = 0.$$

We claim that $W + S_J = V$. It suffices to show $S_i \subseteq W + S_J$ for every $i \in I$. If $S_i \not\subseteq W + S_J$, simplicity gives

$$S_i \cap (W + S_J) = 0.$$

Then, for $J' = J \cup \{i\}$,

$$S_{J'} = S_J \oplus S_i, \qquad S_{J'} \cap W = 0,$$

contrary to maximality. This proves the asserted decomposition. Taking $W = 0$ shows that V is a direct sum of simple subrepresentations. For general W, the projection $V \to W$ afforded by the decomposition expresses W as a sum of simple images of the S_i; applying the same argument to that sum expresses W as a direct sum. □

Proof of Lemma 3.25 using Lemma 3.26. Let χ occur on an H-stable line $L \subseteq V$ for some $(V, r_V) \in \mathrm{Rep}_k G$. It suffices to construct a representation (W, r_W) of G containing a line L_1 such that

1. H acts on L_1 through $m\chi$ for some $m > 0$;
2. L_1 is a direct summand of W as an H-representation.

Indeed, if $W = L_1 \oplus M$ as H-representations, then $L_1^\vee \simeq \mathrm{Ann}_{W^\vee}(M) \subseteq W^\vee$ is an H-stable line with character $-m\chi$.

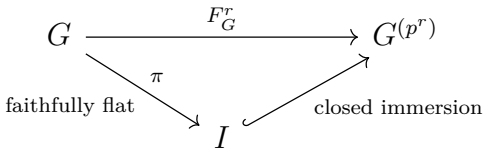
Case 1: $G(k)$ is schematically dense in G (for example, G is smooth). Let $W \subseteq V$ be the sum of all H-stable one-dimensional subspaces of V. In particular, $L \subseteq W$. We claim that W is a G-subrepresentation.

Since $H \trianglelefteq G$, conjugation by every $g \in G(k)$ preserves H, so g sends an H-stable line to another H-stable line. Thus W is stable under $G(k)$, or equivalently,

$$G(k) \subseteq \mathrm{Stab}_G(W)(k).$$

The stabilizer is a closed subgroup scheme of G; schematic density therefore implies $\mathrm{Stab}_G(W) = G$. This proves the claim. As an H-representation, W is a sum of one-dimensional, hence simple, subrepresentations. Lemma 3.26 shows that L is an H-equivariant direct summand of W. We may take $L_1 = L$ and $m = 1$.

Case 2: the general case. If $\mathrm{char}\, k = 0$, Cartier's Theorem 3.13 implies that G is smooth, so Case 1 applies. Suppose $\mathrm{char}\, k = p > 0$. By the remark after Proposition 3.11, for some sufficiently large $r > 0$, the iterated relative Frobenius has a factorization



in which I is smooth. Set

$$q = p^r, \qquad N = \ker(F_G^r) = \ker \pi.$$

Consider the symmetric power $S^qV \in \mathrm{Rep}_k G$. The natural inclusion of the line $L^{\otimes q}$ into S^qV gives an H-stable line with character $q\chi$. We claim that

$$L^{\otimes q} \subseteq (S^qV)^N.$$

Write $L = ke_1$ and extend e_1 to a basis $e_1, \ldots, e_n$ of V. The representation $r_V \colon G \to \mathrm{GL}(V) \simeq \mathrm{GL}_n$ gives the commutative square

$$\begin{array}{ccc} G & \xrightarrow{F_G^r} & G^{(q)} \\ {\scriptstyle r_V}\downarrow & & \downarrow{\scriptstyle r_V^{(q)}} \\ \mathrm{GL}_n & \xrightarrow[F^r_{\mathrm{GL}_n}]{} & \mathrm{GL}_n^{(q)}. \end{array}$$

For every $R \in \mathrm{Alg}_k$ and $g \in N(R)$, write $r_V(g) = (a_{ij})$. Since $F_G^r(g) = e$, naturality of Frobenius gives

$$a_{ij}^q = \delta_{ij}.$$

Here e_i^q denotes the image of $e_i^{\otimes q}$ in S^qV. The binomial identity in characteristic p yields

$$g(e_1^q) = \left(\sum_{i=1}^n a_{i1}e_i\right)^q = \sum_{i=1}^n a_{i1}^q e_i^q = \sum_{i=1}^n \delta_{i1}e_i^q = e_1^q.$$

Thus the claim holds scheme-theoretically.

By Lemma 3.22,

$$U := (S^qV)^N \in \mathrm{Rep}_k G.$$

The subgroup N acts trivially on U, so the representation factors through the quotient map $\pi\colon G \to I$:

$$\begin{array}{ccc} G & \longrightarrow & \mathrm{GL}(U) \\ & \searrow^{\pi} & \nearrow \\ & I \simeq G/N & \end{array}$$

Let $W \subseteq U$ be the sum of all H-stable lines of U. The claim gives $L^{\otimes q} \subseteq W$. Normality of H shows, as in Case 1, that W is stable under $G(k)$. Since $k = \overline{k}$, the faithfully flat morphism $G \to I$ is surjective on k-points: every nonempty fiber has a k-point. Consequently W is stable under $I(k)$. Because I is smooth and k is algebraically closed, $I(k)$ is schematically dense in I. Hence W is stable under I, and therefore under G.

Finally, W is a sum of simple H-representations, so Lemma 3.26 shows that $L^{\otimes q}$ is an H-equivariant direct summand of W. Taking $L_1 = L^{\otimes q}$ and $m = q$ proves Lemma 3.25. □

3.3.1 Weil restriction

Definition 3.27. Let A be a finite-dimensional k-algebra and X an A-scheme. The *Weil restriction*, or *restriction of scalars*, is the functor

$$\begin{aligned} \mathrm{Res}_{A/k} X\colon\ \mathrm{Alg}_k &\longrightarrow \mathrm{Set}, \\ R &\longmapsto X(A \otimes_k R). \end{aligned}$$

When it is representable, its representing k-scheme is denoted by the same symbol.

Theorem 3.28 (Representability of Weil restriction)**.** If A is a finite-dimensional k-algebra and X is quasi-projective over A, then $\mathrm{Res}_{A/k} X$ is represented by a k-scheme.

Example 3.29. Let $m = \dim_k A$ and $X = \mathbb{A}^n_A$. For every $R \in \mathrm{Alg}_k$,

$$(\mathrm{Res}_{A/k}\, \mathbb{A}^n_A)(R) = \mathbb{A}^n_A(A \otimes_k R) = (A \otimes_k R)^n \simeq R^{nm}$$

as sets, functorially in R, after choosing a k-basis of A. Thus

$$\mathrm{Res}_{A/k}\, \mathbb{A}^n_A \simeq \mathbb{A}^{nm}_k.$$

Proposition 3.30 (Adjunction)**.** Whenever $\mathrm{Res}_{A/k}\, X$ exists, there is a natural bijection

$$\mathrm{Hom}_A(Y_A, X) \simeq \mathrm{Hom}_k\big(Y, \mathrm{Res}_{A/k}\, X\big) \tag{1}$$

for every k-scheme Y, where

$$Y_A = Y \times_{\mathrm{Spec}\, k} \mathrm{Spec}\, A.$$

In particular, for $Y = \mathrm{Spec}\, R$,

$$\mathrm{Hom}_{k\text{-schemes}}(\mathrm{Spec}\, R, \mathrm{Res}_{A/k}\, X) \simeq \mathrm{Hom}_{A\text{-schemes}}(\mathrm{Spec}(A \otimes_k R), X).$$

Thus Weil restriction is right adjoint to base change along $k \to A$ on the schemes for which it exists.

Remark. If H is a quasi-projective group scheme of finite type over A, then its Weil restriction is an algebraic group over k, representing the group-valued functor

$$\begin{aligned} \mathrm{Res}_{A/k}\, H\colon\ \mathrm{Alg}_k &\longrightarrow \mathrm{Grp}, \\ R &\longmapsto H(A \otimes_k R). \end{aligned}$$

Its group operations are induced functorially by those of H.

Proposition 3.31 (The affine case)**.** Let A be a finite-dimensional k-algebra and let H be an affine group scheme of finite type over A. Then $\mathrm{Res}_{A/k}\, H$ is a linear algebraic group over k.

Proof. Write

$$H = \mathrm{Spec}\big(A[x_1, \ldots, x_n]/(f_1, \ldots, f_s)\big).$$

For every $R \in \mathrm{Alg}_k$,

$$(\mathrm{Res}_{A/k} H)(R) = \{(x_1, \ldots, x_n) \in (A \otimes_k R)^n \mid f_\ell(x_1, \ldots, x_n) = 0 \text{ for } 1 \le \ell \le s\}.$$

Choose a k-basis $e_1, \ldots, e_m$ of A. Each coordinate can be written uniquely as

$$x_i = \sum_{j=1}^{m} e_j \otimes y_{ij}, \qquad y_{ij} \in R.$$

Using the structure constants of A in this basis, expand

$$f_\ell \left(\sum_j e_j \otimes y_{1j}, \ldots, \sum_j e_j \otimes y_{nj} \right) = \sum_{a=1}^{m} e_a \otimes F_{\ell a}((y_{ij})),$$

where $F_{\ell a} \in k[y_{ij} \mid 1 \le i \le n,\ 1 \le j \le m]$. The equations $f_\ell = 0$ are equivalent to all the equations $F_{\ell a} = 0$. Hence the restriction is represented by the affine scheme of finite type

$$\mathrm{Spec} \frac{k[y_{ij} \mid 1 \le i \le n,\ 1 \le j \le m]}{(F_{\ell a} \mid 1 \le \ell \le s,\ 1 \le a \le m)}.$$

The group structure is the one described above, proving the assertion.[12] □

Example 3.32 (The adjoint of the identity)**.** Let G be an algebraic group over k. Taking $X = G_A$ and $Y = G$ in (1), the identity of G_A corresponds to a morphism

$$\iota_G : G \longrightarrow \mathrm{Res}_{A/k}(G_A).$$

For every $R \in \mathrm{Alg}_k$, its map on points is

$$\iota_G(R) : G(R) \longrightarrow G_A(A \otimes_k R),$$

induced by the k-algebra homomorphism

$$R \longrightarrow A \otimes_k R, \qquad r \longmapsto 1 \otimes r.$$

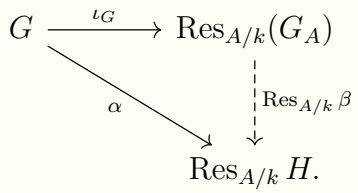
Example 3.33 (Adjunction for homomorphisms)**.** Let G be an algebraic group over k and H an algebraic group over A for which the restriction exists. The adjunction

[12] 2026.10.19 stopped at here.

restricts to

$$\mathrm{Hom}_{A\text{-grp}}(G_A, H) \simeq \mathrm{Hom}_{k\text{-grp}}\big(G, \mathrm{Res}_{A/k} H\big), \qquad \beta \longmapsto \alpha.$$

The corresponding homomorphisms fit into the commutative diagram



Now we can prove Theorem 3.24 for general base field k case.

Proof of Theorem 3.24 over a general base field. Recall that Theorem 3.24 has already been proved over an algebraically closed field. Applying that case over $\overline{k}$ and taking a faithful representation of the quotient gives a representation

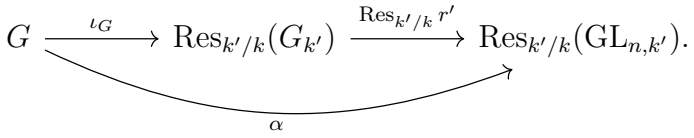
$$r : G_{\overline{k}} \longrightarrow \mathrm{GL}_{n,\overline{k}}, \qquad \ker r = H_{\overline{k}}.$$

There is a finite extension k'/k over which r is defined. Thus there is a homomorphism

$$r' : G_{k'} \longrightarrow \mathrm{GL}_{n,k'}, \qquad r'_{\overline{k}} = r, \qquad \ker r' = H_{k'}.$$

The last equality follows by faithfully flat descent from $\overline{k}$ to k'.

Consider the composite homomorphism of linear algebraic groups



We claim that $\ker \alpha = H$. Indeed, for every $R \in \mathrm{Alg}_k$, the map $\alpha(R)$ is

$$G(R) \xrightarrow{\iota_G(R)} G_{k'}(k' \otimes_k R) \xrightarrow{r'(k' \otimes_k R)} \mathrm{GL}_n(k' \otimes_k R).$$

Since $R \to k' \otimes_k R$ is faithfully flat, subgroup membership descends, and hence

$$\begin{aligned} \ker\big(\alpha(R)\big) &= \iota_G(R)^{-1}\big(\ker(r'(k' \otimes_k R))\big) \\ &= \iota_G(R)^{-1}\big(H_{k'}(k' \otimes_k R)\big) \\ &= H(R). \end{aligned}$$

Finally, factor α through its scheme-theoretic image as

$$G \xrightarrow{q} Q \hookrightarrow \mathrm{Res}_{k'/k}(\mathrm{GL}_{n,k'}).$$

The image factorization for homomorphisms of linear algebraic groups makes q faithfully flat, and $\ker q = \ker \alpha = H$. □

Corollary 3.34. Every normal algebraic subgroup H of a linear algebraic group G is the kernel of a finite-dimensional representation of G.

Proof. Take a faithful representation $Q \hookrightarrow \mathrm{GL}(V)$. The composite

$$G \longrightarrow Q \hookrightarrow \mathrm{GL}(V)$$

has kernel H. □

3.4 Quotient of non-normal subgroups

Definition 3.35 (Quotient by a subgroup)**.** Let G be a linear algebraic group and $H \leq G$ an algebraic subgroup. A *quotient of G by H* is a k-scheme X, equipped with an action

$$\mu : G \times X \longrightarrow X$$

and a point $o \in X(k)$, such that the following hold for every $R \in \mathrm{Alg}_k$. Write

$$\mu_0 : G \longrightarrow X, \qquad g \longmapsto g \cdot o.$$

(a) The nonempty fibres of

$$\mu_0(R) : G(R) \longrightarrow X(R), \qquad g \longmapsto g \cdot o_R,$$

are the cosets $gH(R)$ in $G(R)$. Equivalently, the induced map

$$G(R)/H(R) \longrightarrow X(R)$$

is injective.

(b) Every $x \in X(R)$ lifts to an element of $G(R')$ for some faithfully flat R-algebra R'.

Proposition 3.36. If X is a quotient of G by H, then the canonical morphism $\mu_0 : G \to X$ is faithfully flat.

Proof. It suffices to show that

$$G \times_X \operatorname{Spm} R \longrightarrow \operatorname{Spm} R$$

is faithfully flat for every affine open subset $\operatorname{Spm} R \subseteq X$, corresponding to a point $x \in X(R)$. By condition (b), there is a faithfully flat R-algebra R' and a lift $\widetilde{x} \in G(R')$ of $x_{R'}$. Thus we have a commutative diagram

$$\begin{array}{ccc} \operatorname{Spm} R' & \longrightarrow & \operatorname{Spm} R \\ {\scriptstyle \widetilde{x}}\downarrow & & \downarrow{\scriptstyle x} \\ G & \xrightarrow[\mu_0]{} & X. \end{array}$$

By fppf descent, it suffices to show that

$$G \times_X \operatorname{Spm} R' \longrightarrow \operatorname{Spm} R'$$

is faithfully flat. We claim that

$$G \times_X \operatorname{Spm} R' \simeq H \times_k \operatorname{Spm} R'.$$

Indeed, for every R'-algebra R'', the points of the left-hand side are the elements $g \in G(R'')$ such that

$$\mu_0(R'')(g) = x_{R''}.$$

By condition (a), these are exactly $\widetilde{x}_{R''} H(R'')$. Therefore

$$h \longmapsto \widetilde{x}_{R''} h$$

gives a natural bijection with $H(R'')$, and Yoneda's lemma proves the claim. Under this isomorphism, the morphism to $\operatorname{Spm} R'$ is the projection from $H \times_k \operatorname{Spm} R'$, which is faithfully flat:

$$\begin{array}{ccc} G \times_X \operatorname{Spm} R' & \xrightarrow{\sim} & H \times_k \operatorname{Spm} R' \\ & \searrow & \downarrow{\scriptstyle \operatorname{pr}_2 \text{ faithfully flat}} \\ & & \operatorname{Spm} R'. \end{array}$$

□

Lemma 3.37. If X is a quotient of G by H, then the morphism

$$G \times H \longrightarrow G \times_X G, \qquad (g,h) \longmapsto (g, gh)$$

is an isomorphism.

Proof. For every $R \in \mathrm{Alg}_k$, condition (a) gives

$$\begin{aligned} G(R) \times_{X(R)} G(R) &= \{(g_1, g_2) \mid \mu_0(g_1) = \mu_0(g_2)\} \\ &= \{(g_1, g_2) \mid g_1^{-1} g_2 \in H(R)\}. \end{aligned}$$

Thus the inverse on R-points is

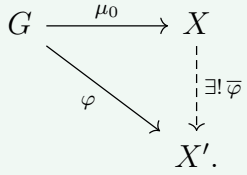
$$(g_1, g_2) \longmapsto (g_1, g_1^{-1} g_2).$$

These maps are natural in R, so the assertion follows from Yoneda's lemma. □

Proposition 3.38 (Universal property)**.** Let (X, μ, o) be a quotient of G by H and let $\varphi : G \to X'$ be a morphism of k-schemes. Suppose that, for every $R \in \mathrm{Alg}_k$, the map $\varphi(R)$ is constant on each coset $gH(R)$. Then there is a unique morphism $\overline{\varphi} : X \to X'$ such that

$$\varphi = \overline{\varphi} \circ \mu_0.$$

Equivalently, the following diagram commutes:



Proof. The morphism $\mu_0 : G \to X$ is faithfully flat and of finite type, hence is an fpqc cover. Descent for morphisms gives an equalizer

$$\mathrm{Hom}_k(X, X') \xrightarrow{\mu_0^*} \mathrm{Hom}_k(G, X') \underset{\mathrm{pr}_2^*}{\overset{\mathrm{pr}_1^*}{\rightrightarrows}} \mathrm{Hom}_k(G \times_X G, X').$$

Under Lemma 3.37, the two projections correspond to

$$\mathrm{pr}_G,\ m_H : G \times H \rightrightarrows G, \qquad m_H(g,h) = gh.$$

The hypothesis says precisely that

$$\varphi \circ \mathrm{pr}_G = \varphi \circ m_H.$$

Thus φ belongs to the equalizer and descends uniquely to X. □

Corollary 3.39. A quotient (X, μ, o) of G by H is uniquely determined up to a unique isomorphism compatible with the quotient maps.

Proof. If X and X_1 are two quotients, their quotient maps have the same cosets as fibres. The universal property gives morphisms in both directions. Uniqueness in that property shows that their composites are the identity morphisms. These isomorphisms also preserve the distinguished point and the G-action. □

Definition 3.40. We denote a quotient X by G/H and call it *the quotient of G by H*.

Proposition 3.41. Let G/H be a quotient as above.

1. $\dim G = \dim H + \dim(G/H)$.
2. If G is smooth over k, then G/H is smooth over k.

Proof. (i) The quotient map is faithfully flat, and after a faithfully flat base change its fibres are isomorphic to H. Thus all geometric fibres have dimension $\dim H$, and the dimension formula gives the assertion.

(ii) Write $X = G/H$. Faithful flatness gives an injection

$$\mathcal{O}_X \longrightarrow (\mu_0)_* \mathcal{O}_G.$$

The same is true after any extension of the base field. If G is smooth, it is geometrically reduced; hence X is geometrically reduced as well. Moreover, $G(\overline{k})$ acts transitively on $X(\overline{k})$: every geometric fibre of the quotient map is nonempty and has a $\overline{k}$-point. Over the perfect field $\overline{k}$, the reduced scheme $X_{\overline{k}}$ has a nonempty smooth locus. Transitivity carries a smooth point to every geometric point, so $X_{\overline{k}}$ is smooth. Therefore X is smooth over k. □

Remark. The quotient G/H is a homogeneous space over $\overline{k}$.

3.5 Construction of quotient

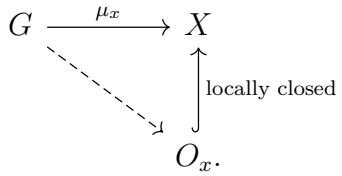
Theorem 3.42 (Existence of quotients)**.** Let G be an algebraic group over k and let $H \leq G$ be an algebraic subgroup. Then G/H exists.

Remark (Scope and idea of the proof)**.** We prove the theorem here only when G is a smooth linear algebraic group.

Idea. Constructing a quotient directly is hard, whereas constructing a stabilizer is easier. If G acts on a k-scheme X and $x \in X(k)$ has stabilizer $G_x = H$, then one expects

$$G/H \simeq O_x = \{g \cdot x \mid g \in G\} \subseteq X.$$

The orbit is locally closed. Give it its reduced subscheme structure; we need to factor the orbit map through it and prove faithful flatness:



Why assume smoothness? In particular, a morphism from the reduced scheme G factors through the reduced structure on its locally closed image. The lemmas below supply the other needed properties.

Definition 3.43 (Stabilizer)**.** Let $\mu : G \times X \to X$ be an action of an algebraic group on a k-scheme and let $x \in X(k)$. The *stabilizer* G_x is the group functor

$$G_x : \mathrm{Alg}_k \longrightarrow \mathrm{Grp}, \qquad R \longmapsto \{g \in G(R) \mid g \cdot x_R = x_R\}.$$

It is represented by the fibre of the orbit map

$$\mu_x : G \longrightarrow X, \qquad g \longmapsto g \cdot x,$$

over x, and is an algebraic subgroup of G.

Lemma 3.44. Let $\mu : G \times X \to X$ be an action and let $x \in X(k)$. Then (X, μ, x) is a quotient of G by G_x if and only if

$$\mu_x : G \longrightarrow X, \qquad g \longmapsto g \cdot x,$$

is faithfully flat.

Proof. For every $R \in \mathrm{Alg}_k$,

$$g_1 \cdot x_R = g_2 \cdot x_R \quad \Longleftrightarrow \quad g_1^{-1} g_2 \in G_x(R),$$

so condition (a) in Definition 3.35 always holds. If μ_x is faithfully flat, the pullback along any $y : \operatorname{Spm} R \to X$ is an fpqc cover of $\operatorname{Spm} R$. Taking a finite affine open cover of that pullback yields a faithfully flat R-algebra R' and a lift of $y_{R'}$ to $G(R')$. Thus condition (b) holds. Conversely, a quotient map is faithfully flat by Proposition 3.36. □

Lemma 3.45 (Equivariant images)**.** Let G be an algebraic group over k, and let X and Y be k-schemes on which G acts. Let $f : X \to Y$ be a G-equivariant morphism. Suppose that $G(\overline{k})$ acts transitively on $X(\overline{k})$.

1. The image $f(|X|)$ is locally closed in $|Y|$.
2. If X is reduced, then f factors as

$$X \xrightarrow{\text{faithfully flat}} f(X)_{\mathrm{red}} \xhookrightarrow{\text{immersion}} Y,$$

where $f(X)_{\mathrm{red}}$ denotes the image equipped with its reduced locally closed subscheme structure.

Proof sketch. (i) We may check the topological assertion after extending scalars to $\overline{k}$: the projection $|Y_{\overline{k}}| \to |Y|$ is a quotient map, as is its restriction over every closed subset. Thus assume $k = \overline{k}$. By Chevalley's theorem, $f(|X|)$ is constructible and contains a dense open subset U of its closure. Equivariance and transitivity imply that $G(k)$ acts transitively on the closed points of $f(|X|)$. The translates gU are open in the closure and contained in the image. Their union contains every closed point of the image, hence all of the constructible image. Thus the image is open in its closure.

(ii) Put $Z = f(X)_{\mathrm{red}}$. Since X is reduced, f factors through Z. Generic flatness over the reduced scheme Z gives a dense open subset $U \subseteq Z$ over which f is flat. After base change to $\overline{k}$, equivariance and transitivity translate this flatness locus to every closed point of $Z_{\overline{k}}$. These translations preserve the scheme-theoretic image, which commutes with this flat base change. Consequently the translated open subsets cover $Z_{\overline{k}}$, and f is flat everywhere. Flatness descends to k, and $X \to Z$ is surjective by the definition of Z. It is therefore faithfully flat. □

Applied to the orbit map $\mu_x : G \to X$, with G acting on itself by left multiplication, part (i) shows that μ_x has locally closed image.

Definition 3.46 (Orbit)**.** For an action $\mu : G \times X \to X$ and a point $x \in X(k)$, the *orbit* O_x is the locally closed subset $\mu_x(|G|)$ equipped with its reduced subscheme structure.

Proposition 3.47. Let $\mu : G \times X \to X$ be an action, let $x \in X(k)$, and suppose that G is reduced.

1. The orbit O_x is stable under G, and the orbit map

$$\mu_x : G \longrightarrow O_x, \qquad g \longmapsto g \cdot x,$$

is faithfully flat. In particular,

$$G/G_x \simeq O_x.$$

2. If G is smooth over k, then O_x is smooth over k.

Proof. (i) The faithful flatness follows from Lemma 3.45. To see that the action restricts to O_x, pull back the morphism $G \times O_x \to X$ along the faithfully flat cover

$$G \times G \longrightarrow G \times O_x, \qquad (g, g') \longmapsto (g, g' \cdot x).$$

On this cover, the action is $(g, g') \mapsto (gg') \cdot x$, which factors through O_x. This factorization descends, so O_x is G-stable. Lemma 3.44 now gives $G/G_x \simeq O_x$.

(ii) This follows from (i) and Proposition 3.41(ii). □

Proof of Definition 3.43 for a smooth linear algebraic group. Let $H \leq G$. By Proposition 3.47, it suffices to construct an action $G \times X \to X$ and a point $x \in X(k)$ such that $H = G_x$.

By Chevalley's theorem, there are a finite-dimensional representation

$$r : G \longrightarrow \mathrm{GL}(V)$$

and a one-dimensional subspace $L \subseteq V$ such that

$$H = \mathrm{Stab}_G(L).$$

Writing $n = \dim_k V$, the action on the underlying affine space

$$G \times \mathbb{A}^n \longrightarrow \mathbb{A}^n$$

induces an action on projective space

$$G \times \mathbb{P}(V) \longrightarrow \mathbb{P}(V) \simeq \mathbb{P}^{n-1}.$$

Here $\mathbb{P}(V)$ parametrizes lines in V. For the point $x = [L] \in \mathbb{P}(V)(k)$, the scheme-theoretic stabilizer is

$$G_x = \mathrm{Stab}_G(L) = H.$$

Since G is smooth, it is reduced, and therefore

$$G/H \simeq O_{[L]} \subseteq \mathbb{P}(V).$$

This is the required quotient in the smooth linear case. □

A Homework

A.1 Week 1

Exercise A.1. Show that the two definitions (algebraic geometry defintion and functor defintion) of algebraic groups coincide.

Proof. In the class we proved that for any algebraic group G in the algebraic geometry sense, it induces

$$\widetilde{G} : \mathrm{Alg}_k \longrightarrow \mathrm{Grp}, \qquad R \longmapsto G(R),$$

and $\widetilde{G}$ is an algebraic group in the functor sense.

It suffices to show that, for any functor $\widetilde{G} : \mathrm{Alg}_k \to \mathrm{Grp}$ such that $\widetilde{G}$, seen as $\mathrm{Alg}_k \to \mathrm{Set}$, is representable by a k-scheme G, then G is an algebraic group over k in the algebraic geometry sense.

Lemma A.2. [GW20, Corollary 4.7] Let X and Y be S-schemes. Then it is equivalent to give the following data.

1. A morphism of S-schemes from X to Y.
2. For all S-schemes T a map $f(T) : X_S(T) \to Y_S(T)$ of sets which is functorial in T, i.e., for all morphisms $u : T' \to T$ of S-schemes the following diagram is commutative:
$$\begin{array}{ccc} X_S(T) & \xrightarrow{f(T)} & Y_S(T) \\ {\scriptstyle X_S(u)}\downarrow & & \downarrow{\scriptstyle Y_S(u)} \\ X_S(T') & \xrightarrow[f(T')]{} & Y_S(T'). \end{array}$$
3. For all affine S-schemes $T = \operatorname{Spec} B$ a map $f(T) : X_S(T) \to Y_S(T)$ of sets which is functorial in B.

- **Construction of $G \times G \xrightarrow{m} G$.** By Lemma A.2, it is equivalent to give, for all $R \in \mathrm{Alg}_k$, a map
$$(G \times G)(R) \xrightarrow{m(R)} G(R)$$

and, for all k-algebra homomorphisms $R \to T$, to have the following diagram commute:

$$\begin{array}{ccc} (G \times G)(R) & \xrightarrow{m(R)} & G(R) \\ \downarrow & & \downarrow \\ (G \times G)(T) & \xrightarrow[m(T)]{} & G(T). \end{array}$$

Because it is well-known that $(G \times G)(R) = G(R) \times G(R)$ and $G(R) \cong \widetilde{G}(R)$ is a group, $m(R)$ can just be the multiplication of $\widetilde{G}(R)$. Because the vertical maps are group homomorphisms, hence it commutes.

- **Construction of $e : \operatorname{Spec} k \to G$.** It just corresponds to $G(k) \cong \widetilde{G}(k)$, namely the element $1_{\widetilde{G}(k)}$.
- **Construction of $i : G \to G$.** By Lemma A.2, it is equivalent to give, for all $R \in \mathrm{Alg}_k$, a map

$$G(R) \xrightarrow{i(R)} G(R)$$

and, for all k-algebra homomorphisms $R \to T$, to have the following diagram commute:

$$\begin{array}{ccc} G(R) & \xrightarrow{i(R)} & G(R) \\ \downarrow & & \downarrow \\ G(T) & \xrightarrow[i(T)]{} & G(T). \end{array}$$

Since $G(R) \cong \widetilde{G}(R)$ is a group, $i(R)$ can just be the inverse map of $\widetilde{G}(R)$. Because the vertical maps are group homomorphisms, hence it commutes.

- **Associativity.** To prove that

$$\begin{array}{ccc} G \times G \times G & \xrightarrow{(m,\mathrm{id}_G)} & G \times G \\ {\scriptstyle (\mathrm{id}_G,m)}\downarrow & & \downarrow{\scriptstyle m} \\ G \times G & \xrightarrow[m]{} & G \end{array}$$

commutes, i.e.,

$$G \times G \times G \xrightarrow{(m,\mathrm{id}_G)} G \times G \xrightarrow{m} G$$

and

$$G \times G \times G \xrightarrow{(\mathrm{id}_G,m)} G \times G \xrightarrow{m} G$$

give the same morphism, by Lemma A.2 it suffices to show that for every $R \in \mathrm{Alg}_k$,

$$(G \times G \times G)(R) \xrightarrow{(m(R),\mathrm{id}_{G(R)})} (G \times G)(R) \xrightarrow{m(R)} G(R)$$

and

$$(G \times G \times G)(R) \xrightarrow{(\mathrm{id}_{G(R)},m(R))} (G \times G)(R) \xrightarrow{m(R)} G(R)$$

give the same map. It is obvious by $G(R) \cong \widetilde{G}(R)$ being a group and the construction of $m(R)$, i.e., it is just the associativity on $\widetilde{G}(R)$.

- **Existence of unit.** To show

$$\begin{array}{ccc} G \cong G \times \operatorname{Spec} k & \xrightarrow{(\mathrm{id}_G,e)} & G \times G \\ & \searrow^{\mathrm{id}_G} & \downarrow m \\ & & G \end{array}$$

commutes, it suffices to show for every $R \in \mathrm{Alg}_k$ that

$$\begin{array}{ccc} G(R) & \xrightarrow{g \mapsto (g,1)} & G(R) \times G(R) \\ & \searrow^{\mathrm{id}_{G(R)}} & \downarrow m(R) \\ & & G(R) \end{array}$$

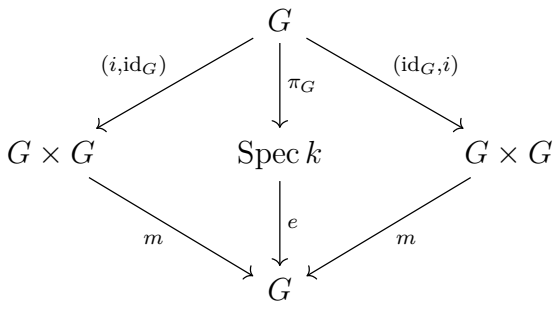
commutes, obviously.

Similarly,

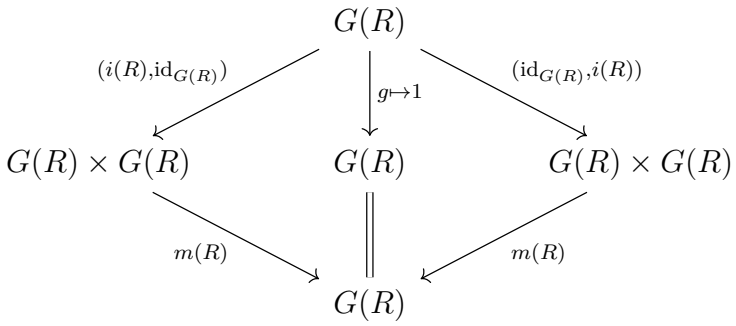
$$\begin{array}{ccc} G & \xrightarrow{(e,\mathrm{id}_G)} & G \times G \\ & \searrow^{\mathrm{id}_G} & \downarrow m \\ & & G \end{array}$$

commutes can be proved similarly.

- **Inverse property.** To show the following diagram commutes,



it suffices to show, for every $R \in \mathrm{Alg}_k$, the following diagram of maps commutes:



namely $g \mapsto (g^{-1}, g) \mapsto 1$ and $g \mapsto (g, g^{-1}) \mapsto 1$, obviously.

□

Exercise A.3. (a) Show that GL_n, B_n and U_n are linear algebraic groups.

(b) Show that U_n and B_n are both algebraic subgroups of GL_n.

(c) write down $m^\#$, $i^\#$ and $e^\#$ for GL_2.

(d) Define the functor $\mathrm{SL}_n : \mathrm{Alg}_k \to (\mathrm{gps})$ such that

$$\mathrm{SL}_n(R) = \{A \in \mathrm{GL}_n(R) \mid \det(A) = 1\}.$$

Show that it's a linear algebraic group, called special linear group.

(e) Compute the dimenson of all these algebraic groups GL_n, SL_n, B_n and U_n. Here dimension of an algebraic group is defined to be the dimension of the underlying scheme over k.

Proof of (a). **Claim.** See GL_n, B_n, U_n as functors from Alg_k to Grp as

$$\begin{aligned}\mathrm{GL}_n(R) &= \{(x_{ij}) \in M_n(R) \mid \det(x_{ij}) \in R^\times\},\\ B_n(R) &= \{(x_{ij}) \in \mathrm{GL}_n(R) \mid x_{ij} = 0 \text{ for } i > j\},\\ U_n(R) &= \{(x_{ij}) \in B_n(R) \mid x_{ii} = 1 \text{ for all } i\}.\end{aligned}$$

GL_n (B_n, U_n) is represented by affine $\mathrm{GL}_{n,k}$ $(B_{n,k}, U_{n,k})$. In fact,

$$\mathrm{GL}_{n,k} = \operatorname{Spm} k[X_{ij}]_{\det(X_{ij})},$$

where $\det(X_{ij})$ is a polynomial in $k[X_{ij}]$ and the subscript denotes localization,

$$B_{n,k} = \operatorname{Spm}\left(k[X_{ij}]_{\det(X_{ij})}/(X_{ij})_{i>j}\right) = \operatorname{Spm}\left((k[X_{ij}]_{i\le j})_{X_{11}X_{22}\cdots X_{nn}}\right),$$

and

$$U_{n,k} = \operatorname{Spm}\left(\frac{(k[X_{ij}]_{i\le j})_{X_{11}X_{22}\cdots X_{nn}}}{(X_{11}-1, X_{22}-1, \ldots, X_{nn}-1)}\right) \cong \operatorname{Spm}(k[X_{ij}]_{i<j}).$$

Reason: it suffices to show that

$$\mathrm{GL}_n(R) \cong \mathrm{Hom}_{\mathrm{Sch}}(\operatorname{Spm} R, \mathrm{GL}_{n,k}) \cong \mathrm{Hom}_{\mathrm{Alg}_k}(k[X_{ij}]_{\det(X_{ij})}, R), \tag{1}$$

$$B_n(R) \cong \mathrm{Hom}_{\mathrm{Alg}_k}\left((k[X_{ij}]_{i\le j})_{X_{11}\cdots X_{nn}}, R\right), \tag{2}$$

$$U_n(R) \cong \mathrm{Hom}_{\mathrm{Alg}_k}(k[X_{ij}]_{i<j}, R). \tag{3}$$

Notice that

$$M_n(R) \cong \mathrm{Hom}_{\mathrm{Alg}_k}(k[X_{ij}], R), \qquad (r_{ij}) \longleftrightarrow (X_{ij} \mapsto r_{ij}).$$

Thus (1) and (2) follow from the universal property of localization:

$$\begin{array}{ccc} k[X_{ij}] & \xrightarrow{X_{ij}\mapsto r_{ij}} & R \\ \downarrow & \nearrow & \\ k[X_{ij}]_{\det(X_{ij})} & & \end{array} \qquad \text{with } \det(r_{ij}) \in R^\times,$$

and (3) is direct. □

Proof of (b). By our construction in (a),

$$U_n \hookrightarrow B_n \hookrightarrow \mathrm{GL}_n.$$

Show $U_n \hookrightarrow B_n \hookrightarrow \mathrm{GL}_n$ are homomorphisms of algebraic groups in the functorial sense; i.e.,

for every $R \in \mathrm{Alg}_k$,

$$U_n(R) \hookrightarrow B_n(R) \hookrightarrow \mathrm{GL}_n(R)$$

are homomorphisms of groups (inclusion). It is natural.

But there is a detail: the inclusion $B_n(R) \to \mathrm{GL}_n(R)$ for any $R \in \mathrm{Alg}_k$, which is obviously functorial on Alg_k, just corresponds to the natural closed immersion $B_n \hookrightarrow \mathrm{GL}_n$ in Lemma A.2.

Reason: The natural closed immersion $B_n \hookrightarrow \mathrm{GL}_n$ corresponds to the natural quotient map

$$S \longrightarrow S/I, \qquad S = k[X_{ij}]_{\det(X_{ij})}, \qquad I \text{ generated by } (X_{ij})_{i>j}.$$

For every $R \in \mathrm{Alg}_k$, the induced

$$\mathrm{Mor}(\mathrm{Spm}\, R, B_n) \longrightarrow \mathrm{Mor}(\mathrm{Spm}\, R, \mathrm{GL}_n)$$

corresponds to

$$\begin{array}{ccc} S & \xrightarrow{\text{natural}} & S/I \\ & \searrow & \downarrow \\ & & R, \end{array}$$

and corresponds to $B_n(R) \to \mathrm{GL}_n(R)$ by

$$(r_{ij}) \longmapsto (r_{ij}),$$

that is,

$$\begin{pmatrix} X_{ij} \mapsto r_{ij} \\ S/I \to R \end{pmatrix} \longmapsto \begin{pmatrix} X_{ij} \mapsto r_{ij} \\ S \to R \end{pmatrix},$$

which is just the natural inclusion. □

Solution of (c).

$$m^{\#} : k[x,y,z,w]_{(xw-yz)} \longrightarrow k[x,y,z,w]_{(xw-yz)} \otimes k[x,y,z,w]_{(xw-yz)}.$$

See an element in $k[x,y,z,w]_{(xw-yz)}$ as a function on GL_2. For example, if $m^{\#}(x) = \sum_i f_i \otimes g_i$, then for

$$\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}, \quad \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \in \mathrm{GL}_2,$$

we have

$$a_1a_2 + b_1c_2 = x\left(\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}\begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}\right) = \sum_i f_i\left(\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}\right) g_i\left(\begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}\right).$$

Thus

$$m^{\#}(x) = x \otimes x + y \otimes z.$$

Similarly,

$$\begin{aligned} m^{\#}(y) &= x \otimes y + y \otimes w, \\ m^{\#}(z) &= z \otimes x + w \otimes z, \\ m^{\#}(w) &= z \otimes y + w \otimes w. \end{aligned}$$

Also

$$e^{\#} : k[x,y,z,w]_{(xw-yz)} \longrightarrow k, \qquad x, w \longmapsto 1, \quad y, z \longmapsto 0,$$

corresponding to $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, and

$$i^{\#} : k[x,y,z,w]_{(xw-yz)} \longrightarrow k[x,y,z,w]_{(xw-yz)},$$

with

$$\begin{aligned} x &\longmapsto (xw-yz)^{-1}w, \\ y &\longmapsto -(xw-yz)^{-1}y, \\ z &\longmapsto -(xw-yz)^{-1}z, \\ w &\longmapsto (xw-yz)^{-1}x, \end{aligned} \qquad \text{corresponding to } A \longmapsto A^{-1}.$$

□

Proof of (d). $\mathrm{SL}_n(R)$ are groups for any $R \in \mathrm{Alg}_k$. It suffices to show SL_n, as a functor from Alg_k to Set, is represented by

$$\mathrm{SL}_{n,k} := \mathrm{Spm}\left(\frac{k[X_{ij}]_{\det(X_{ij})}}{(\det(X_{ij})-1)}\right) = \mathrm{Spm}\left(\frac{k[X_{ij}]}{(\det(X_{ij})-1)}\right).$$

It suffices to show that, for any $R \in \mathrm{Alg}_k$,

$$\mathrm{SL}_n(R) \cong \mathrm{Hom}_{\mathrm{Sch}}(\mathrm{Spm}\, R, \mathrm{SL}_n) \cong \mathrm{Hom}_{\mathrm{Alg}_k}\left(\frac{k[X_{ij}]}{(\det(X_{ij})-1)}, R\right),$$

and this is satisfied by

$$(r_{ij}) \longleftrightarrow (X_{ij} \mapsto r_{ij}).$$

□

Solution of (e). GL_n is an open subscheme of $\operatorname{Spm} k[X_{ij}]$, hence

$$\dim \mathrm{GL}_n = n^2.$$

SL_n is a hypersurface of $\operatorname{Spm} k[X_{ij}]$, hence

$$\dim \mathrm{SL}_n = n^2 - 1.$$

B_n is cut out of GL_n by $\frac{n(n-1)}{2}$ functions X_{ij}, $i > j$. Hence

$$\dim B_n = n^2 - \frac{n(n-1)}{2} = \frac{n(n+1)}{2}.$$

Finally,

$$U_n \cong \operatorname{Spm} k[X_{ij}]_{i<j}, \qquad \dim U_n = \frac{n(n-1)}{2}.$$

□

Exercise A.4. For $C \in \mathrm{GL}_n(k)$, consider the functor $G_C : \mathrm{Alg}_k \to (\mathrm{gps})$ with

$$G_C(R) = \{A \in \mathrm{GL}_n(R) \mid A^t C A = C\}.$$

(a). Show that G_C is a linear algebraic group.

(b). When $n = 2m$ is even and $C = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix}$, then G_C is called the symplectic group denoted by Sp_n. Compute its dimension.

(c). For $\mathrm{char}(k) \neq 2$, when $C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & I_m \\ 0 & I_m & 0 \end{pmatrix}$ (resp. $C = \begin{pmatrix} 0 & I_m \\ I_m & 0 \end{pmatrix}$) if $n = 2m+1$ odd (resp. $n = 2m$ even), then G_C is called the orthogonal group denoted by O_n. Compute its dimension.

Proof of (a). $G_C(R)$ is a group for all $R \in \mathrm{Alg}_k$. Write

$$C = (c_{ij}), \qquad c_{ij} \in k.$$

Then

$$(X_{ij})^t (c_{ij})(X_{ij}) - (c_{ij}) =: \big(f_{ij}(X_{11}, \ldots, X_{nn})\big), \qquad f_{ij} \in k[X_{ij}].$$

Claim. $G_C(-)$ is represented by

$$\operatorname{Spm} \frac{k[X_{ij}]}{(f_{ij})},$$

which is affine.

According to the definition, at first it should be

$$\operatorname{Spm}\left(k[X_{ij}]_{\det(X_{ij})}/(f_{ij})\right),$$

but $\det(X_{ij})$ can be removed because $A^tCA = C$ and $C \in \mathrm{GL}_n$ imply $A \in \mathrm{GL}_n$.

Reason: It suffices to show, for any $R \in \mathrm{Alg}_k$,

$$G_C(R) \cong \operatorname{Hom}_{\mathrm{Sch}}\left(\operatorname{Spm} R, \operatorname{Spm} \frac{k[X_{ij}]}{(f_{ij})}\right) \cong \operatorname{Hom}_{\mathrm{Alg}_k}\left(\frac{k[X_{ij}]}{(f_{ij})}, R\right),$$

and it is satisfied by

$$(r_{ij}) \longleftrightarrow (X_{ij} \mapsto r_{ij}).$$

□

Solution of (b). Notice that, because C is anti-symmetric,

$$f_{ij} = -f_{ji}.$$

Therefore $(f_{ij}) \subseteq k[X_{ij}]$ can be generated by at most $\frac{n(n-1)}{2}$ polynomials. Hence

$$\dim \mathrm{Sp}_n \geq n^2 - \frac{n(n-1)}{2} = \frac{n(n+1)}{2}.$$

We want to compute the corank of the Jacobian to get the upper bound of $\dim \mathrm{Sp}_n$. By [Vak25, 13.1.M, 13.1.N], for every $p \in \mathrm{Sp}_n$,

$$\begin{aligned}\dim_{\mathrm{Sp}_n} p &\leq \operatorname{corank}\left(\left(\frac{\partial f_{ij}}{\partial X_{kl}}(p)\right)_{i,j,k,l\leq n}\right) \\ &= \dim_{\kappa(p)}\left\{B = (b_{kl}) \in M_{n\times n}(\kappa(p)) \;\middle|\; \forall i,j,\ \sum_{k,l} \frac{\partial f_{ij}}{\partial X_{kl}}(p)\cdot b_{kl} = 0\right\}.\end{aligned}$$

By

$$f_{ij} = \sum_{r,s} X_{ri} c_{rs} X_{sj} - c_{ij},$$

we have, for every k, l,

$$\frac{\partial f_{ij}}{\partial X_{kl}} = \delta_{il} \sum_s c_{ks} X_{sj} + \delta_{jl} \sum_r X_{ri} c_{rk}.$$

Thus, for every i, j,

$$\begin{aligned} \sum_{k,l} \frac{\partial f_{ij}}{\partial X_{kl}}(p) \cdot b_{kl} &= \sum_{k,s} b_{ki} c_{ks} X_{sj}(p) + \sum_{r,k} X_{ri}(p) c_{rk} b_{kj} \\ &= (B^t C A(p) + A(p)^t C B)_{ij}. \end{aligned}$$

Therefore

$$\begin{aligned} \dim_{\mathrm{Sp}_n} p &\le \dim_{\kappa(p)} \{B \in M_{n\times n}(\kappa(p)) \mid B^t C A(p) + A(p)^t C B = 0\} \\ &= \dim_{\kappa(p)} \{B \in M_{n\times n}(\kappa(p)) \mid B^t C A(p) = (B^t C A(p))^t\} \qquad \text{by } C = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix} \\ &= \dim_{\kappa(p)} \{Y \in M_{n\times n}(\kappa(p)) \mid Y = Y^t\} \qquad \text{by } C, A(p) \in \mathrm{GL}_n(\kappa(p)) \\ &= \frac{n(n+1)}{2}. \end{aligned}$$

Hence

$$\dim \mathrm{Sp}_n = \frac{n(n+1)}{2}.$$

□

Solution of (c). Because C is symmetric,

$$f_{ij} = f_{ji}.$$

Therefore $(f_{ij}) \subseteq k[X_{ij}]$ can be generated by at most $\frac{n(n+1)}{2}$ polynomials. Hence

$$\dim \mathrm{O}_n \ge n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}.$$

By a similar discussion as in (b), for every $p \in \mathrm{O}_n$,

$$\begin{aligned} \dim_{\mathrm{O}_n} p &\le \dim_{\kappa(p)} \{B \in M_{n\times n}(\kappa(p)) \mid B^t C A(p) + A(p)^t C B = 0\} \\ &= \dim_{\kappa(p)} \{B \in M_{n\times n}(\kappa(p)) \mid B^t C A(p) = -(B^t C A(p))^t\} \qquad \text{by } C = C^t \text{ and } \operatorname{char} k \neq 2 \\ &= \dim_{\kappa(p)} \{Y \in M_{n\times n}(\kappa(p)) \mid Y = -Y^t\} \qquad \text{by } C, A(p) \in \mathrm{GL}_n(\kappa(p)) \\ &= \frac{n(n-1)}{2}. \end{aligned}$$

Hence

$$\dim \mathrm{O}_n = \frac{n(n-1)}{2}.$$

□

Exercise A.5. Show that the sequence of homomomorphism of algebraic groups

$$e \to \mathrm{SL}_n \to \mathrm{GL}_n \xrightarrow{\det} \mathbb{G}_m \to e$$

is exact, here the first map is the natural inclusion and the second one is the determinant map.

Proof.

- $\mathrm{SL}_n \cong \mathrm{Ker}(\det)$. For every $R \in \mathrm{Alg}_k$,

  $$K(R) := \ker\big(\mathrm{GL}_n(R) \xrightarrow{\det} \mathbb{G}_m(R) = R^\times\big)$$

  is a group. By definition, $K(R)$ is represented by $\mathrm{Ker}(\mathrm{GL}_n \xrightarrow{\det} \mathbb{G}_m)$. Also, $K(R) = \mathrm{SL}_n(R)$ is represented by SL_n.

- $\mathrm{GL}_n \xrightarrow{\det} \mathbb{G}_m$ is faithfully flat, i.e. flat and surjective by [Vak25, The paragraph after 24.5.B].

  Surjectivity. It suffices to find a section $s : \mathbb{G}_m \to \mathrm{GL}_n$. It suffices to find a k-algebra homomorphism

  $$s^\# : k[X_{ij}]_{\det(X_{ij})} \longrightarrow k[t, t^{-1}]$$

  such that, with

  $$\overset{\#}{\det} : k[t, t^{-1}] \longrightarrow k[X_{ij}]_{\det(X_{ij})}, \qquad t \longmapsto \det(X_{ij}),$$

  we have

  $$s^\# \circ \overset{\#}{\det} = \mathrm{id}_{k[t,t^{-1}]}.$$

  Consider $s^\#$ given by

  $$X_{11} \longmapsto t, \qquad X_{ii} \longmapsto 1 \text{ for } i > 1, \qquad X_{ij} \longmapsto 0 \text{ for } i \neq j.$$

  So

  $$\mathbb{G}_m \xrightarrow{s} \mathrm{GL}_n \xrightarrow{\det} \mathbb{G}_m, \qquad \mathbb{G}_m \xrightarrow{\mathrm{id}} \mathbb{G}_m,$$

  and det is surjective.

Flatness. With

$$k[t,t^{-1}] \xrightarrow{\det^{\#}} k[X_{ij}]_{\det(X_{ij})}$$

injective by the above choice of the section and the fact that $k[X_{ij}]_{\det(X_{ij})}$ is integral, $k[X_{ij}]_{\det(X_{ij})}$ is torsion-free as a $k[t,t^{-1}]$-module. The ring $k[t,t^{-1}]$ is a localization of a PID, hence is a PID. A torsion-free algebra over a PID is flat over it.

□

A.2 Week 2

Exercise A.6. The natural isomorphism of set valued functors

$$\mathrm{SL}_n(R) \times \mathbb{G}_m(R) \longrightarrow \mathrm{GL}_n(R), \qquad (A,r) \longmapsto A \cdot \mathrm{diag}(r,1,\ldots,1)$$

defines an isomorphism of k-algebras

$$\mathcal{O}(\mathrm{GL}_n) \simeq \mathcal{O}(\mathrm{SL}_n) \otimes \mathcal{O}(\mathbb{G}_m).$$

(a) Show that SL_n is connected.

(b) Please describe the map

$$\mathcal{O}(\mathrm{SL}_n) \longrightarrow \mathcal{O}(\mathrm{GL}_n)$$

from the previous isomorphism.

Proof. **(a)** By the given isomorphism and Yoneda,

$$\mathrm{SL}_n \times \mathbb{G}_m \simeq \mathrm{GL}_n.$$

Also

$$\mathrm{GL}_n \simeq \mathrm{Spm}\, k[X_{ij}]_\Delta, \qquad \Delta = \det(X_{ij}).$$

Since $k[X_{ij}]_\Delta$ is an integral domain, GL_n is irreducible, hence connected. Consider

$$\mathrm{GL}_n \xrightarrow{\sim} \mathrm{SL}_n \times \mathbb{G}_m \xrightarrow{\mathrm{pr}_1} \mathrm{SL}_n.$$

It is surjective. The image of a connected space is connected, so SL_n is connected.

(b) The map in question is induced by

$$\mathrm{GL}_n \xrightarrow{\sim} \mathrm{SL}_n \times \mathbb{G}_m \xrightarrow{\mathrm{pr}_1} \mathrm{SL}_n.$$

On coordinate rings it is

$$\mathcal{O}(\mathrm{SL}_n) \longrightarrow \mathcal{O}(\mathrm{SL}_n) \otimes \mathcal{O}(\mathbb{G}_m) \xrightarrow{\sim} \mathcal{O}(\mathrm{GL}_n), \qquad f \longmapsto f \otimes 1.$$

Write

$$\mathcal{O}(\mathrm{SL}_n) = \frac{k[X_{ij}]}{(\Delta - 1)}, \qquad \mathcal{O}(\mathbb{G}_m) = k[T, T^{-1}], \qquad \mathcal{O}(\mathrm{GL}_n) = k[X_{ij}]_\Delta,$$

and let $\overline{X}_{ij}$ denote the image of X_{ij} in $\mathcal{O}(\mathrm{SL}_n)$. Under the given functorial isomorphism,

$$X_{i1} \longmapsto \overline{X}_{i1} \otimes T, \qquad X_{ij} \longmapsto \overline{X}_{ij} \otimes 1 \quad (j \neq 1), \qquad \Delta \longmapsto 1 \otimes T.$$

Therefore

$$\boxed{\overline{X}_{i1} \longmapsto \frac{X_{i1}}{\Delta}, \qquad \overline{X}_{ij} \longmapsto X_{ij} \quad (j \neq 1).}$$

□

Exercise A.7. Let $\mathrm{char}(k) \neq 2$. Let $O' = G_C$ with $C = I_n$ (notation from Week 1 Ex. 3).

(a) Show that

$$O_{n,\overline{k}} \simeq O'_{n,\overline{k}}$$

as algebraic groups over $\overline{k}$.

(b) Show that the determinant map

$$\det : O_n \longrightarrow \mu_2$$

is a homomorphism of algebraic groups, where

$$\mu_2 = \mathrm{Ker}(\mathbb{G}_m \longrightarrow \mathbb{G}_m,\ x \longmapsto x^2).$$

Define

$$\mathrm{SO}_n := \mathrm{Ker}(\det : O_n \to \mu_2).$$

And similarly define

$$\mathrm{SO}'_n := \mathrm{Ker}(\det : O'_n \to \mu_2).$$

(c) For each k-algebra R, let $V(R)$ denote the set of skew-symmetric matrices, i.e.

the matrices $A \in M_n(R)$ such that $A^t = -A$. Show that the functor

$$R \longmapsto V(R)$$

is represented by a finitely generated k-algebra C, and that C is an integral domain.

(d) Show that

$$A \longmapsto (I_n - A)(I_n + A)^{-1}$$

defines a bijection from a nonempty open subset of $\mathrm{SO}'_n(\overline{k})$ onto an open subset of $V(\overline{k})$, with partial inverse

$$B \longmapsto (I_n - B)(I_n + B)^{-1}.$$

(e) Deduce that $\mathrm{SO}'_{n,\overline{k}}$ is connected.

(f) Deduce that SO_n is connected.

Proof. Let J be the symmetric matrix defining O_n in Week 1, namely

$$J = \begin{cases} \begin{pmatrix} 0 & I_m \\ I_m & 0 \end{pmatrix}, & n = 2m, \\ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & I_m \\ 0 & I_m & 0 \end{pmatrix}, & n = 2m+1. \end{cases}$$

(a) Choose $\alpha, \beta \in \overline{k}$ such that

$$\alpha^2 = 2, \qquad \beta^2 = -1.$$

For the 2×2 hyperbolic block we have

$$\begin{pmatrix} \frac{1}{\alpha} & \frac{1}{\alpha} \\ \frac{\beta}{\alpha} & -\frac{\beta}{\alpha} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\alpha} & \frac{\beta}{\alpha} \\ \frac{1}{\alpha} & -\frac{\beta}{\alpha} \end{pmatrix} = I_2.$$

Applying this to the hyperbolic blocks of J gives $Q \in \mathrm{GL}_n(\overline{k})$ such that

$$Q^t J Q = I_n.$$

Hence, for every $R \in \mathrm{Alg}_{\overline{k}}$,

$$O_{n,\overline{k}}(R) = \{A \in \mathrm{GL}_n(R) \mid A^t J A = J\}$$

is naturally identified with

$$\{B \in \mathrm{GL}_n(R) \mid B^t I_n B = I_n\} = O'_{n,\overline{k}}(R)$$

by

$$A \longmapsto Q^{-1} A Q.$$

Indeed, $Q^t J Q = I_n$ implies $J = Q^{-t} Q^{-1}$, and therefore

$$(Q^{-1}AQ)^t (Q^{-1}AQ) = Q^t A^t J A Q = Q^t J Q = I_n.$$

The inverse is $B \mapsto QBQ^{-1}$. This is functorial in R, so by Yoneda

$$O_{n,\overline{k}} \simeq O'_{n,\overline{k}}.$$

(b) It is enough to show that for every $R \in \mathrm{Alg}_k$,

$$\det(R) : O_n(R) \longrightarrow \mu_2(R)$$

is a group homomorphism. We have

$$\mu_2(R) = \mathrm{Ker}\Big(\mathbb{G}_m(R) \xrightarrow{r \mapsto r^2} \mathbb{G}_m(R)\Big) = \{r \in R^\times \mid r^2 = 1\}.$$

For $A \in O_n(R)$, from

$$A^t J A = J$$

we get

$$\det(A)^2 \det(J) = \det(J).$$

Since $\det(J) \in k^\times$, it follows that

$$\det(A)^2 = 1.$$

Thus the image of $O_n(R)$ under determinant is contained in $\mu_2(R)$. Moreover,

$$O_n(R) \xrightarrow{\det} \mathbb{G}_m(R)$$

is a group homomorphism. Hence

$$O_n(R) \longrightarrow \mu_2(R)$$

is a group homomorphism for every R, and therefore

$$\det : O_n \longrightarrow \mu_2$$

is a homomorphism of algebraic groups.

(c) We claim that V is represented by

$$\operatorname{Spm} \frac{k[X_{ij} \mid 1 \le i,j \le n]}{(X_{ij} + X_{ji} \mid 1 \le i,j \le n)}.$$

Indeed, for every $R \in \mathrm{Alg}_k$,

$$\begin{aligned}\operatorname{Hom}_{k\text{-alg}}\left(\frac{k[X_{ij}]}{(X_{ij}+X_{ji})}, R\right) &\simeq \{f \in \operatorname{Hom}_{k\text{-alg}}(k[X_{ij}], R) \mid f(X_{ij}) + f(X_{ji}) = 0 \ \forall i,j\} \\ &\simeq \{(r_{ij}) \in M_n(R) \mid r_{ij} + r_{ji} = 0 \ \forall i,j\} \\ &= V(R).\end{aligned}$$

Thus the claim follows by Yoneda. Since $\mathrm{char}(k) \neq 2$, the relations imply

$$X_{ii} = 0, \qquad X_{ji} = -X_{ij},$$

and hence

$$C := \frac{k[X_{ij}]}{(X_{ij}+X_{ji})} \simeq k[X_{ij} \mid i < j].$$

Therefore C is finitely generated and is an integral domain.

(d) Consider, as classical varieties,

$$U = \left\{A \in \mathrm{SO}'_n(\overline{k}) \mid \det(I_n + A) \neq 0\right\}$$

and

$$W = \left\{B \in V(\overline{k}) \mid \det(I_n + B) \neq 0\right\}.$$

These are open, and they are nonempty since $I_n \in U$ and $0 \in W$. We claim that

$$U \xrightarrow{\sim} W, \qquad A \longmapsto (I_n - A)(I_n + A)^{-1},$$

with inverse

$$B \longmapsto (I_n - B)(I_n + B)^{-1}.$$

First, it is enough to check the bijection set-theoretically. Indeed, the entries of $(I_n - Y)(I_n + Y)^{-1}$ belong to the localization in which $\det(I_n + Y)$ is inverted, so they define the corresponding homomorphism of localized $\overline{k}$-algebras. The other direction is defined similarly. Thus, once the two formulas are checked to be inverse on points, they give inverse morphisms of the two open subschemes.

Let $A \in U$ and put

$$B = (I_n - A)(I_n + A)^{-1}.$$

Since $A^t A = I_n$, we have $A^t = A^{-1}$, and

$$\begin{aligned} B^t &= (I_n + A^t)^{-1}(I_n - A^t) \\ &= (I_n + A^{-1})^{-1}(I_n - A^{-1}) \\ &= (A + I_n)^{-1}(A - I_n) \\ &= (A - I_n)(A + I_n)^{-1} \\ &= -B. \end{aligned}$$

Hence $B \in V(\overline{k})$. Moreover,

$$I_n + B = 2(I_n + A)^{-1},$$

so $\det(I_n + B) \neq 0$, and hence $B \in W$.

Conversely, let $B \in W$ and put

$$A = (I_n - B)(I_n + B)^{-1}.$$

Since $B^t = -B$,

$$\begin{aligned} A^t &= (I_n + B^t)^{-1}(I_n - B^t) \\ &= (I_n - B)^{-1}(I_n + B) \\ &= A^{-1}. \end{aligned}$$

Also

$$\det(A) = \frac{\det(I_n - B)}{\det(I_n + B)} = \frac{\det((I_n - B)^t)}{\det(I_n + B)} = \frac{\det(I_n + B)}{\det(I_n + B)} = 1.$$

Thus $A \in \mathrm{SO}'_n(\overline{k})$. Moreover,

$$I_n + A = 2(I_n + B)^{-1},$$

so $A \in U$.

Finally, if $B = (I_n - A)(I_n + A)^{-1}$, then

$$I_n - B = 2A(I_n + A)^{-1}, \qquad I_n + B = 2(I_n + A)^{-1},$$

and hence

$$(I_n - B)(I_n + B)^{-1} = A.$$

The other composition is the same. Therefore $U \simeq W$.

(e) We use the Cartan–Dieudonné theorem: for an n-dimensional nondegenerate symmetric bilinear space over a field of characteristic not 2, every element of the orthogonal group is a composition of at most n reflections.

For a nonisotropic vector v, write

$$S_v(x) = x - \frac{2\langle x, v\rangle}{\langle v, v\rangle} v.$$

Its eigenvalues are $-1, 1, \ldots, 1$, so $\det(S_v) = -1$.

Let U be the open subset from part **(d)**. We claim that for any nonisotropic vectors u, v,

$$S_u S_v \in U(\overline{k})^2.$$

Because $\overline{k}$ is infinite, we can choose a nonisotropic vector w such that

$$\langle u, w\rangle \neq 0, \qquad \langle v, w\rangle \neq 0.$$

Indeed, we choose w outside the union of the three proper closed subsets cut out by

$$\langle w, w\rangle, \qquad \langle u, w\rangle, \qquad \langle v, w\rangle.$$

We show that $S_u S_w \in U(\overline{k})$. Suppose, to the contrary, that -1 is an eigenvalue of $S_u S_w$. Choose $x \neq 0$ with

$$S_u S_w x = -x.$$

Since $S_u^2 = 1$, this gives

$$S_w x = -S_u x.$$

Using the definitions of the two reflections,

$$x - \frac{2\langle x, w\rangle}{\langle w, w\rangle} w = -x + \frac{2\langle x, u\rangle}{\langle u, u\rangle} u,$$

so

$$x = \frac{\langle x, u\rangle}{\langle u, u\rangle}u + \frac{\langle x, w\rangle}{\langle w, w\rangle}w.$$

Pairing this equality first with u and then with w gives

$$\frac{\langle x, w\rangle}{\langle w, w\rangle}\langle u, w\rangle = 0, \qquad \frac{\langle x, u\rangle}{\langle u, u\rangle}\langle u, w\rangle = 0.$$

Since $x \neq 0$, the two coefficients on the left cannot both be zero. Hence $\langle u, w\rangle = 0$, a contradiction. Therefore -1 is not an eigenvalue of $S_u S_w$, so

$$\det(I_n + S_u S_w) \neq 0.$$

Since $\det(S_u S_w) = 1$, we have $S_u S_w \in U(\overline{k})$. Similarly, $S_w S_v \in U(\overline{k})$. Hence

$$S_u S_v = (S_u S_w)(S_w S_v) \in U(\overline{k})^2.$$

Now let $A \in \mathrm{SO}'_n(\overline{k})$. By Cartan–Dieudonné,

$$A = S_{v_1} \cdots S_{v_\ell}.$$

Since

$$1 = \det(A) = (-1)^\ell,$$

the integer ℓ is even. Pairing the reflections and applying the claim shows that

$$A \in \langle U(\overline{k})\rangle.$$

Therefore

$$\mathrm{SO}'_n(\overline{k}) = \langle U(\overline{k})\rangle.$$

By part **(d)**, U is isomorphic to the nonempty open subset W of V. By part **(c)**, V is irreducible. Hence U is irreducible, and therefore connected. Since $I_n \in U$, we have

$$U \subseteq (\mathrm{SO}'_{n,\overline{k}})^\circ.$$

The identity component is a subgroup, so it contains the subgroup generated by $U(\overline{k})$. As this subgroup is all of $\mathrm{SO}'_n(\overline{k})$, every $\overline{k}$-point lies in the identity component. Thus

$$\mathrm{SO}'_{n,\overline{k}} = (\mathrm{SO}'_{n,\overline{k}})^\circ,$$

and $\mathrm{SO}'_{n,\overline{k}}$ is connected.

(f) From part **(a)**, for every $R \in \mathrm{Alg}_{\overline{k}}$,

$$O_{n,\overline{k}}(R) \simeq O'_{n,\overline{k}}(R).$$

This isomorphism is given by conjugation, hence preserves determinant. Therefore it restricts functorially to

$$\mathrm{SO}_{n,\overline{k}}(R) \simeq \mathrm{SO}'_{n,\overline{k}}(R),$$

and by Yoneda

$$\mathrm{SO}_{n,\overline{k}} \simeq \mathrm{SO}'_{n,\overline{k}}.$$

By part **(e)**, $\mathrm{SO}'_{n,\overline{k}}$ is connected. Hence SO_n is geometrically connected, and in particular connected. □

Exercise A.8. Let k be algebraically closed.

(a) Let G be a closed subgroup of GL_n over k. Then $\mathbb{A}^n$ has a natural action of G. Determine the orbits and closed orbits for

$$G = \mathrm{GL}_n, T_n, \mathrm{SL}_n.$$

(b) The natural action of $G = \mathrm{GL}_2$ on $\mathbb{A}^2$ induces an action of G on $\mathbb{P}^1$ (as $\mathbb{P}^1$ is the quotient of $\mathbb{A}^2 \setminus \{(0,0)\}$). Please determine the orbits for the action of G on $\mathbb{P}^1$ and the diagonal action of G on $\mathbb{P}^1 \times \mathbb{P}^1$. Please also indicate the closed orbits.

Proof. **(a)** For $G = \mathrm{GL}_n$, the orbits are

$$\mathbb{A}^n \setminus \{0\}, \qquad \{0\}.$$

Indeed, GL_n acts transitively on the nonzero vectors. The only closed orbit is $\{0\}$.

For $G = T_n$, the orbits are indexed by subsets

$$S \subseteq \{1, \ldots, n\}.$$

For such an S, the corresponding orbit is

$$\mathcal{O}_S = \{(x_1, \ldots, x_n) \in \mathbb{A}^n \mid x_i \neq 0 \iff i \in S\}.$$

Equivalently,

$$\mathcal{O}_S \simeq \mathrm{Spm}\left(k[X_i \mid i \in S]_{\prod_{i \in S} X_i}\right) \subseteq \mathrm{Spm}\, k[X_i \mid i \in S] \subseteq \mathbb{A}^n.$$

Its closure is the corresponding coordinate subspace, so the only closed orbit is $\{0\}$.

For $G = \mathrm{SL}_n$, when $n = 1$, the group SL_1 is trivial. Hence every point is an orbit, and every orbit is closed. When $n \geq 2$, SL_n is transitive on $\mathbb{A}^n \setminus \{0\}$: extend a nonzero vector to a basis and rescale one of the remaining basis vectors so that the resulting matrix has determinant 1. Thus the orbits are

$$\mathbb{A}^n \setminus \{0\}, \qquad \{0\},$$

and the only closed orbit is $\{0\}$.

(b) By the discussion above, the orbit of $G = \mathrm{GL}_2$ on $\mathbb{P}^1$ is all of

$$\mathbb{P}^1.$$

Hence it is closed.

For the diagonal action of G on $\mathbb{P}^1 \times \mathbb{P}^1$, there are two orbits:

$$\Delta, \qquad (\mathbb{P}^1 \times \mathbb{P}^1) \setminus \Delta,$$

where

$$\Delta = \mathrm{im}\Big(\mathbb{P}^1 \xrightarrow{\mathrm{diag}} \mathbb{P}^1 \times \mathbb{P}^1\Big).$$

Indeed, Δ is one orbit by transitivity on $\mathbb{P}^1$. If $x \neq y$, nonzero representatives of the two distinct lines form a basis of k^2, so every ordered pair of distinct points belongs to the same orbit. Since $\mathbb{P}^1$ is complete, the diagonal morphism is proper and its image Δ is closed. Therefore Δ is the only closed orbit. □

A.3 Week 3

Exercise A.9. Let $V = k^2$ be the natural representation of $G = \mathrm{GL}_2$.

(a) Let $V^\vee$ be the dual representation of V. Determine the $\mathcal{O}(G)$-comodule of $V^\vee$.

(b) Determine the $\mathcal{O}(G)$-comodule of the representation $V \otimes V$.

Proof. (a) Let $\{e_i\}_{i=1}^2$ be the standard basis of V, and let $\{e_i^\vee\}_{i=1}^2$ be its dual basis, so $e_i^\vee(e_j) = \delta_{ij}$. Write

$$\rho(e_j^\vee) = \sum_{i=1}^{2} e_i^\vee \otimes f_{ij}, \qquad f_{ij} \in \mathcal{O}(G).$$

This corresponds to the dual representation $G \longrightarrow \mathrm{GL}_{V^\vee}$. For a k-algebra R and $g \in G(R)$,

its action on $V^\vee \otimes_k R$ is given by

$$(ge_j^\vee)(w) = e_j^\vee(g^{-1}w).$$

Consequently, evaluating the coaction at g and then at e_k gives

$$f_{kj}(g) = (ge_j^\vee)(e_k) = e_j^\vee(g^{-1}e_k) = (g^{-1})_{jk}.$$

Put $\Delta = \det(T_{ij}) = T_{11}T_{22} - T_{12}T_{21}$. For $n = 2$,

$$g^{-1} = \frac{1}{\Delta(g)} \begin{pmatrix} T_{22}(g) & -T_{12}(g) \\ -T_{21}(g) & T_{11}(g) \end{pmatrix}.$$

Thus

$$f_{11} = \frac{T_{22}}{\Delta}, \qquad f_{12} = -\frac{T_{21}}{\Delta}, \qquad f_{21} = -\frac{T_{12}}{\Delta}, \qquad f_{22} = \frac{T_{11}}{\Delta},$$

and the requested coaction is

$$\rho(e_1^\vee) = e_1^\vee \otimes \frac{T_{22}}{\Delta} - e_2^\vee \otimes \frac{T_{12}}{\Delta},$$
$$\rho(e_2^\vee) = -e_1^\vee \otimes \frac{T_{21}}{\Delta} + e_2^\vee \otimes \frac{T_{11}}{\Delta}.$$

(b) Write

$$\rho(e_i \otimes e_j) = \sum_{r,s=1}^{2} (e_r \otimes e_s) \otimes f_{ij}^{rs}.$$

For every k-algebra R and $g \in G(R)$, evaluation gives

$$\sum_{r,s} (e_r \otimes e_s) f_{ij}^{rs}(g) = g(e_i \otimes e_j) = ge_i \otimes ge_j = \sum_{r,s} (e_r \otimes e_s) g_{ri} g_{sj}.$$

Therefore $f_{ij}^{rs} = T_{ri}T_{sj}$, and hence

$$\boxed{\rho(e_i \otimes e_j) = \sum_{r,s=1}^{2} (e_r \otimes e_s) \otimes T_{ri}T_{sj}.}$$

□

Exercise A.10. Let $G = \mathrm{GL}_2$. Let $V = \mathcal{O}(G)$ be the regular representation of G. Let $f \in V$ such that $f(g) = a_{11}a_{12}$ for any $g = (a_{ij})_{1 \le i,j \le 2}$. Find a finite dimensional

subrepresentation W of V such that $f \in W$.

Proof. For the regular representation the coaction is the comultiplication

$$m^{\#} : \mathcal{O}(G) \longrightarrow \mathcal{O}(G) \otimes \mathcal{O}(G), \qquad T_{ij} \longmapsto \sum_{r=1}^{2} T_{ir} \otimes T_{rj}.$$

Here $f = T_{11}T_{12}$, so

$$\begin{aligned} m^{\#}(f) &= (T_{11} \otimes T_{11} + T_{12} \otimes T_{21})(T_{11} \otimes T_{12} + T_{12} \otimes T_{22}) \\ &= T_{11}^2 \otimes T_{11}T_{12} + T_{11}T_{12} \otimes (T_{21}T_{12} + T_{11}T_{22}) + T_{12}^2 \otimes T_{21}T_{22}. \end{aligned}$$

$$\boxed{W = \operatorname{span}_k\{T_{11}^2, T_{11}T_{12}, T_{12}^2\}.}$$

Clearly $f \in W$. Moreover,

$$\begin{aligned} m^{\#}(T_{11}^2) &= T_{11}^2 \otimes T_{11}^2 + 2T_{11}T_{12} \otimes T_{11}T_{21} + T_{12}^2 \otimes T_{21}^2, \\ m^{\#}(T_{12}^2) &= T_{11}^2 \otimes T_{12}^2 + 2T_{11}T_{12} \otimes T_{12}T_{22} + T_{12}^2 \otimes T_{22}^2. \end{aligned}$$

Together with the displayed formula for $m^{\#}(f)$, these identities show that $m^{\#}(W) \subseteq W \otimes \mathcal{O}(G)$. Thus W is a finite dimensional subrepresentation containing f. □

Exercise A.11. Let G be a connected algebraic group over a field k of characteristic zero, and let H be an algebraic subgroup of G. Show that H is normal in G if and only if, for every representation (V, r) of G and character χ of H, the eigenspace V_χ is stable under G.

Proof. Suppose first that H is normal in G. We will prove, in comodule language, that

$$\rho_r(V_\chi) \subseteq V_\chi \otimes_k \mathcal{O}(G). \tag{1}$$

It is first enough to prove that every character of H is invariant under conjugation by G. Consider the conjugation morphism

$$c : H \times G \longrightarrow H, \qquad (h, g) \longmapsto g^{-1}hg,$$

and put $a = \chi^{\#}(T) \in \mathcal{O}(H)$. Its comorphism $c^{\#} : \mathcal{O}(H) \to \mathcal{O}(H) \otimes \mathcal{O}(G)$ is the $\mathcal{O}(G)$-comodule structure on $\mathcal{O}(H)$. As in the finite dimensional subcomodule construction, choose

a finite dimensional subcomodule U containing a, with a k-basis $u_1, \dots, u_N$, and write

$$c^{\#}(a) = \sum_{i=1}^{N} u_i \otimes b_i, \qquad b_i \in \mathcal{O}(G).$$

The coefficients b_i define a morphism $\varphi : G_{\bar{k}} \to U_{\bar{k}} \simeq \mathbb{A}^N_{\bar{k}}$. For $g \in G(\bar{k})$, the corresponding specialization is

$$c^{\#}_g(a) = \sum_{i=1}^{N} u_i \, b_i(g) \quad \text{in } \mathcal{O}(H)_{\bar{k}}$$

is group-like: it is the element corresponding to the character $h \mapsto \chi(g^{-1}hg)$. Distinct group-like elements are linearly independent. Therefore only finitely many group-like elements can belong to $U_{\bar{k}}$. Thus $\varphi\big(G(\bar{k})\big)$ is finite. Since G is connected, it is geometrically irreducible; in characteristic zero $G_{\bar{k}}$ is reduced, so $G(\bar{k})$ is Zariski dense in $G_{\bar{k}}$. It follows that φ is constant. At the identity it has value a; therefore $c^{\#}(a) = a \otimes 1$. Hence the comorphisms of $\chi \circ c$ and $\chi \circ \mathrm{pr}_H$ agree on T, and so

$$\chi \circ c = \chi \circ \mathrm{pr}_H, \qquad \text{i.e.} \qquad \chi_R(g^{-1}hg) = \chi_R(h) \quad (R \in \mathrm{Alg}_k).$$

Now let $v \in V_\chi$, $R \in \mathrm{Alg}_k$, $g \in G(R)$, and $h \in H(R)$. Normality gives $g^{-1}hg \in H(R)$, and hence

$$h(gv) = g\big((g^{-1}hg)v\big) = gv\,\chi_R(g^{-1}hg) = gv\,\chi_R(h). \tag{2}$$

Write $\rho_r(v) = \sum_i v_i \otimes f_i$ with the $f_i \in \mathcal{O}(G)$ linearly independent. Applying (2) to the universal points of H and G[13]gives

$$\sum_i \rho_{r|_H}(v_i) \otimes f_i = \sum_i (v_i \otimes \chi^{\#}(T)) \otimes f_i \quad \text{in } V \otimes \mathcal{O}(H) \otimes \mathcal{O}(G).$$

Since the f_i are linearly independent, $\rho_{r|_H}(v_i) = v_i \otimes \chi^{\#}(T)$ for every i. Thus $v_i \in V_\chi$, proving (1).

Conversely, assume that V_χ is G-stable for every representation of G and every character χ of H. By Chevalley's theorem, choose a finite dimensional representation (V, r) and a one dimensional subspace $L \subseteq V$ with $\mathrm{Stab}_G(L) = H$. The action of H on L is a one dimensional representation, hence a character

$$\chi : H \longrightarrow \mathrm{GL}(L) \simeq \mathbb{G}_m.$$

[13]It just means taking $R = \mathcal{O}(H) \otimes \mathcal{O}(G)$ and $g \in G(R)$ corresponding to $\mathcal{O}(G) \to \mathcal{O}(H) \otimes \mathcal{O}(G) : \alpha \mapsto 1 \otimes \alpha$ and $h \in H(R)$ corresponding to $\mathcal{O}(H) \to \mathcal{O}(H) \otimes \mathcal{O}(G) : \beta \mapsto \beta \otimes 1$.

Write $L = kv$. The restricted coaction is

$$\rho_{r|_H}(v) = v \otimes \chi^{\#}(T),$$

so $v \in V_\chi$.

By the assumed G-stability of V_χ, for all $R \in \mathrm{Alg}_k$, $g \in G(R)$ and $h \in H(R)$,

$$r_R(h)r_R(g)(v \otimes 1) = r_R(g)(v \otimes 1)\chi_R(h).$$

Applying $r_R(g^{-1})$ gives

$$r_R(g^{-1}hg)(v \otimes 1) = (v \otimes 1)\chi_R(h).$$

Because $\chi_R(h) \in R^\times$, this says that $g^{-1}hg$ stabilizes $L \otimes R$. As $\mathrm{Stab}_G(L) = H$, we have $g^{-1}hg \in H(R)$. Thus $H(R)$ is normal in $G(R)$ for every R, and H is normal in G. The connectedness assumption was used in the forward implication. □

Exercise A.12. Find a finite dimension representation V of GL_n and a 1 dimensional subspace W of V such that $\mathrm{Stab}_G(W) = \mathrm{SL}_n$.

Proof. Consider

$$V = ke \oplus \bigwedge^n k^n,$$

with GL_n acting trivially on the first summand and by det on the second. This gives a representation $r : \mathrm{GL}_n \to \mathrm{GL}(V)$. Let $\omega = e_1 \wedge \cdots \wedge e_n$ be the standard volume vector and take

$$W = k(e + \omega) \subseteq ke \oplus \bigwedge^n k^n.$$

For a k-algebra R, write $g = (g_{ij}) \in \mathrm{GL}_n(R)$. Then

$$r_R(g)(e + \omega) = e + \det(g)\omega.$$

This generates the same rank-one summand as $e + \omega$ exactly when $\det(g) = 1$ (compare the first coordinates). Therefore the stabilizer functor is represented by

$$\mathcal{O}(\mathrm{GL}_n)/(\det -1) = \mathcal{O}(\mathrm{SL}_n),$$

so $\mathrm{Stab}_G(W) = \mathrm{SL}_n$. □

A.4 Week 4-5

Exercise A.13. Show that all finite dimensional representations of $\mathbb{G}_m \times \mathbb{G}_m$ are direct sum of eigenspaces.

Solution. Let $r : \mathbb{G}_m \times \mathbb{G}_m \to \mathrm{GL}_V$ be a representation, where V is a finite dimensional k-vector space. We want to show that

$$V = \bigoplus_{\chi:\mathbb{G}_m\times\mathbb{G}_m\to\mathbb{G}_m} V_\chi.$$

Recall that

$$V_\chi = \{v \in V \mid \rho_r(v) = v \otimes \chi^\#(T)\}.$$

For every character χ, the induced map is

$$\chi^\# : k[T, T^{-1}] \longrightarrow k[x, x^{-1}, y, y^{-1}].$$

Since T must map to a unit, initially $\chi^\#(T) = cx^m y^n$ for some $c \in k^\times$ and $m, n \in \mathbb{Z}$. As $\chi(e) = e$, evaluating at the identity gives $c = 1$. Denote by $\chi_{m,n}$ the character with $\chi^\#_{m,n}(T) = x^m y^n$.

Write the coaction as

$$\rho_r : V \longrightarrow V \otimes \mathcal{O}(\mathbb{G}_m \times \mathbb{G}_m), \qquad \rho_r(v) = \sum_{m,n\in\mathbb{Z}} v_{m,n} \otimes x^m y^n,$$

where the sum is finite. We claim that $v_{m,n} \in V_{\chi_{m,n}}$. The comodule identity says that

$$(\rho_r \otimes \mathrm{id}) \circ \rho_r = (\mathrm{id} \otimes m^\#) \circ \rho_r.$$

Recall that $m^\#(f)(g_1, g_2) = f(g_1 g_2)$. In our situation,

$$m^\#(x^m y^n)((a, b), (c, d)) = (ac)^m (bd)^n,$$

so $m^\#(x^m y^n) = x^m y^n \otimes x^m y^n$. Therefore

$$\sum_{m,n} \rho_r(v_{m,n}) \otimes x^m y^n = \sum_{m,n} v_{m,n} \otimes x^m y^n \otimes x^m y^n.$$

The Laurent monomials $\{x^m y^n\}_{m,n\in\mathbb{Z}}$ are k-linearly independent. Hence

$$\rho_r(v_{m,n}) = v_{m,n} \otimes x^m y^n, \qquad v_{m,n} \in V_{\chi_{m,n}}.$$

Thus $V = \sum_{m,n} V_{\chi_{m,n}}$. A sum of character eigenspaces is direct, so

$$V = \bigoplus_{m,n\in\mathbb{Z}} V_{\chi_{m,n}}.$$

Only finitely many summands are nonzero because $\dim_k V < \infty$. □

Exercise A.14. Let $G = \mathrm{SL}_2$ and let B be the closed subgroup of G of the upper triangular matrices. Let U be the closed subgroup of the unipotent matrices of B.

1. Show that every character of $U \simeq \mathbb{G}_a$ is trivial.
2. Let $G = \mathrm{SL}_2$ and let B be the closed subgroup of G of the upper triangular matrices. Deduce the characters of B over k.
3. Consider the regular representation $\mathcal{O}(G)$ of G. Consider $\mathcal{O}(G)$ as a representation of U. Compute the eigenspace $\mathcal{O}(G)^U$ of U in $\mathcal{O}(G)$.
4. Determine the characters of B that occurs in the regular representation of G. Deduce the possible characters of B that occurs in some finite dimensional representation of G.

Solution. **(a)** Let $\chi : \mathbb{G}_a \to \mathbb{G}_m$ be a character. Its comorphism is

$$\chi^{\#} : k[T, T^{-1}] \longrightarrow k[x].$$

Since $\chi^{\#}(T) \in k[x]^\times = k^\times$ and

$$\chi^{\#}(T)(e_{\mathbb{G}_a}) = T(\chi(e_{\mathbb{G}_a})) = T(e_{\mathbb{G}_m}) = 1,$$

we have $\chi^{\#}(T) = 1$. Thus χ factors through $\mathbb{G}_a \to \operatorname{Spec} k \xrightarrow{e} \mathbb{G}_m$ and is trivial.

(b) We have

$$B = \operatorname{Spec}\big(k[x, y, z]/(xy - 1)\big), \qquad \begin{pmatrix} x & z \\ 0 & y \end{pmatrix}, \quad xy = 1.$$

A character $\chi : B \to \mathbb{G}_m$ corresponds to

$$\chi^{\#} : k[T, T^{-1}] \longrightarrow k[x, y, z]/(xy - 1).$$

The element $\chi^{\#}(T)$ must be a unit. Combining this with $\chi^{\#}(T)(e_B) = T(e_{\mathbb{G}_m}) = 1$, we

obtain

$$\chi^{\#}(T) = \bar{x}^{\,n} \quad (n \in \mathbb{Z}), \qquad \bar{x} = \bar{y}^{-1}.$$

Geometrically, every character has the form

$$\chi_n : \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} \longmapsto a^n, \qquad n \in \mathbb{Z}.$$

(c) Let $\iota : U \hookrightarrow G$ be the inclusion and let $r : U \xrightarrow{\iota} G \to \mathrm{GL}_{\mathcal{O}(G)}$ be the restriction of the regular representation. Its coaction is

$$\rho_r = (\mathrm{id} \otimes \iota^{\#}) \circ m^{\#} : \mathcal{O}(G) \longrightarrow \mathcal{O}(G) \otimes \mathcal{O}(U).$$

Thus

$$\mathcal{O}(G)^U = \{v \in \mathcal{O}(G) \mid \rho_r(v) = v \otimes 1\}.$$

For $\alpha \in G(k)$, $\beta \in U(k)$, and $v \in \mathcal{O}(G)^U$,

$$v(\alpha) = (v \otimes 1)(\alpha, \beta) = \rho_r(v)(\alpha, \beta) = m^{\#}(v)(\alpha, \beta) = v(\alpha\beta).$$

Consequently, for every $\lambda \in k$,

$$v\begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix} = v\begin{pmatrix} t_{11} & t_{11}\lambda + t_{12} \\ t_{21} & t_{21}\lambda + t_{22} \end{pmatrix}.$$

We claim that

$$\mathcal{O}(G)^U = k[\bar{T}_{11}, \bar{T}_{21}] \subseteq \mathcal{O}(G) = k[T_{ij}]/(\det(T_{ij}) - 1),$$

the subring generated by $\bar{T}_{11}$ and $\bar{T}_{21}$.

First,

$$\mathcal{O}(G)_{\bar{T}_{11}} = k[\bar{T}_{11}, \bar{T}_{11}^{-1}, \bar{T}_{12}, \bar{T}_{21}].$$

View v as an element of this ring; on $D(\bar{T}_{11})$, this means

$$v(t_{11}, t_{12}, t_{21}) = v\left(t_{11}, t_{12}, t_{21}, \frac{t_{12}t_{21} + 1}{t_{11}}\right).$$

Then

$$v(t_{11}, t_{12}, t_{21}) = v(t_{11}, t_{12} + \lambda t_{11}, t_{21}) \quad \text{for every } \lambda \in k.$$

Hence $v \in k[\bar{T}_{11}, \bar{T}_{11}^{-1}, \bar{T}_{21}]$. Similarly, view v as an element of

$$\mathcal{O}(G)_{\bar{T}_{21}} = k[\bar{T}_{21}, \bar{T}_{21}^{-1}, \bar{T}_{11}, \bar{T}_{22}].$$

Here

$$v(t_{11}, t_{21}, t_{22}) = v\left(t_{11}, \frac{t_{11}t_{22} - 1}{t_{21}}, t_{21}, t_{22}\right),$$

and

$$v(t_{11}, t_{21}, t_{22}) = v(t_{11}, t_{21}, t_{21}\lambda + t_{22}).$$

Thus $v \in k[\bar{T}_{11}, \bar{T}_{21}, \bar{T}_{21}^{-1}]$. Therefore

$$\begin{aligned} v &\in k[\bar{T}_{11}, \bar{T}_{21}, \bar{T}_{21}^{-1}] \cap k[\bar{T}_{11}, \bar{T}_{21}, \bar{T}_{11}^{-1}] \cap \mathcal{O}(G) \\ &= k[\bar{T}_{11}, \bar{T}_{21}], \end{aligned}$$

which proves the claim.

(d) First, let $v \in \mathcal{O}(G)_{\chi_n}$. For $\begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix} \in G(k)$ and $\begin{pmatrix} \lambda & \mu \\ 0 & \lambda^{-1} \end{pmatrix} \in B(k)$, the eigenspace condition gives

$$\begin{aligned} \lambda^n v\begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix} &= (v \otimes \chi_n^{\#}(T))\left(\begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix}, \begin{pmatrix} \lambda & \mu \\ 0 & \lambda^{-1} \end{pmatrix}\right) \\ &= \rho_r(v)\left(\begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix}, \begin{pmatrix} \lambda & \mu \\ 0 & \lambda^{-1} \end{pmatrix}\right) \\ &= m^{\#}(v)\left(\begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix}, \begin{pmatrix} \lambda & \mu \\ 0 & \lambda^{-1} \end{pmatrix}\right) \\ &= v\begin{pmatrix} t_{11}\lambda & t_{11}\mu + t_{12}\lambda^{-1} \\ t_{21}\lambda & t_{21}\mu + t_{22}\lambda^{-1} \end{pmatrix}. \end{aligned}$$

The same argument as above gives $v \in k[\bar{T}_{11}, \bar{T}_{21}]$. Analyzing the dependence on λ, v must be homogeneous of degree n. Thus

$$\mathcal{O}(G)_{\chi_n} = \begin{cases} 0, & n < 0, \\ \bigoplus\limits_{\substack{p,q \geq 0 \\ p+q=n}} k\bar{T}_{11}^p \bar{T}_{21}^q, & n \geq 0. \end{cases}$$

Second, for $n \geq 0$, we can extend this eigenspace to a G-stable, finite dimensional k-subspace V of $\mathcal{O}(G)$. Then $B \to G \to \mathrm{GL}_V$ admits χ_n. Conversely, if $B \to G \to \mathrm{GL}_V$ admits χ_n for some finite dimensional representation (V, r), then (V, r) can be embedded into

$$\left(\mathcal{O}(G)^{\oplus \dim_k V}, (m^{\#})^{\oplus \dim_k V}\right).$$

By the preceding computation, we must have $n \geq 0$. Therefore χ_n occurs in some finite dimensional representation of G if and only if $n \geq 0$. □

Exercise A.15. Let p be a prime number.

1. Let G be an abstract finite group. For $x \in G$, show that there exists unique elements $x_{p'}, x_p \in G$ such that $x = x_{p'}x_p = x_p x_{p'}$ with the order of $x_{p'}$ coprime to p and the order of x_p a power of p.
2. Let $G = \mathrm{GL}_n(\mathbb{F}_p)$. For $x \in G$, show that $x = x_{p'}x_p$ is the Jordan decomposition of x with $x_{p'}$ the semisimple part and x_p the unipotent part.
3. Determine the Jordan decomposition for $\begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{F}_5)$.

Solution. **(a)** Write $\mathrm{ord}(x) = p^m n$, where $\gcd(p, n) = 1$. Choose $a, b \in \mathbb{Z}$ such that $p^m a + nb = 1$.

Existence. We have

$$x = x^{p^m a}x^{nb} = x^{nb}x^{p^m a}.$$

Moreover,

$$\mathrm{ord}(x^{p^m a}) = \frac{p^m n}{\gcd(p^m a, p^m n)} = n, \qquad \mathrm{ord}(x^{nb}) = \frac{p^m n}{\gcd(nb, p^m n)} = p^m,$$

since $\gcd(a, n) = 1$ and $\gcd(p^m, b) = 1$. Set $x_p = x^{nb}$ and $x_{p'} = x^{p^m a}$.

Uniqueness. Suppose that

$$x = x_p x_{p'} = x_{p'}x_p = y_p y_{p'} = y_{p'}y_p$$

is another such decomposition. Then

$$xy_p = y_p y_{p'} y_p = y_p y_p y_{p'} = y_p x,$$

and similarly $xy_{p'} = y_{p'}x$. Since $x_p = x^{nb}$ and $x_{p'} = x^{p^m a}$, these elements also commute with y_p and $y_{p'}$. Thus $x_p, x_{p'}, y_p, y_{p'}$, and x all commute. Since $\gcd(\mathrm{ord}(y_p), \mathrm{ord}(y_{p'})) = 1$,

$$\mathrm{ord}(x) = \mathrm{ord}(y_p)\,\mathrm{ord}(y_{p'}),$$

so $\operatorname{ord}(y_p) = \operatorname{ord}(x_p)$ and $\operatorname{ord}(y_{p'}) = \operatorname{ord}(x_{p'})$. Consequently,

$$\begin{aligned}\operatorname{ord}(y_p^{-1}x_p) \mid \operatorname{lcm}(\operatorname{ord}(y_p), \operatorname{ord}(x_p)) = \operatorname{ord}(x_p),\\ \operatorname{ord}(y_{p'}x_{p'}^{-1}) \mid \operatorname{ord}(x_{p'}).\end{aligned}$$

But $y_p^{-1}x_p = y_{p'}x_{p'}^{-1}$ by the decompositions. Its order divides both a power of p and an integer coprime to p, so

$$y_p^{-1}x_p = y_{p'}x_{p'}^{-1} = e.$$

Hence $y_p = x_p$ and $y_{p'} = x_{p'}$.

(b) The element x_p is unipotent, since in characteristic p,

$$(x_p - \mathrm{id})^{p^m} = x_p^{p^m} - \mathrm{id} = \mathrm{id} - \mathrm{id} = 0.$$

The element $x_{p'}$ satisfies $x_{p'}^r - \mathrm{id} = 0$ for some r with $\gcd(r, p) = 1$. Let $P(T) \in \mathbb{F}_p[T]$ be its minimal polynomial. Then $P(T) \mid T^r - 1$. The polynomial $T^r - 1$ has no multiple roots in $\overline{\mathbb{F}}_p$ because $\gcd(r, p) = 1$. Thus $P(T)$ has no multiple roots in $\overline{\mathbb{F}}_p$, so $x_{p'}$ is diagonalizable over $\overline{\mathbb{F}}_p$, i.e., $x_{p'}$ is semisimple.

(c) Put $x = \begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix}$. All the following matrix equalities are in $\mathbb{F}_5$:

$$\begin{aligned}x^2 &= \begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix} = \begin{pmatrix} 10 & 12 \\ 12 & 18 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 2 & 3 \end{pmatrix},\\ x^3 &= \begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} 0 & 2 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 6 & 11 \\ 6 & 15 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix},\\ x^4 &= \begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 6 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}.\end{aligned}$$

Then

$$x^5 = \begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 7 & 10 \\ 15 & 12 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix},$$

and hence

$$x^{20} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}^4 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Consider

$$x_p = x^{-4} = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}^{-1} = \begin{pmatrix} 3 & 4 \\ 4 & 4 \end{pmatrix}, \qquad x_{p'} = x^5 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

□

A.5 Week 6

Exercise A.16. Show that the semisimple elements (respectively unipotent elements) of $\mathrm{SL}_n(k)$ with $n \geq 2$, do not form subgroups.

Solution. We first consider $n = 2$.

Semisimple elements. Assume $k \neq \mathbb{F}_2, \mathbb{F}_3$, and choose $a \in k^\times \setminus \{1, -1\}$. Consider

$$A = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}, \qquad B = \begin{pmatrix} a^{-1} & a^{-1} \\ 0 & a \end{pmatrix}.$$

Since $a \neq a^{-1}$, both A and B are diagonalizable, hence semisimple elements of $\mathrm{SL}_2(k)$. However,

$$AB = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is not semisimple. Thus the semisimple elements do not form a subgroup.

Unipotent elements. Consider

$$U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad V = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Both U and V are unipotent, whereas

$$UV = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \det(UV - I_2) = -1.$$

Hence $UV - I_2$ is invertible, so UV is not unipotent. Therefore the unipotent elements do not form a subgroup either.

For $n \geq 3$, take the block diagonal matrices

$$A' = \begin{pmatrix} A & 0 \\ 0 & I_{n-2} \end{pmatrix}, \quad B' = \begin{pmatrix} B & 0 \\ 0 & I_{n-2} \end{pmatrix},$$
$$U' = \begin{pmatrix} U & 0 \\ 0 & I_{n-2} \end{pmatrix}, \quad V' = \begin{pmatrix} V & 0 \\ 0 & I_{n-2} \end{pmatrix}.$$

These give the same counterexamples in $\mathrm{SL}_n(k)$. □

Exercise A.17. Let k be an algebraically closed field of characteristic 0. Let $x \in \mathrm{GL}_n(k)$ be unipotent with $x \neq e$,

1. Show that x is of infinite order.
2. Show that the unique smallest closed subgroup G of GL_n over k with $x \in G(k)$ is isomorphic to $\mathbb{G}_a$.
3. Deduce that if H is a connected 1-dimensional closed subgroup of GL_n over k such that $H(k)$ contains an unipotent element $x \neq e$, then all elements in $H(k)$ are unipotent.
4. Let H be a closed subgroup of GL_n over k such that all elements in $H(k)$ are unipotent. Show that H is connected.

Solution. **(a)** Suppose that $x^r = e$ for some $r \in \mathbb{Z}_{>0}$, and let $p(T)$ be the minimal polynomial of x. Then

$$p(T) \mid T^r - 1.$$

On the other hand, since x is unipotent, $(x-e)^m = 0$ for some $m \in \mathbb{Z}_{>0}$, so

$$p(T) \mid (T-1)^m.$$

Since $\mathrm{char}(k) = 0$, the polynomial $T^r - 1$ has no repeated roots. Hence $p(T) = T - 1$, which gives $x = e$, a contradiction. Therefore x has infinite order.

(b) For the unipotent matrix x, put

$$\log x = \sum_{i=1}^{\infty} \frac{(-1)^{i+1}}{i}(x-e)^i.$$

This is in fact a finite sum. If $(x-e)^m = 0$, put

$$P(U) = \sum_{i=1}^{m-1} \frac{(-1)^{i+1}}{i}(U-1)^i \in k[U], \qquad \log x = P(x).$$

Then

$$\sum_{j=0}^{m-1} \frac{(\log x)^j}{j!} = \exp(\log x) = x.$$

Thus there is also a polynomial $Q(U) \in k[U]$ such that $x = Q(\log x)$.

Define $\phi : \mathbb{G}_a \longrightarrow \mathrm{GL}_{n,k}$ by the coordinate-ring homomorphism

$$\phi^{\#} : k[T_{ij}, \Delta^{-1}] \longrightarrow k[T], \qquad \Delta = \det(T_{ij}),$$
$$T_{ij} \longmapsto \sum_{\ell=0}^{m-1} \frac{((\log x)^{\ell})_{ij}}{\ell!} T^{\ell}.$$

In other words, $\phi(T) = \exp(T \log x)$.

Well-definedness of $\phi^{\#}$. It suffices to show that $\phi^{\#}(\Delta) = 1$. Since $\log x$ is nilpotent, there is a matrix $y \in \mathrm{GL}_n(k)$ for which $y(\log x)y^{-1}$ is strictly upper triangular. Consequently,

$$\begin{aligned} \phi^{\#}(\Delta) &= \det\big(\exp(T \log x)\big) \\ &= \det\big(\exp(Ty(\log x)y^{-1})\big) = 1, \end{aligned}$$

because the last exponential is upper triangular with all diagonal entries equal to 1.

Surjectivity of $\phi^{\#}$. Since $e \neq x = \exp(\log x)$, we have $\log x \neq 0$. Choose indices $p, q \in \{1, \ldots, n\}$ with $(\log x)_{pq} \neq 0$, and write $\mathcal{T} = (T_{ij})$. The logarithm polynomial applied to the generic matrix gives

$$\frac{\big(P(\mathcal{T})\big)_{pq}}{(\log x)_{pq}} \xrightarrow{\ \phi^{\#}\ } \frac{\big(P(\exp(T \log x))\big)_{pq}}{(\log x)_{pq}} = \frac{T(\log x)_{pq}}{(\log x)_{pq}} = T.$$

Therefore $\phi^{\#}$ is surjective, and ϕ is a closed immersion. The identity

$$\exp(S \log x) \exp(T \log x) = \exp((S + T) \log x)$$

shows that ϕ is a homomorphism of algebraic groups. Also, $\phi(1) = x$.

We now prove that its image is the smallest closed subgroup containing x. Let $H \hookrightarrow \mathrm{GL}_{n,k}$ be any closed subgroup with $x \in H(k)$. First, $\phi^{-1}(H) = \mathbb{G}_a$ set-theoretically. Indeed, for every $\ell \in \mathbb{Z}$,

$$\phi(\ell) = x^{\ell} \in H(k).$$

Thus $\phi^{-1}(H)$ contains the integer points of $\mathbb{G}_a$. These are infinitely many distinct points because $\mathrm{char}(k) = 0$. Since k is algebraically closed and $\phi^{-1}(H)$ is Zariski closed in $\mathbb{G}_a$, it must be all of $\mathbb{G}_a$ set-theoretically. The groups in question are smooth in characteristic 0, and ϕ therefore factors through H as a homomorphism of algebraic groups. It follows that $G = \phi(\mathbb{G}_a)$ is the unique smallest closed subgroup containing x, and $G \cong \mathbb{G}_a$.

(c) Under the smoothness condition, connectedness of an algebraic group over k is

equivalent to irreducibility. By (b), there is a closed immersion

$$\mathbb{G}_a \hookrightarrow H.$$

Both groups are irreducible and 1-dimensional, so this closed subgroup must be all of H. Thus $H \cong \mathbb{G}_a$ via the construction in (b). Every $h \in H(k)$ is consequently of the form $\exp(t \log x)$ for some $t \in k$. After the same conjugation as above, this is the exponential of a strictly upper triangular matrix, and hence is unipotent.

(d) Let $h \in H(k)$ with $h \neq e$. By (b), there is a closed immersion

$$\mathbb{G}_a \hookrightarrow H, \qquad 0 \longmapsto e, \quad 1 \longmapsto h.$$

Its image is connected, so h belongs to the identity component H^0. Therefore

$$H^0(k) = H(k).$$

Since k is algebraically closed, it follows that $H = H^0$. Hence H is connected. □

Exercise A.18. Let k be a field of characteristic 2. Let G (resp. μ_2, resp. α_2) be the kernel of relative Frobenius on SL_2 (resp. $\mathbb{G}_m$, resp. $\mathbb{G}_a$) over k.

1. Describe μ_2 and α_2 as algebraic groups.
2. Show that G is finite and connected but not smooth.
3. Show that there is an exact sequence

$$1 \longrightarrow \mu_2 \longrightarrow G \longrightarrow \alpha_2 \times \alpha_2 \longrightarrow 1.$$

Solution. **(a)** For every $R \in \mathrm{Alg}_k$, the relative Frobenius on $\mathbb{G}_m$ is given by

$$F_{\mathbb{G}_m/k}(R) : \mathbb{G}_m(R) \longrightarrow \mathbb{G}_m^{(2)}(R), \qquad r \longmapsto r^2.$$

On coordinate rings, this is the map

$$k[T, T^{-1}] \otimes_{k, F_k} k \longrightarrow k[T, T^{-1}],$$
$$T \otimes 1 \longmapsto T^2,$$

followed, for an R-point, by evaluation $T \mapsto r$. Thus

$$\mu_2(R) = \{r \in R^\times \mid r^2 = 1\},$$

with the group law induced by multiplication, and

$$\mu_2 = \operatorname{Spec} \frac{k[T]}{(T^2-1)} = \operatorname{Spec} \frac{k[T]}{((T+1)^2)}.$$

Similarly, the relative Frobenius $F_{\mathbb{G}_a/k}(R) : \mathbb{G}_a(R) \longrightarrow \mathbb{G}_a^{(2)}(R)$ sends $r \longmapsto r^2$, corresponding to

$$k[T] \otimes_{k,F_k} k \longrightarrow k[T], \qquad T \otimes 1 \longmapsto T^2.$$

Consequently, with the group law induced by addition,

$$\alpha_2(R) = \{r \in R \mid r^2 = 0\}, \qquad \alpha_2 = \operatorname{Spec} \frac{k[T]}{(T^2)}.$$

(b) For every $R \in \mathrm{Alg}_k$, the relative Frobenius on SL_2 sends (r_{ij}) to (r_{ij}^2). On coordinate rings, it sends

$$\begin{aligned}(k[T_{ij}]/(\Delta-1)) \otimes_{k,F_k} k &\longrightarrow k[T_{ij}]/(\Delta-1),\\ T_{ij} \otimes 1 &\longmapsto T_{ij}^2,\end{aligned}$$

where $\Delta = T_{11}T_{22} - T_{12}T_{21}$. Hence

$$G(R) = \{(r_{ij}) \in \mathrm{SL}_2(R) \mid r_{ij}^2 = \delta_{ij}\}.$$

It follows that

$$G \cong \operatorname{Spec} \frac{k[x,y,z,w]}{(xw-yz-1,\ x^2-1,\ y^2,\ z^2,\ w^2-1)}.$$

The coordinate ring is finite-dimensional over k, since all four generators satisfy the displayed quadratic relations. Its reduction is

$$G_{\mathrm{red}} \cong \operatorname{Spec} \frac{k[x,w]}{(x+1,w+1)} \cong \operatorname{Spec} k,$$

a single point. Therefore G is finite and connected. It is nonreduced, so it is not smooth.

(c) Consider $\varphi : G \longrightarrow \alpha_2 \times \alpha_2$ given on coordinate rings by

$$\begin{aligned}\varphi^{\#} : \frac{k[s,t]}{(s^2,t^2)} &\longrightarrow \frac{k[x,y,z,w]}{(xw-yz-1,\ (x+1)^2,\ y^2,\ z^2,\ (w+1)^2)},\\ s &\longmapsto xy,\\ t &\longmapsto xz.\end{aligned}$$

For every $R \in \mathrm{Alg}_k$, the corresponding map is

$$\varphi_R : G(R) \longrightarrow \alpha_2(R) \times \alpha_2(R), \qquad (r_{ij}) \longmapsto (r_{11}r_{12}, r_{11}r_{21}).$$

This is a homomorphism of algebraic groups. To verify this, let

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad M' = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \quad \text{belong to } G(R).$$

In particular,

$$a^2 = d^2 = (a')^2 = (d')^2 = 1, \qquad b^2 = c^2 = (b')^2 = (c')^2 = 0,$$

and $ad + bc = a'd' + b'c' = 1$. The two coordinates of $\varphi_R(MM')$ are

$$\begin{aligned} (aa' + bc')(ab' + bd') &= a^2a'b' + ab(a'd' + b'c') + b^2c'd' \\ &= a'b' + ab, \\ (aa' + bc')(ca' + dc') &= ac(a')^2 + a'c'(ad + bc) + bd(c')^2 \\ &= ac + a'c'. \end{aligned}$$

Thus $\varphi_R(MM') = \varphi_R(M) + \varphi_R(M')$, using $\mathrm{char}(k) = 2$.

The kernel. For every $R \in \mathrm{Alg}_k$,

$$\ker(\varphi_R) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \;\middle|\; \begin{array}{c} a, b, c, d \in R, \quad ad + bc = 1, \\ a^2 = d^2 = 1, \quad b^2 = c^2 = 0, \\ ab = ac = 0 \end{array} \right\}.$$

For such a matrix, the relations give

$$d = a^2d = a + abc = a, \qquad b = b(ad + bc) = 0, \qquad c = c(ad + bc) = 0.$$

Hence, functorially in R,

$$\ker(\varphi_R) = \left\{ \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \;\middle|\; a^2 = 1 \right\} \cong \mu_2(R).$$

The map $\mu_2 \hookrightarrow G$ is therefore the scalar-matrix inclusion $a \mapsto aI_2$.

Flatness and surjectivity. Put

$$A = \frac{k[s,t]}{(s^2, t^2)}.$$

We claim that, as an A-algebra through $\varphi^{\#}$,

$$\mathcal{O}(G) \cong \frac{A[u]}{(u^2+1)}.$$

Indeed, consider the homomorphism

$$\begin{aligned} \frac{k[s,t,u]}{(s^2,t^2,u^2+1)} &\longrightarrow \frac{k[x,y,z,w]}{(xw+yz+1,\ x^2+1,\ y^2,\ z^2,\ w^2+1)}, \\ s &\longmapsto xy, \\ t &\longmapsto xz, \\ u &\longmapsto x. \end{aligned}$$

Its inverse is given by

$$x \longmapsto u, \qquad y \longmapsto us, \qquad z \longmapsto ut, \qquad w \longmapsto u(st+1).$$

These maps are inverse to one another, giving the claimed isomorphism. In particular, $\mathcal{O}(G)$ is free over A with basis $1, u$, so φ is flat. Moreover, the A-algebra homomorphisms

$$A \longrightarrow \frac{A[u]}{(u^2+1)} \longrightarrow A, \qquad u \longmapsto 1$$

have composite id_A. They give a section

$$\alpha_2 \times \alpha_2 \longrightarrow G \longrightarrow \alpha_2 \times \alpha_2,$$

so φ is surjective. Together with the kernel calculation, this proves that

$$1 \longrightarrow \mu_2 \longrightarrow G \xrightarrow{\varphi} \alpha_2 \times \alpha_2 \longrightarrow 1$$

is exact. □

A.6 Week 7

Exercise A.19. (a) Describe Deligne torus $\mathbb{S} := \mathrm{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m$ as a linear algebraic group.

(b) Show that there is an isomorphism $\mathbb{S}_{\mathbb{C}} \simeq \mathbb{G}_m \times \mathbb{G}_m$ of algebraic groups.

(c) Let V be a finite dimensional $\mathbb{R}$-vector space. A real Hodge structure on V is a

decomposition $V_{\mathbb{C}} = \sum_{p,q\in\mathbb{Z}} V^{p,q}$ of sub-$\mathbb{C}$-vector spaces such that $\overline{V^{p,q}} = V^{q,p}$ for all $p, q \in \mathbb{Z}$. Show that there is a natural bijection

$$\{\text{real Hodge structures on } V\} \simeq \{\text{representations: } \mathbb{S} \longrightarrow \mathrm{GL}_V\}.$$

Solution. **(a)** For every $\mathbb{R}$-algebra R, we have

$$\begin{aligned}\big(\mathrm{Res}_{\mathbb{C}/\mathbb{R}}\,\mathbb{G}_m\big)(R) &\simeq \mathrm{Hom}_{\mathbb{C}\text{-alg}}\left(\frac{\mathbb{C}[x_1,x_2]}{(x_1x_2-1)}, R\otimes_{\mathbb{R}}\mathbb{C}\right)\\ &= \left\{(x_1,x_2)\in(R\otimes_{\mathbb{R}}\mathbb{C})^2 \;\middle|\; x_1x_2-1=0\right\}.\end{aligned}$$

Write

$$x_1 = r_{11}\otimes 1 + r_{12}\otimes i, \qquad x_2 = r_{21}\otimes 1 + r_{22}\otimes i.$$

Then

$$x_1x_2 - 1 = (r_{11}r_{21} - r_{12}r_{22} - 1)\otimes 1 + (r_{11}r_{22} + r_{12}r_{21})\otimes i.$$

Consequently, the above functor is naturally isomorphic to

$$\left\{(r_{11},r_{12},r_{21},r_{22})\in R^4 \;\middle|\; \begin{aligned} r_{11}r_{21} - r_{12}r_{22} - 1 &= 0,\\ r_{11}r_{22} + r_{12}r_{21} &= 0\end{aligned}\right\}.$$

Thus

$$\mathbb{S} = \mathrm{Spm}\,\frac{\mathbb{R}[x,y,z,w]}{(xz-yw-1,\ xw+yz)}.$$

(b) Base change gives

$$\mathbb{S}_{\mathbb{C}} = \mathrm{Spm}\,\frac{\mathbb{C}[x,y,z,w]}{(xz-yw-1,\ xw+yz)}.$$

Consider the homomorphism

$$\begin{aligned}\frac{\mathbb{C}[x,y,z,w]}{(xy-1,\ zw-1)} &\longrightarrow \frac{\mathbb{C}[x,y,z,w]}{(xz-yw-1,\ xw+yz)},\\ x &\longmapsto x+iy, & y &\longmapsto z+iw,\\ z &\longmapsto x-iy, & w &\longmapsto z-iw.\end{aligned}$$

The inverse can be constructed easily, so this induces an isomorphism of schemes

$$F : \mathbb{S}_{\mathbb{C}} \longrightarrow \mathbb{G}_m\times\mathbb{G}_m.$$

We check that it respects the group structures. The multiplication

$$(x_1 \otimes 1 + x_2 \otimes i)(y_1 \otimes 1 + y_2 \otimes i) = (x_1y_1 - x_2y_2) \otimes 1 + (x_1y_2 + x_2y_1) \otimes i$$

gives the group law on $\mathrm{Res}_{\mathbb{C}/\mathbb{R}}\,\mathbb{G}_m$. For a $\mathbb{C}$-algebra R, the induced map is

$$F_R(x, y, z, w) = (x + iy,\ z + iw,\ x - iy,\ z - iw),$$

where the target is written in the coordinates $\mathbb{C}[x, y, z, w]/(xy - 1, zw - 1)$. For two points (x, y, z, w) and (x', y', z', w'), multiplication in $\mathbb{S}_{\mathbb{C}}(R)$ gives

$$(xx' - yy',\ xy' + yx',\ zz' - ww',\ zw' + wz').$$

Applying F_R gives

$$\begin{pmatrix} xx' - yy' + i(xy' + yx') \\ zz' - ww' + i(zw' + wz') \\ xx' - yy' - i(xy' + yx') \\ zz' - ww' - i(zw' + wz') \end{pmatrix}.$$

On the other hand, first applying F_R to each point and then multiplying in $(\mathbb{G}_m \times \mathbb{G}_m)(R)$ gives

$$\begin{pmatrix} (x + iy)(x' + iy') \\ (z + iw)(z' + iw') \\ (x - iy)(x' - iy') \\ (z - iw)(z' - iw') \end{pmatrix},$$

which is the same expression. Thus the diagram

$$\begin{array}{ccc} \mathbb{S}_{\mathbb{C}} \times \mathbb{S}_{\mathbb{C}} & \xrightarrow{\quad m_{\mathbb{S}_{\mathbb{C}}} \quad} & \mathbb{S}_{\mathbb{C}} \\ {\scriptstyle F\times F}\downarrow & & \downarrow{\scriptstyle F} \\ (\mathbb{G}_m \times \mathbb{G}_m) \times (\mathbb{G}_m \times \mathbb{G}_m) & \xrightarrow[m_{\mathbb{G}_m \times \mathbb{G}_m}]{} & \mathbb{G}_m \times \mathbb{G}_m \end{array}$$

commutes on R-points for every $\mathbb{C}$-algebra R. Hence F is an isomorphism of linear algebraic groups, not merely of schemes.

(c) We construct maps

$$\{\text{real Hodge structures on } V\} \underset{\Psi}{\overset{\Phi}{\rightleftarrows}} \{\mathbb{S} \longrightarrow \mathrm{GL}_V\}$$

and show that they are inverse to one another.

Construction of Φ. Suppose that

$$V_{\mathbb{C}} = \bigoplus_{p,q\in\mathbb{Z}} V^{p,q}, \qquad V^{p,q} = \overline{V^{q,p}}.$$

For a $\mathbb{C}$-algebra R, consider the action

$$\begin{aligned}(\mathbb{G}_m \times \mathbb{G}_m)(R) \times V_{\mathbb{C}}(R) &\longrightarrow V_{\mathbb{C}}(R),\\ \left((r_1, r_2), \sum_{p,q} v_{p,q}\right) &\longmapsto \sum_{p,q} v_{p,q} r_1^p r_2^q,\end{aligned}$$

where $r_1, r_2 \in R^\times$ and $v_{p,q} \in V^{p,q} \otimes_{\mathbb{C}} R$. Recall that the isomorphism in (b), when expressed using the two unit coordinates, sends

$$(a, b, c, d) \longmapsto (a + bi,\ a - bi), \qquad ac - bd = 1, \quad ad + bc = 0.$$

Inspired by this, for an $\mathbb{R}$-algebra R we consider the following action of $\mathbb{S}(R)$ on $V(R)$. If $v \in V$ and $v_{\mathbb{C}} = \sum_{p,q} v_{p,q}$ with $v_{p,q} \in V^{p,q}$, set

$$(a, b, c, d) \cdot (v \otimes 1) = \sum_{p,q} (v_{p,q} \otimes 1)(a \otimes 1 + b \otimes i)^p (a \otimes 1 - b \otimes i)^q,$$

and extend R-linearly.

We verify that this is well-defined. The displayed sum initially belongs to $V \otimes_{\mathbb{R}} R \otimes_{\mathbb{R}} \mathbb{C}$; we must check that it lies in $V \otimes_{\mathbb{R}} R$. Since $v \in V$, we have $v = \overline{v}$, so

$$\sum_{p,q} v_{p,q} = \sum_{p,q} \overline{v_{p,q}}.$$

As $\overline{V^{p,q}} = V^{q,p}$, this implies

$$v_{p,q} = \overline{v_{q,p}} \qquad \text{for all } p, q \in \mathbb{Z}.$$

Therefore

$$\begin{aligned}
&\overline{\sum_{p,q}(v_{p,q}\otimes 1)(a\otimes 1+b\otimes i)^p(a\otimes 1-b\otimes i)^q}\\
&\quad=\sum_{p,q}(\overline{v_{p,q}}\otimes 1)(a\otimes 1-b\otimes i)^p(a\otimes 1+b\otimes i)^q\\
&\quad=\sum_{p,q}(v_{q,p}\otimes 1)(a\otimes 1-b\otimes i)^p(a\otimes 1+b\otimes i)^q\\
&\quad=\sum_{p,q}(v_{p,q}\otimes 1)(a\otimes 1+b\otimes i)^p(a\otimes 1-b\otimes i)^q.
\end{aligned}$$

Thus the required inclusion holds. The R-linearity is obvious, and the group action is inherited from the previously constructed action

$$\mathbb{S}_{\mathbb{C}}\times V_{\mathbb{C}}\simeq(\mathbb{G}_m\times\mathbb{G}_m)\times V_{\mathbb{C}}\longrightarrow V_{\mathbb{C}}.$$

This defines Φ.

Construction of Ψ. Given a representation $r:\mathbb{S}\longrightarrow \mathrm{GL}_V$, we obtain a representation

$$\mathbb{G}_m\times\mathbb{G}_m\simeq\mathbb{S}_{\mathbb{C}}\xrightarrow{r_{\mathbb{C}}}\mathrm{GL}_{V_{\mathbb{C}}}.$$

As proved before, this induces a decomposition

$$V_{\mathbb{C}}=\bigoplus_{p,q\in\mathbb{Z}}V^{p,q},$$

where $V^{p,q}$ is the eigenspace corresponding to the character

$$\begin{aligned}
\chi^{p,q}:\mathbb{G}_m\times\mathbb{G}_m&\longrightarrow\mathbb{G}_m,\\
(z_1,z_2)&\longmapsto z_1^pz_2^q.
\end{aligned}$$

We need to show that $V^{p,q}=\overline{V^{q,p}}$. Write

$$v=v_1\otimes 1+v_2\otimes i\in V^{p,q}\subseteq V\otimes_{\mathbb{R}}\mathbb{C},\qquad v_1,v_2\in V.$$

Then, for $z_1,z_2\in\mathbb{C}^\times$,

$$(z_1,z_2)v=vz_1^pz_2^q.$$

In particular, we consider $z_1=z=a+bi$ and $z_2=\overline{z}=a-bi$ with $a,b\in\mathbb{R}$.

Recall that

$$\begin{aligned} \mathbb{S}_{\mathbb{C}} &\xrightarrow{\sim} \mathbb{G}_m \times \mathbb{G}_m, \\ (a,b,c,d) &\longmapsto (a+bi,\ a-bi), \end{aligned} \qquad ac-bd=1, \quad ad+bc=0.$$

For $(a,b,c,d) \in \mathbb{S}(\mathbb{R})$ and $w \in V$, we have

$$(z,\overline{z})(w \otimes 1) = \big(r(a,b,c,d)w\big) \otimes 1 \in V \otimes_{\mathbb{R}} \mathbb{R}.$$

Consequently,

$$\begin{aligned} (z,\overline{z})v &= (z,\overline{z})(v_1 \otimes 1) + (z,\overline{z})(v_2 \otimes i) \\ &= v z^p \overline{z}^q, \end{aligned}$$

where the first summand lies in $V \otimes_{\mathbb{R}} \mathbb{R}\{1\}$ and the second in $V \otimes_{\mathbb{R}} \mathbb{R}\{i\}$. Taking complex conjugates gives

$$\begin{aligned} \overline{v}\,\overline{z}^p z^q &= \overline{v z^p \overline{z}^q} \\ &= \overline{(z,\overline{z})(v_1 \otimes 1) + (z,\overline{z})(v_2 \otimes i)} \\ &= (z,\overline{z})(v_1 \otimes 1) - (z,\overline{z})(v_2 \otimes i) \\ &= (z,\overline{z})(v_1 \otimes 1 - v_2 \otimes i) \\ &= (z,\overline{z})\overline{v}. \end{aligned}$$

Write $\overline{v} = \sum_{r,s} w_{r,s}$ with $w_{r,s} \in V^{r,s}$. Then

$$(z,\overline{z})\overline{v} = \sum_{r,s} z^r \overline{z}^s w_{r,s} = \overline{v}\, z^q \overline{z}^p.$$

Hence $\overline{v}$ can only belong to $V^{q,p}$. Thus

$$\overline{V^{p,q}} \subseteq V^{q,p}.$$

Interchanging p and q yields

$$V^{p,q} \subseteq \overline{V^{q,p}} \subseteq V^{p,q},$$

so $V^{p,q} = \overline{V^{q,p}}$, as required. This defines Ψ.

By construction, Φ and Ψ are inverse: in both constructions, $V^{p,q}$ is precisely the eigenspace of $\chi^{p,q}$. They therefore give the desired natural bijection. $\square$

Exercise A.20. Let $\mathbb{G}_m \subset \mathrm{GL}_n$ be the normal subgroup consisting of scalar matrices. Let $\mathrm{PGL}_n := \mathrm{GL}_n / \mathbb{G}_m$ be the quotient of GL_n by $\mathbb{G}_m$.

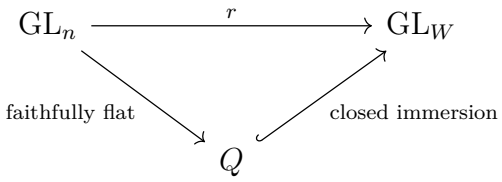
(a) Describe PGL_2 as a linear algebraic group.

(b) Show that the quotient map $\mathrm{GL}_n \longrightarrow \mathrm{PGL}_n$ restricts to a quotient map $\mathrm{SL}_n \longrightarrow \mathrm{PGL}_n$ with kernel μ_n (which is kernel of the multiplication by n homomorphism on $\mathbb{G}_m$).

Solution. **(a)** Consider $W = M_{n\times n}(k)$ and the representation

$$r : \mathrm{GL}_n \longrightarrow \mathrm{GL}_W, \qquad r(R)(g) : X \longmapsto gXg^{-1},$$

for $R \in \mathrm{Alg}_k$, $g \in \mathrm{GL}_n(R)$ and $X \in M_{n\times n}(R)$. A natural fact is that $\ker r = \mathbb{G}_m$. Factoring r through its image gives



and we identify Q with $\mathrm{GL}_n / \mathbb{G}_m = \mathrm{PGL}_n$.

In particular, when $n = 2$, using the basis $E_{11}, E_{12}, E_{21}, E_{22}$ of W, we have

$$r : \mathrm{GL}_2 \longrightarrow \mathrm{GL}_W \simeq \mathrm{GL}_4,$$
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \longmapsto \frac{1}{\Delta} \begin{pmatrix} ad & -ac & bd & -bc \\ -ab & a^2 & -b^2 & ab \\ cd & -c^2 & d^2 & -cd \\ -bc & ac & -bd & ad \end{pmatrix}, \qquad \Delta = ad - bc.$$

Equivalently, writing $D = \det(t_{ij})$ for the determinant on GL_4, the comorphism is

$$r^{\#} : k[t_{11}, t_{12}, \ldots, t_{44}, D^{-1}] \longrightarrow k[x, y, z, w, \Delta^{-1}],$$
$$t_{ij} \longmapsto f_{ij}(x, y, z, w),$$

where the f_{ij} are the entries of the matrix above, with $(a, b, c, d) = (x, y, z, w)$ and $\Delta =$

$xw - yz$. The ring $\mathcal{O}(\mathrm{GL}_4)/\ker r^\#$ is integral. Consequently,

$$\begin{aligned}\mathcal{O}(\mathrm{PGL}_2) &\simeq \mathcal{O}(\mathrm{GL}_4)/\ker r^\# \simeq \operatorname{im} r^\# \\ &\simeq k[x^2, y^2, z^2, w^2, xy, xz, xw, yz, yw, zw, \Delta^{-1}],\end{aligned}$$

that is, the degree 2 Veronese algebra of $k[x, y, z, w, \Delta^{-1}]$.

(b) The composite

$$\mathrm{SL}_n \longrightarrow \mathrm{GL}_n \longrightarrow \mathrm{PGL}_n \hookrightarrow \mathrm{GL}_W$$

obviously has kernel μ_n. To prove that $\mathrm{SL}_n \longrightarrow \mathrm{PGL}_n$ is a quotient map, it suffices to prove that it is faithfully flat. Recall Proposition 3.16: if $q : G \longrightarrow Q$ is a homomorphism of linear algebraic groups with $G = \operatorname{Spm} B$ and $Q = \operatorname{Spm} A$, then q is a quotient map if and only if $q^\# : A \longrightarrow B$ is injective. Thus it suffices to prove that

$$\mathcal{O}(\mathrm{PGL}_n) \longrightarrow \mathcal{O}(\mathrm{SL}_n)$$

is injective.

Put

$$A = \mathcal{O}(\mathrm{GL}_n) = k[X_{11}, \ldots, X_{nn}, \Delta^{-1}], \qquad \Delta = \det(X_{ij}), \qquad B = \frac{A[T]}{(T^n - \Delta)}.$$

Consider

$$\mathcal{O}(\mathrm{GL}_W) \longrightarrow \mathcal{O}(\mathrm{GL}_n) \underset{\varphi_2^\#}{\overset{\varphi_1^\#}{\rightrightarrows}} B, \tag{$*$}$$

where

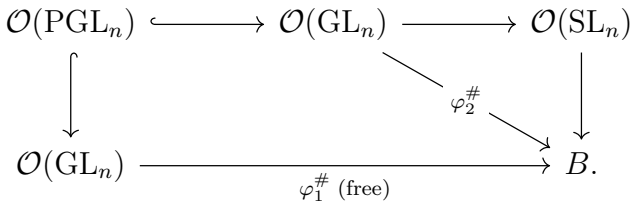
$$\varphi_1^\#(X_{ij}) = X_{ij}, \qquad \varphi_2^\#(X_{ij}) = T^{-1}X_{ij}.$$

On points, these correspond to

$$\begin{aligned}&\operatorname{Spm} B \rightrightarrows \mathrm{GL}_n \longrightarrow \mathrm{GL}_W, \\ &((a_{ij}), \alpha) \longmapsto (a_{ij}), \qquad\qquad \alpha^n = \det(a_{ij}). \\ &((a_{ij}), \alpha) \longmapsto \alpha^{-1}(a_{ij}),\end{aligned}$$

Both composites give the same conjugation action, so the first map in $(*)$ equalizes the two arrows.

Also, $\varphi_2^{\#}$ factors through $\mathcal{O}(\mathrm{SL}_n)$. We therefore have the commutative diagram



It follows from this diagram that $\mathcal{O}(\mathrm{PGL}_n) \longrightarrow \mathcal{O}(\mathrm{SL}_n)$ is injective. This proves the claim. □

Exercise A.21. Let $U \subset B \subset \mathrm{GL}_2$ be the closed subgroup of GL_2 consisting of upper triangular unipotent matrices and upper triangular matrices respectively. Describe the quotients SL_2/U, GL_2/U and B/U.

Solution. **(1) The quotient SL_2/U.** Consider the natural action

$$\mu : \mathrm{SL}_2 \times (\mathbb{A}^2 \setminus \{0\}) \longrightarrow \mathbb{A}^2 \setminus \{0\}$$

and the point $o = (1, 0) \in (\mathbb{A}^2 \setminus \{0\})(k)$. For every $R \in \mathrm{Alg}_k$, the orbit map is

$$\mu_o(R) : \mathrm{SL}_2(R) \longrightarrow (\mathbb{A}^2 \setminus \{0\})(R),$$
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \longmapsto \begin{pmatrix} a \\ c \end{pmatrix}.$$

Recall[14]that

$$(\mathbb{A}^2 \setminus \{0\})(R) = \{(r_1, r_2) \in R^2 \mid r_1 R + r_2 R = R\}.$$

It is easy to see that the induced map

$$\mathrm{SL}_2(R)/U(R) \longrightarrow (\mathbb{A}^2 \setminus \{0\})(R)$$

is injective. For $(\alpha, \beta) \in (\mathbb{A}^2 \setminus \{0\})(R)$, choose $\alpha', \beta' \in R$ such that $\alpha\alpha' + \beta\beta' = 1$. Then

$$\begin{pmatrix} \alpha & -\beta' \\ \beta & \alpha' \end{pmatrix} \in \mathrm{SL}_2(R)$$

maps to $\binom{\alpha}{\beta}$. Thus every element of $(\mathbb{A}^2 \setminus \{0\})(R)$ lifts to $\mathrm{SL}_2(R)$. Consequently by Definition 3.35,

$$\boxed{\mathrm{SL}_2/U \simeq \mathbb{A}^2 \setminus \{0\}.}$$

(2) The quotient GL_2/U**.** Consider

$$\begin{aligned} \mathrm{GL}_2 &\longrightarrow (\mathbb{A}^2 \setminus \{0\}) \times \mathbb{G}_m, \\ \begin{pmatrix} a & b \\ c & d \end{pmatrix} &\longmapsto \left(\begin{pmatrix} a \\ c \end{pmatrix}, \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} \right). \end{aligned}$$

For the corresponding action of GL_2 on $(\mathbb{A}^2 \setminus \{0\}) \times \mathbb{G}_m$, consider the fibre of the orbit map at $((1,0),1)$. It is easy to see that the induced map

$$\mathrm{GL}_2(R)/U(R) \longrightarrow (\mathbb{A}^2 \setminus \{0\})(R) \times \mathbb{G}_m(R)$$

is injective. Given $(\alpha, \beta) \in (\mathbb{A}^2 \setminus \{0\})(R)$ and $\lambda \in \mathbb{G}_m(R)$, choose $\alpha', \beta' \in R$ with $\alpha\alpha' + \beta\beta' = 1$. Then

$$\begin{pmatrix} \alpha & -\lambda\beta' \\ \beta & \lambda\alpha' \end{pmatrix}$$

has determinant λ and maps to $\left(\binom{\alpha}{\beta}, \lambda\right)$. The lifting condition holds, and hence

$$\boxed{\mathrm{GL}_2/U \simeq (\mathbb{A}^2 \setminus \{0\}) \times \mathbb{G}_m.}$$

(3) The quotient B/U**.** Consider

$$\begin{aligned} B &\longrightarrow \mathbb{G}_m \times \mathbb{G}_m, \\ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} &\longmapsto (a, d). \end{aligned}$$

[14] **Addition.** We justify the identity

$$(\mathbb{A}^2 \setminus \{0\})(R) = \{(\alpha, \beta) \in R^2 \mid \alpha R + \beta R = R\}.$$

The left-hand side is

$$\left\{\varphi : \operatorname{Spec} R \longrightarrow \mathbb{A}^2 \;\middle|\; \varphi^{-1}(\{0\}) \text{ is empty}\right\}.$$

Equivalently, it consists of the homomorphisms $k[x,y] \longrightarrow R$ for which no prime ideal of R lies over (x,y). For a morphism $\varphi : \operatorname{Spec} R \longrightarrow \mathbb{A}^2$ with

$$\varphi^{\#}(x) = \alpha, \qquad \varphi^{\#}(y) = \beta,$$

we have

$$\varphi^{-1}(\{0\}) \simeq \operatorname{Spec}\left(R \otimes_{k[x,y]} \frac{k[x,y]}{(x,y)}\right) \simeq \operatorname{Spec} \frac{R}{(\alpha,\beta)}.$$

Thus

$$\varphi^{-1}(\{0\}) = \varnothing \quad \Longleftrightarrow \quad R/(\alpha,\beta) = 0 \quad \Longleftrightarrow \quad \alpha R + \beta R = R,$$

as claimed.

For the corresponding action of B on $\mathbb{G}_m \times \mathbb{G}_m$, consider the fibre of the orbit map at $(1,1)$. The induced map

$$B(R)/U(R) \longrightarrow (\mathbb{G}_m \times \mathbb{G}_m)(R)$$

is obviously injective. Moreover,

$$\begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$$

lifts any $(a,d) \in (\mathbb{G}_m \times \mathbb{G}_m)(R)$ to $B(R)$. Therefore

$$\boxed{B/U \simeq \mathbb{G}_m \times \mathbb{G}_m.}$$

□

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